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Chapter 1

Methods and Preliminaries

1.1 The Unitary renormalisation group method

The URG method was introduced and formalised in refs. [1–4]. This section is adapted from those references and expanded wherever required.

Description of the problem

We are given a Hamiltonian $\mathcal{H}$ which is not completely diagonal in the occupation number basis of the electrons, $\hat{n}_k$: $[\mathcal{H}, n_k] \neq 0$. k labels any set of quantum numbers depending on the system. For spin-less Fermions it can be the momentum of the particle, while for spin-full Fermions it can be the set of momentum and spin. There are terms that scatter electrons from one quantum number k to another quantum number k' .

We take a general Hamiltonian,

$$\mathcal{H} = H_e \hat{n}_{q\beta} + H_h (1 - \hat{n}_{q\beta}) + c_{q\beta}^\dagger T + T^\dagger c_{q\beta} \quad (1.1)$$

Formally, we can decompose the entire Hamiltonian in the subspace of the electron we want to decouple ($q\beta$).

$$\mathcal{H} = \begin{pmatrix} |1\rangle & |0\rangle \\ H_1 & T \\ T^\dagger & H_0 \end{pmatrix} \quad (1.2)$$

The basis in which this matrix is written is $\{|1\rangle, |0\rangle\}$ where $|i\rangle$ is the set of all states where $\hat{n}_{q\beta} = i$. The aim of one step of the URG is to find a unitary transformation U such that the new Hamiltonian $U\mathcal{H}U^\dagger$ is diagonal in this already-chosen basis.

$$\tilde{\mathcal{H}} \equiv U\mathcal{H}U^\dagger = \begin{pmatrix} |1\rangle & |0\rangle \\ \tilde{H}_1 & 0 \\ 0 & \tilde{H}_0 \end{pmatrix} \quad (1.3)$$

U_q is defined by

$$\tilde{\mathcal{H}} = U_q \mathcal{H} U_q^\dagger \text{ such that } [\tilde{\mathcal{H}}, n_q] = 0 \quad (1.4)$$

It is clear that U is the diagonalizing matrix for $\mathcal{H}$. Hence we can frame this problem as an eigenvalue equation as well. Let $|\psi_1\rangle, |\psi_0\rangle$ be the basis in which the original Hamiltonian $\mathcal{H}$ has no off-diagonal terms corresponding to $q\beta$. Hence, we can write

$$\mathcal{H} |\psi_i\rangle = \tilde{H}_i |\psi_i\rangle, i \in \{0, 1\} \quad (1.5)$$

Since $|\psi_i\rangle$ is the set of eigenstates of $\mathcal{H}$ and $|i\rangle$ is the set of eigenstates in which $U\mathcal{H}U^\dagger$ has no off-diagonal terms corresponding to $q\beta$, we can relate $|\psi_i\rangle$ and $|i\rangle$ by the same transformation : $|\psi_i\rangle = U^\dagger |i\rangle$. We can expand the state $|\psi_i\rangle$ in the subspace of $q\beta$:

$$|\psi_i\rangle = \sum_{j=0,1} |j\rangle \langle j | \psi_i \rangle \equiv |1\rangle |\phi_1^i\rangle + |0\rangle |\phi_0^i\rangle \quad (1.6)$$

where $|\phi_j^i\rangle = \langle j | \psi_i \rangle$. If we substitute the expansion 1.2 into the eigenvalue equation 1.5, we get

$$\left[H_e \hat{n}_{q\beta} + H_h (1 - \hat{n}_{q\beta}) + c_{q\beta}^\dagger T + T^\dagger c_{q\beta} \right] |\psi_i\rangle = \tilde{H}_i |\psi_i\rangle \quad (1.7)$$

The diagonal parts $H_e = \text{tr} [\mathcal{H} \hat{n}_{q\beta}]$ and $H_h = \text{tr} [\mathcal{H} (1 - \hat{n}_{q\beta})]$ can be separated into a purely diagonal part $\mathcal{H}^d$ that contains the single-particle energies and the multi-particle correlation energies or Hartree-like contributions, and an off-diagonal part $\mathcal{H}^i$ that scatters between the remaining degrees of freedom $k\sigma \neq q\beta$. That is,

$$H_e \hat{n}_{q\beta} + H_h (1 - \hat{n}_{q\beta}) = \mathcal{H}^d + \mathcal{H}^i$$

This gives

$$\left[c_{q\beta}^\dagger T + T^\dagger c_{q\beta} \right] |\psi_i\rangle = \left(\tilde{H}_i - \mathcal{H}^i - \mathcal{H}^d \right) |\psi_i\rangle \quad (1.8)$$

Obtaining the decoupling transformation

We now define a new operator $\hat{\omega}_i = \tilde{H}_i - \mathcal{H}^i$, such that

$$\left[c_{q\beta}^\dagger T + T^\dagger c_{q\beta} \right] |\psi_i\rangle = \left(\hat{\omega}_i - \mathcal{H}^d \right) |\psi_i\rangle \quad (1.9)$$

From the definition of $\hat{\omega}_i$, we can see that it is Hermitian and has no term that scatters in the subspace of $q\beta$, so it is diagonal in $q\beta$ and we can expand it as $\hat{\omega}_i = \hat{\omega}_i^1 \hat{n}_{q\beta} + \hat{\omega}_i^0 (1 - \hat{n}_{q\beta})$. Using the expansion 1.6, we can write

$$\hat{\omega}_i |\psi_i\rangle = \hat{\omega}_i^1 |1\rangle |\phi_1^i\rangle + \hat{\omega}_i^0 |0\rangle |\phi_0^i\rangle \quad (1.10)$$

Since the only requirement on $|\psi_i\rangle$ is that it diagonalize the Hamiltonian in the subspace of $q\beta$, there is freedom in the choice of this state. We can exploit this freedom and choose the $|\phi_0^i\rangle$ to be an eigenstates of $\hat{\omega}_i^{1,0}$ corresponding to real eigenvalues $\omega_i^{1,0}$:

$$\left[\mathcal{H}^d + c_{q\beta}^\dagger T + T^\dagger c_{q\beta} \right] |\psi_i(\omega_i)\rangle = \left(\omega_i^1 - \mathcal{H}^d \right) |1\rangle |\phi_1^i\rangle + \left(\omega_i^0 - \mathcal{H}^d \right) |0\rangle |\phi_0^i\rangle \quad (1.11)$$

If we now substitute the expansion 1.6 and gather the terms that result in $\hat{n}_{q\beta} = 1$, we get

$$c_{q\beta}^\dagger T |0\rangle |\phi_0^i\rangle = (\omega_i^1 - \mathcal{H}^d) |1\rangle |\phi_1^i\rangle \quad (1.12)$$

Similarly, gathering the terms that result in $\hat{n}_{q\beta} = 0$ gives

$$T^\dagger c_{q\beta} |1\rangle |\phi_1^i\rangle = (\omega_i^0 - \mathcal{H}^d) |0\rangle |\phi_0^i\rangle \quad (1.13)$$

We now define two many-particle transition operators:

$$\begin{aligned} \eta^\dagger(\omega_i^1) &= \frac{1}{\omega_i^1 - \mathcal{H}^d} c_{q\beta}^\dagger T \equiv G_1 c_{q\beta}^\dagger T \\ \eta(\omega_i^0) &= \frac{1}{\omega_i^0 - \mathcal{H}^d} T^\dagger c_{q\beta} \equiv G_0 T^\dagger c_{q\beta} \end{aligned} \quad (1.14)$$

where G_j is the propagator $\frac{1}{\omega_i^j - \mathcal{H}^d}$. We can write this compactly as

$$\eta(\hat{\omega}) = G T^\dagger c_{q\beta} = \frac{1}{\hat{\omega} - \mathcal{H}^d} T^\dagger c_{q\beta} \quad (1.15)$$

where $\hat{\omega}_i = \omega_i^0 (1 - \hat{n}_{q\beta}) + \omega_i^1 \hat{n}_{q\beta} = \begin{pmatrix} \omega_i^1 & \\ & \omega_i^0 \end{pmatrix}$ is a 2x2 matrix and $\mathcal{H}^d = \mathcal{H}_0^d (1 - \hat{n}_{q\beta}) + \mathcal{H}_1^d \hat{n}_{q\beta}$ and $G = (\hat{\omega} - \mathcal{H}^d)^{-1}$. It is easy to check that this reproduces the previous forms of η_0 and $\eta_1^\dagger$. We will later find that it is important to demand that these two be Hermitian conjugates of each other; that constraint is imposed on the denominators:

$$\eta^\dagger(\omega_i^0) = \eta^\dagger(\omega_i^1) \implies \frac{1}{\omega_i^1 - \mathcal{H}^d} c_{q\beta}^\dagger T = c_{q\beta}^\dagger T \frac{1}{\omega_i^0 - \mathcal{H}^d} \quad (1.16)$$

Henceforth we will assume that this constraint has been imposed.

In terms of these operators, eq. 1.13 becomes

$$|1\rangle |\phi_1^i\rangle = \eta^\dagger |0\rangle |\phi_0^i\rangle, \quad |0\rangle |\phi_0^i\rangle = \eta |1\rangle |\phi_1^i\rangle \quad (1.17)$$

These allow us to write

$$|\psi_1\rangle = |1\rangle |\phi_1^i\rangle + |0\rangle |\phi_0^i\rangle = (1 + \eta) |1\rangle |\phi_1^i\rangle, \quad |\psi_0\rangle = (1 + \eta^\dagger) |0\rangle |\phi_0^i\rangle \quad (1.18)$$

Recalling that $|\psi_i\rangle = U^\dagger |i\rangle$, we can read off the required transformation:

$$U_1 = 1 + \eta \quad (1.19)$$

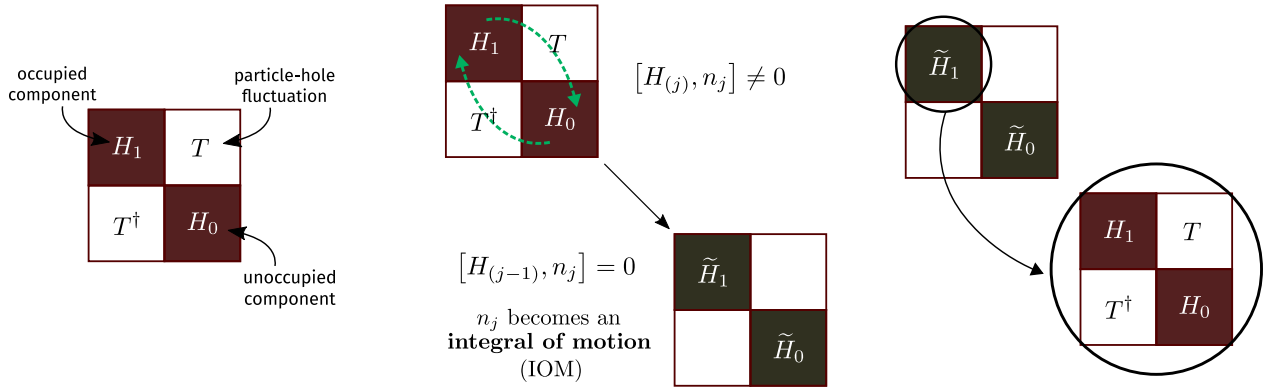


Figure 1.1: Three steps of the URG: Decompose the Hamiltonian in a 2×2 matrix, apply the unitary operator to rotate it, then repeat these steps with one of the rotated blocks.

Properties of the many-body transition operators

The operators η have some important properties. First is the Fermionic nature:

$$\eta^2 = \eta^{\dagger 2} = 0 \quad [c^{\dagger 2} = c^2 = 0] \quad (1.20)$$

Second is:

$$\begin{aligned} |1\rangle |\phi_1^i\rangle &= \eta^\dagger |0\rangle |\phi_0^i\rangle = \eta^\dagger \eta |1\rangle |\phi_1^i\rangle \implies \eta^\dagger \eta = \hat{n}_{q\beta} \\ |0\rangle |\phi_0^i\rangle &= \eta |1\rangle |\phi_1^i\rangle = \eta \eta^\dagger |0\rangle |\phi_0^i\rangle \implies \eta \eta^\dagger = 1 - \hat{n}_{q\beta} \end{aligned} \quad (1.21)$$

and hence the anticommutator

$$\implies \{\eta, \eta^\dagger\} = 1 \quad (1.22)$$

Note that the three equations in 1.21 work only when applied on the eigenstate $|\psi_i\rangle$ and not any arbitrary state.

$$\begin{aligned} \eta^\dagger \eta |\psi_i\rangle &= |1\rangle |\phi_1^i\rangle = \hat{n}_{q\beta} |\psi_i\rangle \\ \eta \eta^\dagger |\psi_i\rangle &= |0\rangle |\phi_0^i\rangle = (1 - \hat{n}_{q\beta}) |\psi_i\rangle \\ \{\eta^\dagger, \eta\} |\psi_i\rangle &= |\psi_i\rangle \end{aligned}$$

Form of the unitary operators

Although we have found the correct similarity transformations U_i (eqs. 1.19), we need to convert them into a unitary transformation. Say we are trying to rotate the eigenstate $|\psi_1\rangle$ into the state $|1\rangle$. We can then work with the transformation

$$U_1 = 1 + \eta \quad (1.23)$$

In this form, this transformation is not unitary. It can however be written in an exponential form:

$$U_1 = e^\eta \quad (1.24)$$

using the fact that $\eta^2 = 0$. It is shown in ref. [5] that corresponding to a similarity transformation e^ω , there exists a unitary transformation e^G where

$$G = \tanh^{-1} (\omega - \omega^\dagger) \quad (1.25)$$

Applying that to the problem at hand gives

$$U_1^\dagger = \exp \left\{ \tanh^{-1} (\eta - \eta^\dagger) \right\} \quad (1.26)$$

Let $x = \tanh y$. Then,

$$x = \frac{e^{2y} + 1}{e^{2y} - 1} \implies y = \frac{1}{2} \log \frac{1+x}{1-x} \implies e^y = e^{\tanh^{-1} x} = \sqrt{\frac{1+x}{1-x}} \quad (1.27)$$

Therefore,

$$\exp \left\{ \tanh^{-1} (\eta - \eta^\dagger) \right\} = \frac{1 + \eta - \eta^\dagger}{\sqrt{(1 + \eta^\dagger - \eta)(1 - \eta^\dagger + \eta)}} = \frac{1 + \eta - \eta^\dagger}{\sqrt{1 + \{\eta, \eta^\dagger\}}} = \frac{1}{\sqrt{2}} (1 + \eta - \eta^\dagger) \quad (1.28)$$

The *unitary* operator that transforms the entangled eigenstate $|\psi_1\rangle$ to the state $|1\rangle$ is thus

$$U_1 = \frac{1}{\sqrt{2}} (1 + \eta^\dagger - \eta) \quad (1.29)$$

It can also be written as $\exp \left\{ \frac{\pi}{4} (\eta^\dagger - \eta) \right\}$ because

$$\begin{aligned} \exp \left\{ \frac{\pi}{4} (\eta^\dagger - \eta) \right\} &= 1 + (\eta^\dagger - \eta) \frac{\pi}{4} + \frac{1}{2!} (\eta^\dagger - \eta)^2 \left(\frac{\pi}{4} \right)^2 + \frac{1}{3!} (\eta^\dagger - \eta)^3 \left(\frac{\pi}{4} \right)^3 + \dots \\ &= 1 + (\eta^\dagger - \eta) \frac{\pi}{4} - \frac{1}{2!} \left(\frac{\pi}{4} \right)^2 - \frac{1}{3!} (\eta^\dagger - \eta) \left(\frac{\pi}{4} \right)^3 + \frac{1}{4!} \left(\frac{\pi}{4} \right)^4 + \dots \\ &= \cos \frac{\pi}{4} + (\eta^\dagger - \eta) \sin \frac{\pi}{4} \\ &= \frac{1}{\sqrt{2}} (1 + \eta^\dagger - \eta) \end{aligned} \quad (1.30)$$

There we used

$$(\eta^\dagger - \eta)^2 = \eta^{\dagger 2} + \eta^2 - \{\eta^\dagger, \eta\} = -1 \quad \left[\because \eta^2 = \eta^{\dagger 2} = 0 \right] \quad (1.31)$$

and hence

$$(\eta^\dagger - \eta)^3 = -1 (\eta^\dagger - \eta) \quad (1.32)$$

and so on.

Effective Hamiltonian

We can now compute the form of the effective Hamiltonian that comes about when we apply U_1 - that is - when we rotate one exact eigenstate $|\psi_1\rangle$ into the occupied Fock space basis $|1\rangle$. From eq. 1.29,

$$\begin{aligned}
 U_1 \mathcal{H} U_1^\dagger &= \frac{1}{2} \left(1 + \eta^\dagger - \eta \right) \mathcal{H} \left(1 + \eta - \eta^\dagger \right) \\
 &= \frac{1}{2} \left(1 + \eta^\dagger - \eta \right) \left(\mathcal{H} + \mathcal{H}\eta - \mathcal{H}\eta^\dagger \right) \\
 &= \frac{1}{2} \left(\mathcal{H} + \mathcal{H}\eta - \mathcal{H}\eta^\dagger + \eta^\dagger \mathcal{H} + \eta^\dagger \mathcal{H}\eta - \eta^\dagger \mathcal{H}\eta^\dagger - \eta \mathcal{H} - \eta \mathcal{H}\eta + \eta \mathcal{H}\eta^\dagger \right) \\
 &= \frac{1}{2} \left(\mathcal{H}^d + \mathcal{H}^i + \mathcal{H}^l + \mathcal{H}\eta - \mathcal{H}\eta^\dagger + \eta^\dagger \mathcal{H} + \eta^\dagger \mathcal{H}\eta - \eta^\dagger \mathcal{H}\eta^\dagger - \eta \mathcal{H} - \eta \mathcal{H}\eta + \eta \mathcal{H}\eta^\dagger \right) \\
 &= \frac{1}{2} \left(\mathcal{H}^d + \mathcal{H}^i + \mathcal{H}^l + [\eta^\dagger - \eta, \mathcal{H}] + \eta^\dagger \mathcal{H}\eta - \eta^\dagger \mathcal{H}\eta^\dagger - \eta \mathcal{H}\eta + \eta \mathcal{H}\eta^\dagger \right)
 \end{aligned} \tag{1.33}$$

In the last two lines, we expanded the Hamiltonian into the three parts $\mathcal{H}^d$, $\mathcal{H}^i$ and a third piece $\mathcal{H}^l \equiv c_{q\beta}^\dagger T + T^\dagger c_{q\beta}$.

For reasons that will become apparent, we will split the terms into two groups:

$$\tilde{\mathcal{H}} = \frac{1}{2} \left(\underbrace{\mathcal{H}^d + \mathcal{H}^i + [\eta^\dagger - \eta, \mathcal{H}] + \eta^\dagger \mathcal{H}\eta + \eta \mathcal{H}\eta^\dagger}_{\text{group 1}} + \overbrace{\mathcal{H}^l - \eta^\dagger \mathcal{H}\eta^\dagger - \eta \mathcal{H}\eta}^{\text{group 2}} \right) \tag{1.34}$$

Group 2 can be easily shown to be 0. Note that terms that have two η or two $\eta^\dagger$ sandwiching a $\mathcal{H}$ can only be nonzero if the intervening $\mathcal{H}$ has an odd number of creation or destruction operators.

$$\eta \mathcal{H} \eta = \eta c_q^\dagger T \eta \tag{1.35}$$

and

$$\eta^\dagger \mathcal{H} \eta^\dagger = \eta^\dagger T^\dagger c_q \eta^\dagger \tag{1.36}$$

Group 2 becomes

$$\text{group 2} = \mathcal{H}^l - \eta^\dagger T^\dagger c_q \eta^\dagger - \eta c_q^\dagger T \eta = c_q^\dagger T + T^\dagger c_q - \eta^\dagger T^\dagger c_q \eta^\dagger - \eta c_q^\dagger T \eta \tag{1.37}$$

To simplify this, we use the relation

$$\begin{aligned}
 \eta c_q^\dagger T \eta &= \frac{1}{\omega_i^0 - \mathcal{H}^d} T^\dagger c_q c_q^\dagger T \eta \\
 &= T^\dagger c_q \frac{1}{\omega_i^1 - \mathcal{H}^d} c_q^\dagger T \eta \quad [\text{eq. 1.16}] \\
 &= T^\dagger c_q \eta^\dagger \eta \quad [\text{eq. 1.15}] \\
 &= T^\dagger c_q \hat{n}_q \quad [\text{eq. 1.21}]
 \end{aligned} \tag{1.38}$$

which gives

$$\eta c_q^\dagger T \eta = T^\dagger c_q \quad (1.39)$$

Taking the Hermitian conjugate of eq. 1.39 gives

$$\eta^\dagger T^\dagger c_q \eta^\dagger = c_q^\dagger T \quad (1.40)$$

Substituting the expressions 1.39 and 1.40 into the expression for group 2, 1.37, shows that it vanishes. This leaves us only with group 1:

$$\tilde{\mathcal{H}} = \frac{1}{2} \left(\mathcal{H}^d + \mathcal{H}^i + \overbrace{\eta^\dagger \mathcal{H} \eta + \eta \mathcal{H} \eta^\dagger}^{\text{group A}} + \underbrace{[\eta^\dagger - \eta, \mathcal{H}]}_{\text{group B}} \right) \quad (1.41)$$

Group A simplifies in the following way. First note that $\eta^\dagger \mathcal{H}^I \eta = \eta^\dagger \mathcal{H}^I \eta = 0$ must be 0 because it will involve consecutive $c_{q\beta}$ or consecutive $c_{q\beta}^\dagger$. We are therefore left with the diagonal part of $\mathcal{H}$, which is $H_e \hat{n}_{q\beta} + H_h (1 - \hat{n}_{q\beta})$.

$$\eta^\dagger [H_e \hat{n}_{q\beta} + H_h (1 - \hat{n}_{q\beta})] \eta + \eta [H_e \hat{n}_{q\beta} + H_h (1 - \hat{n}_{q\beta})] \eta^\dagger = \eta^\dagger H_h \eta + \eta H_e \eta^\dagger \quad (1.42)$$

This can be shown to be equal to the diagonal part:

$$\text{group A} = \eta^\dagger H_h \eta + \eta H_e \eta^\dagger = H_e \hat{n}_{q\beta} + H_h (1 - \hat{n}_{q\beta}) = \mathcal{H}^d + \mathcal{H}^i \quad (1.43)$$

It can also be shown that

$$\text{group B} = [\eta^\dagger - \eta, \mathcal{H}] = 2 [c_{q\beta}^\dagger T, \eta] \quad (1.44)$$

Putting it all together,

$$\tilde{\mathcal{H}} = \mathcal{H}^d + \mathcal{H}^i + [c_{q\beta}^\dagger T, \eta] \quad (1.45)$$

The renormalizing in the Hamiltonian is

$$\Delta \mathcal{H} = \tilde{\mathcal{H}} - \mathcal{H}^d - \mathcal{H}^i = [c_{q\beta}^\dagger T, \eta] \quad (1.46)$$

Because of eq. 1.44, it can also be written as

$$\Delta \mathcal{H} = \frac{1}{2} [\eta^\dagger - \eta, \mathcal{H}_X] = \frac{1}{2} [\eta^\dagger - \eta, \mathcal{H}] \quad (1.47)$$

This form will be useful later when we make the connection with one-shot Schrieffer-Wolff transformation and CUT RG.

To check that the renormalised Hamiltonian indeed commutes with $\hat{n}_{q\beta}$,

$$\begin{aligned} [\tilde{\mathcal{H}}, \hat{n}_{q\beta}] &= [[c_{q\beta}^\dagger T, \eta], \hat{n}_{q\beta}] = [c_{q\beta}^\dagger T \eta, \hat{n}_{q\beta}] - [\eta c_{q\beta}^\dagger T, \hat{n}_{q\beta}] \\ &= c_{q\beta}^\dagger T \eta \hat{n}_{q\beta} - \hat{n}_{q\beta} c_{q\beta}^\dagger T \eta \quad \left[2^{\text{nd}} [\cdot] \text{ is 0, } \because c_{q\beta}^\dagger \hat{n}_{q\beta} = \hat{n}_{q\beta} \eta = 0 \right] \\ &= c_{q\beta}^\dagger T \eta - c_{q\beta}^\dagger T \eta = 0 \end{aligned} \quad (1.48)$$

Fixed point condition

Within the URG, it is a prescription that the fixed point is reached when the denominator of the RG equation vanishes. This is equivalent to either $\omega_i^1 = \mathcal{H}_1^d$ or $\omega_i^0 = \mathcal{H}_0^d$. This shows that at the fixed point, one of the eigenvalues of $\hat{\omega}_i$ matches the corresponding eigenvalue of the diagonal blocks. This also leads to the vanishing of the off-diagonal block, because eqs. 1.12 and 1.13 gives

$$c_{q\beta}^\dagger T |0\rangle |\phi_0^i\rangle = \left(\omega_i^1 - \mathcal{H}_1^d \right) |1\rangle |\phi_1^i\rangle = 0 \implies c_{q\beta}^\dagger T = 0 \quad (1.49)$$

Multiple off-diagonal terms

There is a subtle assumption in the definitions eq. 1.14. In order for η to be the Hermitian conjugate of $\eta^\dagger$, $\mathcal{H}_d$ cannot have any information that relates to the structure of T . To see why, say the total off-diagonal term is composed of two parts: $T = T_1 + T_2$.

$$\begin{aligned} \eta &= \frac{1}{\omega_0 - \mathcal{H}_d} \left(T_1^\dagger + T_2^\dagger \right) c = \left[\frac{1}{\omega^0 - E_1^0} T_1^\dagger c + \frac{1}{\omega^0 - E_2^0} T_2^\dagger c \right] \\ \eta^\dagger &= \frac{1}{\omega^1 - \mathcal{H}_d} c^\dagger (T_1 + T_2) = \left[\frac{1}{\omega^1 - E_1^1} c^\dagger T_1 + \frac{1}{\omega^1 - E_2^1} c^\dagger T_2 \right] \end{aligned} \quad (1.50)$$

where $\mathcal{H}_d T_i^\dagger c = E_i^0 T_i^\dagger c$ and $\mathcal{H}_d c^\dagger T_i = E_i^1 c^\dagger T_i$. We can now see that in order for $\eta = (\eta^\dagger)^\dagger$ to hold, two conditions must be met:

$$\omega^0 - E_1^0 = \omega^1 - E_1^1, \quad \omega^0 - E_2^0 = \omega^1 - E_2^1 \quad (1.51)$$

This will not hold generally. The correct solution is to realize that each such off-diagonal term T_i will come with its own quantum fluctuation scale ω_i .

$$\begin{aligned} \eta &= \sum_i \frac{1}{\omega_i^0 - E_i^0} T_i^\dagger c \\ \eta^\dagger &= \sum_i \frac{1}{\omega_i^1 - E_i^1} c^\dagger T_i \end{aligned} \quad (1.52)$$

If we now impose the condition that $\eta = (\eta^\dagger)^\dagger$, we get the relations

$$\omega_i^0 - \omega_i^1 = E_i^0 - E_i^1 \quad (1.53)$$

and so

$$\eta^\dagger - \eta = \sum_i \frac{1}{\omega_i^0 - E_i^0} \left(c^\dagger T_i - T_i^\dagger c \right) \quad (1.54)$$

The expression for the renormalization will not be just $[c^\dagger T, \eta]$ in this case. That form will be non-Hermitian. The correct form is obtained from the more general form $[\eta^\dagger - \eta, \mathcal{H}_X]$:

$$\begin{aligned}\Delta\mathcal{H} &= \frac{1}{2} [\eta^\dagger - \eta, c^\dagger T + T^\dagger c] = \frac{1}{2} \sum_{ij} \frac{1}{\omega_i^0 - E_i^0} [c^\dagger T_i - T_i^\dagger c, c^\dagger T_j + T_j^\dagger c] \\ &= \frac{1}{2} \sum_{ij} \frac{1}{\omega_i^0 - E_i^0} \left[\hat{n} (T_i T_j^\dagger + T_j T_i^\dagger) - (1 - \hat{n}) (T_i^\dagger T_j + T_j^\dagger T_i) \right] \\ &= \frac{1}{2} \sum_{ij} \left(\frac{1}{\omega_i^0 - E_i^0} + \frac{1}{\omega_j^0 - E_j^0} \right) [\hat{n} T_i T_j^\dagger - (1 - \hat{n}) T_i^\dagger T_j]\end{aligned}\tag{1.55}$$

Equivalence of the two unitaries and preservation of partial trace

In the subsection 1.1, we determined the form of the operator U_1 that unitarily decouples the node $q\beta$ from the other degrees of freedom. Eq. 1.29 was derived by reading off the transformation of $|1\rangle$ to $|\psi_1\rangle$, the first equation in 1.18. We could easily have chosen the other equation in the same equation set,

$$|\psi_0\rangle = (1 + \eta^\dagger) |0\rangle |\phi_0^i\rangle$$

which gives a similarity transformation $1 + \eta^\dagger$ and hence a unitary

$$U_0 = \frac{1}{\sqrt{2}} (1 + \eta - \eta^\dagger)\tag{1.56}$$

This η will however be different from the η in eq. 1.29. The reason is, in order to get U_1 , we must start from the eigenvalue equation $\mathcal{H} |\psi_1\rangle = \tilde{H}_1 |\psi_1\rangle$. This means that the corresponding $\hat{\omega}$ will be defined as $\hat{\omega}_1 = \tilde{H}_1 - \mathcal{H}^i$. On the other hand, in order to get U_0 we must start with $\mathcal{H} |\psi_0\rangle = \tilde{H}_0 |\psi_0\rangle$, and hence this $\hat{\omega}$ will be $\hat{\omega}_0 = \tilde{H}_0 - \mathcal{H}^i$. This difference in the $\hat{\omega}$ will define two different sets of η :

$$\begin{aligned}\text{Starting from } |\psi_1\rangle: \eta_1 &= \frac{1}{\omega_1^0 - \mathcal{H}^d} T^\dagger c_{q\beta} \quad \text{and} \quad \eta_1^\dagger = \frac{1}{\omega_1^1 - \mathcal{H}^d} T^\dagger c_{q\beta} \\ \text{Starting from } |\psi_0\rangle: \eta_0 &= \frac{1}{\omega_0^0 - \mathcal{H}^d} T^\dagger c_{q\beta} \quad \text{and} \quad \eta_0^\dagger = \frac{1}{\omega_0^1 - \mathcal{H}^d} T^\dagger c_{q\beta}\end{aligned}\tag{1.57}$$

The ω_j^i eigenvalues have both upper and lower indices. The upper index i signifies which eigenstate it relates to - $\omega_j |i\rangle = \omega_j^i |i\rangle$. The lower index refers to the exact eigenstate we started with - starting with $\mathcal{H} |\psi_j\rangle = \tilde{H}_j |\psi_j\rangle$ leads to ω_j . The two unitaries are

$$\begin{aligned}U_1 &= \frac{1}{\sqrt{2}} (1 + \eta_1^\dagger - \eta_1) \\ U_0 &= \frac{1}{\sqrt{2}} (1 + \eta_0 - \eta_0^\dagger)\end{aligned}\tag{1.58}$$

Since the two unitaries should give the same effective Hamiltonian, we require $U_1 = U_0$. That requires $\eta_1 = -\eta_0$. Comparing the expressions of the η s, we get

$$\omega_1^0 - \mathcal{H}_0^d = -(\omega_0^0 - \mathcal{H}_0^d) \quad (1.59)$$

This is the constraint that ensures that both unitaries give the same effective Hamiltonian. The condition $\eta_1 + \eta_0 = 0$, when expressed without resolving $\hat{\omega}$ into its eigenvalues can also be shown to be a statement of the preservation of the partial trace under the RG flow.

$$\begin{aligned} \eta_1 &= \frac{1}{\tilde{H}_1 - \mathcal{H}^i - \mathcal{H}^d} T^\dagger c_{q\beta}, \quad \eta_0 = \frac{1}{\tilde{H}_0 - \mathcal{H}^i - \mathcal{H}^d} T^\dagger c_{q\beta} \\ \implies \eta_1 + \eta_0 &= \left[\frac{1}{\tilde{H}_1 - \mathcal{H}^i - \mathcal{H}^d} + \frac{1}{\tilde{H}_0 - \mathcal{H}^i - \mathcal{H}^d} \right] T^\dagger c_{q\beta} = 0 \\ \implies \tilde{H}_1 - \mathcal{H}^i - \mathcal{H}^d &= -[\tilde{H}_0 - \mathcal{H}^i - \mathcal{H}^d] \\ \implies \tilde{H}_1 + \tilde{H}_0 &= 2\mathcal{H}_0 \end{aligned} \quad (1.60)$$

$\mathcal{H}_0 = \mathcal{H}^i + \mathcal{H}^d$ is the total diagonal part of the bare model. To match the dimensions, we must take $\tilde{H}_1 = E_1 \otimes I$ and similarly $\tilde{H}_0 = E_0 \otimes I$, where the rotated Hamiltonian is

$$\tilde{H} = \begin{pmatrix} E_1 & 0 \\ 0 & E_0 \end{pmatrix} \quad (1.61)$$

Therefore, the trace of the rotated Hamiltonian is $t_{\text{new}} = E_1 + E_0$. The trace of the LHS in the final equation of 1.60 is $\text{tr}(\tilde{H}_1 + \tilde{H}_0) = \text{tr}(E_1 \otimes I + E_0 \otimes I) = 2(E_1 + E_0) = 2t_{\text{new}}$. The trace of the RHS in final equation of 1.60 is $2 \times \text{tr}(\mathcal{H}_0) = 2t_{\text{old}}$ where $t_{\text{old}} = \text{tr}(\mathcal{H}_0)$ is the trace of the old Hamiltonian. Equating the LHS and RHS gives $t_{\text{new}} = t_{\text{old}}$.

Complete generator for the unitary transformation

Given some operator O_0 , we can generate a family of unitarily-connected operators O_j using a unitary operator $U(t)$:

$$O_j = U_j O(0) U_j^\dagger, \quad j = 1, 2, \dots \quad (1.62)$$

The discrete change equation for O_j can be represented in the form of a commutator:

$$\Delta O_j \equiv O_{j+1} - O_j = [O_j, S_j] \quad (1.63)$$

where

$$S_j = U_j \Delta U_j^\dagger. \quad (1.64)$$

Note that because $\Delta(U_j U_j^\dagger) = 0$, we have $(\Delta U_j) U_j^\dagger = -U_j (\Delta U_j^\dagger)$ and so S_j is anti-Hermitian. To verify that eq. 1.62 is indeed the solution of eq. 1.63, we differentiate eq. 1.62:

$$O_{j+1} - O_j = \Delta U_j O(0) U_j^\dagger + U_j O(0) \Delta U_j^\dagger = \Delta U_j U_j^\dagger O_j + O_j U_j \Delta U_j^\dagger = [O_j, S_j] \quad (1.65)$$

This shows that given a family of operators eq. 1.62 connected through U_j , we can obtain a generator S_j that defines the flow equation of O_j .

Since the URG is unitary, we should be able to obtain such a generator for it as well. From the expression of the unitary transformation of URG:

$$U_j = \frac{1}{\sqrt{2}} \left(1 + \eta_j^\dagger - \eta_j \right) \quad (1.66)$$

From the definition of the generator S_j , we then get

$$S_j = \frac{1}{2} \left(1 + \eta_j^\dagger - \eta_j \right) \left(\eta_{j+1} - \eta_{j+1}^\dagger - \eta_j + \eta_j^\dagger \right) \quad (1.67)$$

The operators η_j and its hermitean conjugate can be thought of as angular momentum creation and annihilation operators acting on the 2×2 Hilbert space of the occupied and vacant states $|1\rangle |\phi_1\rangle, |0\rangle |\phi_0\rangle$:

$$\eta_j |1\rangle |\phi_1\rangle = |0\rangle |\phi_0\rangle, \quad \eta_j |0\rangle |\phi_0\rangle = 0, \quad \eta_j^\dagger |0\rangle |\phi_0\rangle = |1\rangle |\phi_1\rangle, \quad \eta_j^\dagger |1\rangle |\phi_1\rangle = 0, \quad (1.68)$$

$$(1.69)$$

To check whether they have the correct algebra, we design the three spin operators $S^i, i = \{x, y, z\}$.

$$\begin{aligned} S^x &= \frac{1}{2} (S^+ + S^-) = \frac{1}{2} (\eta_j^\dagger + \eta_j) \\ S^y &= \frac{1}{2i} (S^+ - S^-) = \frac{1}{2i} (\eta_j^\dagger - \eta_j) \\ S^z &= \hat{n} - \frac{1}{2} \end{aligned} \quad (1.70)$$

The commutation relations give

$$\begin{aligned} [S^x, S^y] &= \frac{1}{4i} [\eta_j^\dagger + \eta_j, \eta_j^\dagger - \eta_j] = \frac{1}{2i} [\eta_j, \eta_j^\dagger] = \frac{1}{2i} (1 - \hat{n} - \hat{n}) = \frac{-1}{i} \left(\hat{n} - \frac{1}{2} \right) = iS^z \\ [S^y, S^z] &= \frac{1}{2i} \left[\eta_j^\dagger - \eta_j, \hat{n} - \frac{1}{2} \right] = \frac{1}{2i} [-\eta_j \hat{n} - \hat{n} \eta_j^\dagger] = \frac{1}{2i} (-\eta_j - \eta_j^\dagger) = \frac{i}{2} (\eta_j^\dagger + \eta_j) = iS^x \\ [S^z, S^x] &= \frac{1}{2} \left[\hat{n} - \frac{1}{2}, \eta_j^\dagger + \eta_j \right] = \frac{1}{2} [\hat{n} \eta_j^\dagger - \eta_j \hat{n}] = \frac{1}{2} (\eta_j^\dagger - \eta_j) = i \frac{1}{2i} (\eta_j^\dagger - \eta_j) = iS^y \end{aligned} \quad (1.71)$$

These operators therefore satisfy the commutation algebra of angular momentum operators $[S^i, S^j] = i\epsilon^{ijk} S^k$.

A note on the various quantum fluctuation scales ω_i^j

At a particular step of the URG, there are two quantum fluctuation energy scales associated with each sector. If we rotate $|\psi_1\rangle$ to $|1\rangle$ (particle/occupied sector), the corresponding unitary will be a

function of $\omega_1^{0,1}$. If we, on the other hand, rotate $|\psi_0\rangle$ to $|0\rangle$ (hole/unoccupied sector), the unitary will be a function of $\omega_0^{0,1}$. The superscript j signifies whether this particular ω_i^j is an eigenvalue corresponding to $|1, \phi_i\rangle$ or $|0, \phi_i\rangle$. ω_i^0 occurs in the many-body transition operator η , because η is preceded by c and hence it picks out the eigenstate $|0, \phi_i\rangle$. On the other hand, ω_i^1 occurs in the many-body transition operator $\eta^\dagger$, because that is preceded by $c^\dagger$. This constrains these two values, because we must have $\eta(\omega_i^0) = (\eta^\dagger(\omega_i^1))^\dagger$ (eq. 1.16), for each value of i , giving us two constraints in total. The subscript i signifies whether ω_i^j is a part of the particle sector unitary $U_1(\omega_1^j)$ or the hole sector unitary $U_0(\omega_0^j)$. As mentioned in the previous section, since both ways are equivalent, we must have $U_1 = U_0$ which leads to the constraints $\eta(\omega_0^j) = -\eta(\omega_1^j)$. All the independent constraints are listed below.

$$\begin{aligned}\omega_1^0 - \omega_1^1 &= \mathcal{H}_d^0 - \mathcal{H}_d^1 \\ \omega_0^0 - \omega_0^1 &= \mathcal{H}_d^0 - \mathcal{H}_d^1 \\ \omega_1^0 + \omega_0^0 &= 2\mathcal{H}_d^0\end{aligned}\tag{1.72}$$

The first two come from $\eta(\omega_i^0) = (\eta^\dagger(\omega_i^1))^\dagger$ while the last comes from $\eta(\omega_0^j) = -\eta(\omega_1^j)$. These are the only independent relations. Other relations like the one between ω_1^0 and ω_0^1 can be derived from these. This means that we have four ω and three constraints, such that each step of the URG is characterized by just a single independent quantum fluctuation scale.

Prescription

Given a Hamiltonian

$$\mathcal{H} = \mathcal{H}_1 + \mathcal{H}_0 + c^\dagger T + T^\dagger c\tag{1.73}$$

the goal is to look at the renormalization of the various couplings in the Hamiltonian as we decouple high energy electron states. Typically we have a shell of electrons at some energy D . During the process, we make one simplification. We assume that there is only one electron on that shell at a time, say with quantum numbers q, σ , and calculate the renormalization of the various couplings due to this electron. We then sum the momentum q over the shell and the spin β , and this gives the total renormalization due to decoupling the entire shell.

From eq. 1.45, the first two terms in the rotated Hamiltonian are just the diagonal parts of the bare Hamiltonian; they are unchanged in that part. The renormalization comes from the third term. For one electron $q\beta$ on the shell, the renormalization is

$$\Delta\mathcal{H} = \left[c_{q\beta}^\dagger \text{Tr}(\mathcal{H}c_{q\beta}), \eta \right] = c_{q\beta}^\dagger \text{Tr}(\mathcal{H}c_{q\beta}) \eta - \eta c_{q\beta}^\dagger \text{Tr}(\mathcal{H}c_{q\beta})\tag{1.74}$$

Since this assumes we have obtained this from U_1 , it is fair to tag the η with a suitable label:

$$\Delta\mathcal{H} = c_{q\beta}^\dagger \text{Tr}(\mathcal{H}c_{q\beta}) \eta_1 - \eta_1 c_{q\beta}^\dagger \text{Tr}(\mathcal{H}c_{q\beta})\tag{1.75}$$

It is clear that the first term takes into account virtual excitations that start from a filled state ($\hat{n}_{q\beta} = 1$ initially) - such a term is said to be a part of the *particle sector*.

$$\Delta_1\mathcal{H} = c_{q\beta}^\dagger \text{Tr}(\mathcal{H}c_{q\beta}) \eta_1\tag{1.76}$$

The second term, on the other hand, considers excitations that start from an empty state. They constitute the *hole sector*.

$$\Delta_0 \mathcal{H} = -\eta_1 c_{q\beta}^\dagger \text{Tr}(\mathcal{H} c_{q\beta}) \quad (1.77)$$

To write the total renormalization in a particle-hole symmetric form, we can use the relation $\eta_0 = -\eta_1$, such that both the terms will now come with a positive sign:

$$\Delta \mathcal{H} = c_{q\beta}^\dagger \text{Tr}(\mathcal{H} c_{q\beta}) \eta_1 + \eta_0 c_{q\beta}^\dagger \text{Tr}(\mathcal{H} c_{q\beta}) \quad (1.78)$$

We can make one more manipulation: using eq. 1.16, we get

$$\Delta \mathcal{H} = c_{q\beta}^\dagger \text{Tr}(\mathcal{H} c_{q\beta}) \eta_1 + \text{Tr}(c_{q\beta}^\dagger \mathcal{H}) c_{q\beta} \eta_0^\dagger \quad (1.79)$$

This form of the total renormalization is identical to the one we use in the "Poor Man's scaling"-type of renormalization that was used to get the scaling equations in the Kondo and Anderson models [6, 7]. Writing down the forms of η and $\eta^\dagger$ explicitly, we get

$$\Delta \mathcal{H} = c_{q\beta}^\dagger \text{Tr}(\mathcal{H} c_{q\beta}) \frac{1}{\omega_1^0 - \mathcal{H}_0^d} \text{Tr}(c_{q\beta}^\dagger \mathcal{H}) c_{q\beta} + \text{Tr}(c_{q\beta}^\dagger \mathcal{H}) c_{q\beta} \frac{1}{\omega_0^1 - \mathcal{H}_1^d} c_{q\beta}^\dagger \text{Tr}(\mathcal{H} c_{q\beta}) \quad (1.80)$$

The renormalization due to the entire shell is obtained by summing over all states on the shell.

$$\Delta \mathcal{H} = \sum_{q\beta} \left[c_{q\beta}^\dagger \text{Tr}(\mathcal{H} c_{q\beta}) \frac{1}{\omega_1^0 - \mathcal{H}_0^d} \text{Tr}(c_{q\beta}^\dagger \mathcal{H}) c_{q\beta} + \text{Tr}(c_{q\beta}^\dagger \mathcal{H}) c_{q\beta} \frac{1}{\omega_0^1 - \mathcal{H}_1^d} c_{q\beta}^\dagger \text{Tr}(\mathcal{H} c_{q\beta}) \right] \quad (1.81)$$

These equations will now need to be simplified. For example, in the particle sector, we can set $\hat{n}_{q\beta} = 0$ in the numerator, because there is no such excitation in the initial state. Similarly, in the hole sector, we can set $\hat{n}_{q\beta} = 1$ because that state was occupied in the initial state. Another simplification we typically employ is that $\mathcal{H}_{0,1}^d$ will, in general, have the energies of all the electrons. But we consider only the energy of the on-shell electrons in the denominator. After integrating out these electrons, we can rearrange the remaining operators to determine which term in the Hamiltonian it renormalizes and what is the renormalization.

At first sight, one might think that we must evaluate lots of traces to obtain the terms in $\Delta \mathcal{H}$. A little thought reveals that the terms in the numerator are simply the off-diagonal terms in the Hamiltonian; $\text{Tr}(c_{q\beta}^\dagger \mathcal{H}) c_{q\beta}$ is the off-diagonal term that has $c_{q\beta}$ in it, and $c_{q\beta}^\dagger \text{Tr}(\mathcal{H} c_{q\beta})$ is the off-diagonal term that has $c_{q\beta}^\dagger$ in it. $\mathcal{H}^D$ is just the diagonal part of the Hamiltonian.

1.2 Example I: URG analysis of the star graph model

The star graph problem has already been analyzed using URG and an extensive study of its entanglement properties has already been carried out, in ref. [?]. Here we focus on just deriving the RG equations. The system consists of N spin-like degrees of freedom (labeled 1 through N) individually talking to a spin at the center (labeled 0). Each spin i ($i \in [0, N]$) has an on-site energy ϵ_i . The coupling strength between 0 and i ($i \in [1, N]$) is J_i . We choose the on-site energies such that $\epsilon_{i+1} > \epsilon_i, i \in [N-1, 1]$. In this way, ϵ_1 is the infrared limit and ϵ_N is the ultraviolet limit.

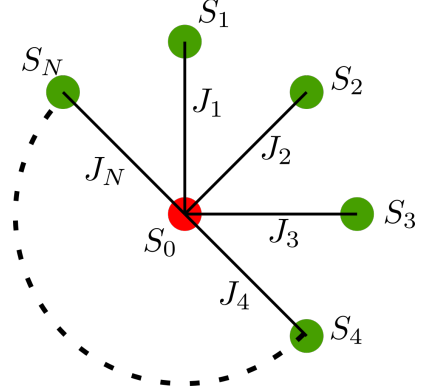


Figure 1.2: Star Graph model

$$\mathcal{H} = \epsilon_0 S_0^z + \sum_{i=1}^N \left[\epsilon_i S_i^z + J_i \vec{S}_0 \cdot \vec{S}_i \right] \quad (2.1)$$

By converting the last term into S^z and $S^\pm$, we can write the Hamiltonian as

$$\mathcal{H} = \epsilon_0 S_0^z + \sum_{i=1}^N \left[\epsilon_i S_i^z + J_i \left(S_0^z S_i^z + \frac{1}{2} (S_0^+ S_i^- + S_0^- S_i^+) \right) \right] \quad (2.2)$$

Calculation of Renormalization

The RG involves decoupling the nodes N through 1, and looking at the resultant renormalization in ϵ_i and J_i . As a simplification, we will ignore the lower nodes in the denominator and keep only the node currently being decoupled, ie node N . Since node 0 is connected to node N , we will keep node 0 in the denominator as well. Making this simplification gives

$$\mathcal{H}^D = \epsilon_0 S_0^z + \epsilon_N S_N^z + J_N S_0^z S_N^z \quad (2.3)$$

The off-diagonal part in the subspace of the node N is

$$\mathcal{H}_X = S_N^+ T + T^\dagger S_N^- = \frac{1}{2} J_N (S_N^+ S_0^- + S_N^- S_0^+) \quad (2.4)$$

The renormalization on doing one step of the URG is given by

$$\begin{aligned} \Delta \mathcal{H} &= S_N^+ T \frac{1}{\omega_0^1 - \mathcal{H}^D} T^\dagger S_N^- + T^\dagger S_N^- \frac{1}{\omega_1^0 - \mathcal{H}^D} S_N^+ T \\ &= \frac{J_N^2}{4} S_N^+ S_0^- \frac{1}{\omega_0^1 - \epsilon_0 S_0^z - \epsilon_N S_N^z - J_N S_0^z S_N^z} S_0^+ S_N^- + \frac{J_N^2}{4} S_0^+ S_N^- \frac{1}{\omega_1^0 - \epsilon_0 S_0^z - \epsilon_N S_N^z - J_N S_0^z S_N^z} S_N^+ S_0^- \end{aligned} \quad (2.5)$$

There, N refers to the spin being decoupled. The first Greens function has S_0^+ and S_N^- in front of it, so we substitute $S_0^z = \frac{1}{2}$, $S_N^z = -\frac{1}{2}$ in that Greens function. For the other Greens function we do the opposite.

$$\Delta\mathcal{H} = \frac{J_N^2}{4} S_N^+ S_0^- \frac{1}{\omega_1^0 - \frac{1}{2}\epsilon_0 + \frac{1}{2}\epsilon_N + \frac{1}{4}J_N} S_0^+ S_N^- + \frac{J_N^2}{4} S_0^+ S_N^- \frac{1}{\omega_0^1 + \frac{1}{2}\epsilon_0 - \frac{1}{2}\epsilon_N + \frac{1}{4}J_N} S_N^+ S_0^- \quad (2.6)$$

To relate ω_1^0 and ω_0^1 , we use eq. 1.72:

$$\omega_1^0 + \omega_0^1 = \mathcal{H}_0^D + \mathcal{H}_1^D = -\frac{1}{2}J_N \implies \omega_1^0 \equiv \omega, \quad \omega_0^1 \equiv \omega' = -\frac{1}{2}J_N - \omega \quad (2.7)$$

So, the renormalization becomes

$$\Delta\mathcal{H} = \frac{J_N^2}{4} \frac{1}{\omega - \frac{1}{2}\epsilon_0 + \frac{1}{2}\epsilon_N + \frac{1}{4}J_N} (S_N^+ S_0^- S_0^+ S_N^- - S_0^+ S_N^- S_N^+ S_0^-) = \frac{J_N^2}{4} \frac{1}{\omega - \frac{1}{2}\epsilon_0 + \frac{1}{2}\epsilon_N + \frac{1}{4}J_N} (S_N^z - S_0^z) \quad (2.8)$$

There we used $S^+ S^- = \frac{1}{2} + S^z$ and $S^- S^+ = \frac{1}{2} - S^z$.

We can now read off the renormalizations in ϵ_N and ϵ_0 .

$$\begin{aligned} \Delta\epsilon_N &= \frac{1}{4} J_N^2 \frac{1}{\omega - \frac{1}{2}\epsilon_0 + \frac{1}{2}\epsilon_N + \frac{1}{4}J_N} \\ \Delta\epsilon_0 &= -\frac{1}{4} J_N^2 \frac{1}{\omega - \frac{1}{2}\epsilon_0 + \frac{1}{2}\epsilon_N + \frac{1}{4}J_N} \end{aligned} \quad (2.9)$$

Nature of flows

We are interested in looking at the renormalization of the central node energy ϵ_0 , upon removing the nodes N through 1. We will hence concentrate on the second RG equation. We first make some simplifying assumptions: $J_i = J$, $\epsilon_i = \epsilon$ for all $i \in \{1, N\}$.

$$\Delta\epsilon_0 = -\frac{1}{4} J^2 \frac{1}{\omega - \frac{1}{2}\epsilon_0 + \frac{1}{2}\epsilon + \frac{1}{4}J} \quad (2.10)$$

Define $\tilde{\omega} = \omega + \frac{1}{2}\epsilon + \frac{1}{4}J$.

$$\Delta\epsilon_0 = -\frac{1}{4} J^2 \frac{1}{\tilde{\omega} - \frac{1}{2}\epsilon_0} \quad (2.11)$$

Our goal here is to look for a fixed-point condition such that the denominator vanishes at some point of the RG. If we start with a bare of ϵ_0 such that $\tilde{\omega} - \frac{1}{2}\epsilon_0 > 0$, the denominator will be positive and the RG equation will be irrelevant. This means that ϵ_0 will keep on decreasing, and the denominator will keep on becoming more and more positive, meaning there cannot be a fixed point in this situation.

If, on other hand, we start with a bare of ϵ_0 such that $\tilde{\omega} - \frac{1}{2}\epsilon_0 < 0$, the denominator will be negative and the RG equation will be relevant. This means that ϵ_0 will keep on increasing, and the

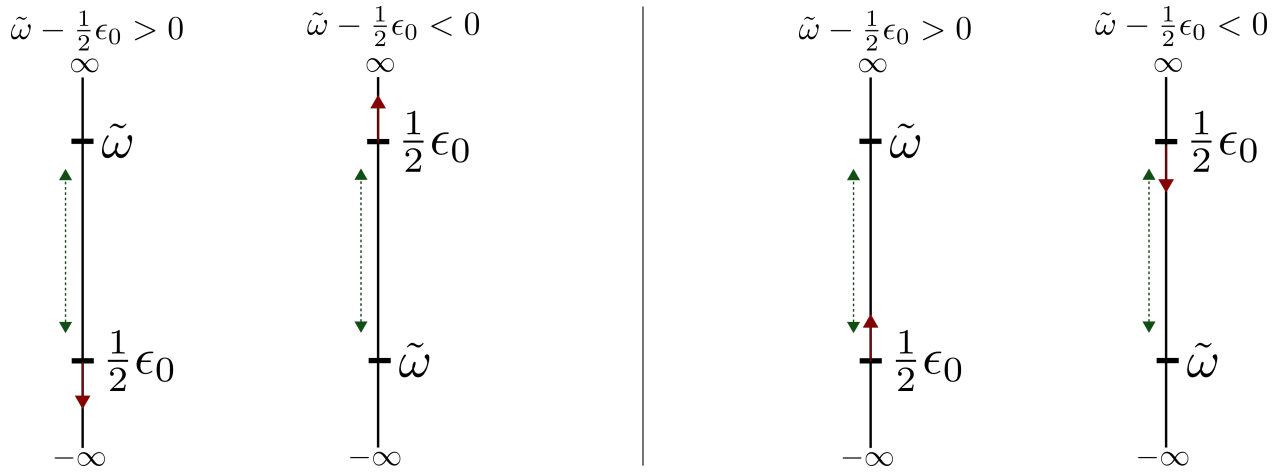


Figure 1.3: Left: RG flow for the two cases. The green line is the distance between the bare values of the two couplings, and hence also the magnitude of the denominator. The red arrow denotes the direction in which ϵ_0 will flow. Upward flow is increase. In both cases, the flow is such that the distance between the two quantities (and hence the magnitude of the denominator) increases. The RG fixed point occurs when the magnitude of the denominator goes to 0. This happens if the distance vanishes. Since the distance necessarily increases, we cannot get a fixed point in this way. Right: RG flow for the two cases with the new $-\tilde{\omega} = \omega' - \frac{1}{2}\epsilon + \frac{1}{4}J$. Now we can see that in both cases, the flow is such that the distance (green dotted line) between the couplings decreases. A fixed point is reached when this distance vanishes.

denominator will keep on becoming more and more negative, meaning there cannot be a fixed point in this situation either. These situations are depicted in figure 1.3.

Since we cannot find a fixed point, we will use a different ω . Instead of ω_1^0 , we will use ω_1^1 . From eq. 1.72, we have

$$\omega_1^0 - \omega_1^1 = \mathcal{H}_0^D - \mathcal{H}_1^D = \epsilon_0 - \epsilon_N = \epsilon_0 - \epsilon \quad (2.12)$$

Defining $\omega_1^1 = \omega'$ and substituting this in eq. 2.10 gives

$$\Delta\epsilon_0 = -\frac{1}{4}J^2 \frac{1}{\omega' - \frac{1}{2}\epsilon + \frac{1}{2}\epsilon_0 + \frac{1}{4}J} \quad (2.13)$$

We again define $-\tilde{\omega} = \omega' - \frac{1}{2}\epsilon + \frac{1}{4}J$.

$$\Delta\epsilon_0 = \frac{1}{4}J^2 \frac{1}{\tilde{\omega} - \frac{1}{2}\epsilon_0} \quad (2.14)$$

We now repeat the exercise of determining the relevance of the flows under various regime. If we start with a bare ϵ_0 such that $\tilde{\omega} + \frac{1}{2}\epsilon_0 > 0$, then the denominator is positive so the renormalization will be irrelevant. ϵ_0 will decrease until we reach $\tilde{\omega} + \frac{1}{2}\epsilon_0 = 0$. This will be a fixed point. However, if we start with a bare ϵ_0 such that $\tilde{\omega} + \frac{1}{2}\epsilon_0 < 0$, then the denominator is negative so the renormalization will be relevant. ϵ_0 will increase until we reach $\tilde{\omega} + \frac{1}{2}\epsilon_0 = 0$. This will again be a fixed point. This new situation is depicted in right panel of figure 1.3.

Effective Hamiltonians

If $\tilde{\omega}$ and ϵ_0 are of the same sign at the bare level, then it is easy to see that since the fixed point is defined by $\tilde{\omega} = \frac{1}{2}\epsilon_0^*$ (* denotes value at fixed point), the effective Hamiltonian at the fixed point will be

$$\mathcal{H}^* = 2\tilde{\omega}S_0^z + \epsilon \sum_i S_i^z + J \sum_i \vec{S}_i \cdot \vec{S}_0, \quad \text{if } \tilde{\omega}\epsilon_0 > 0 \quad (2.15)$$

If, at the bare level, ϵ_0 and $\tilde{\omega}$ are of opposite signs, then ϵ_0 would undergo a change in sign at some point as it flows towards $\tilde{\omega}$. Since we do not expect a coupling to change sign under RG, we will restrict it to 0 in such cases.

$$\mathcal{H}^* = \epsilon \sum_i S_i^z + J \sum_i \vec{S}_i \cdot \vec{S}_0, \quad \text{if } \tilde{\omega}\epsilon_0 < 0 \quad (2.16)$$

Things get much more simpler if we assume the onsite energies of the surrounding nodes are zero.

$$\begin{aligned} \mathcal{H}^* &= 2\tilde{\omega}S_0^z + J \sum_i \vec{S}_i \cdot \vec{S}_0, & \text{if } \tilde{\omega}\epsilon_0 > 0 \\ \mathcal{H}^* &= J \sum_i \vec{S}_i \cdot \vec{S}_0, & \text{if } \tilde{\omega}\epsilon_0 < 0 \end{aligned} \quad (2.17)$$

Fixed points

The fixed points are obtained numerically by solving the RG equation. As mentioned before, there are two types of solutions: The first kind is those in which ϵ_0 and $\tilde{\omega}$ are of the same sign, and the former flows to the latter without crossing the 0 axis. These flows are shown (obtained numerically) in fig. 1.4. The second kind are those where the two couplings have different signs, and so ϵ_0 flows to 0. These are shown in fig. 1.5.

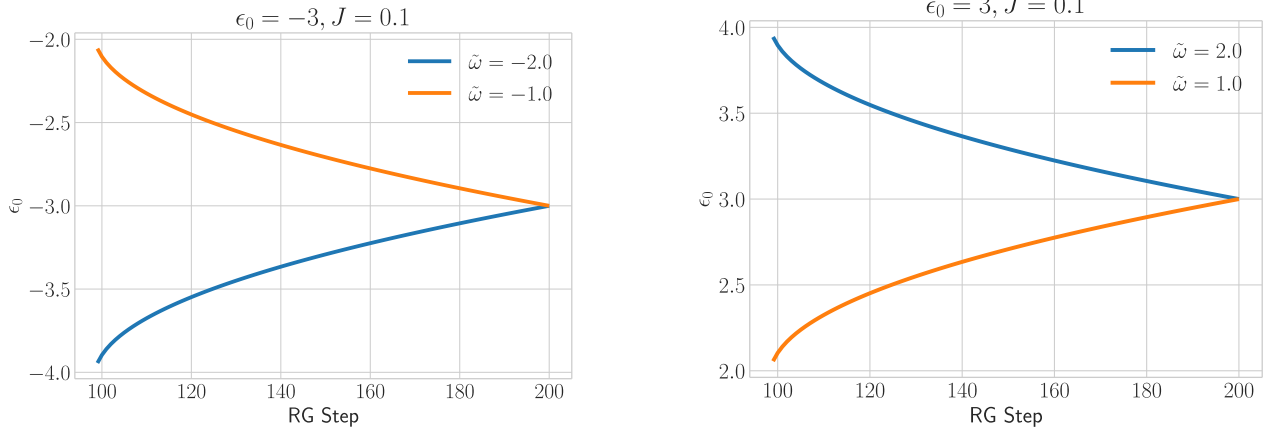


Figure 1.4: Flows where ϵ_0 and $\tilde{\omega}$ have same sign. The left and right panels show flows starting from negative and positive values respectively. The two plots in each panel correspond to different values of $\tilde{\omega}$, one greater than the bare ϵ_0 , the other less than that. The fixed point value is $2\tilde{\omega}$.

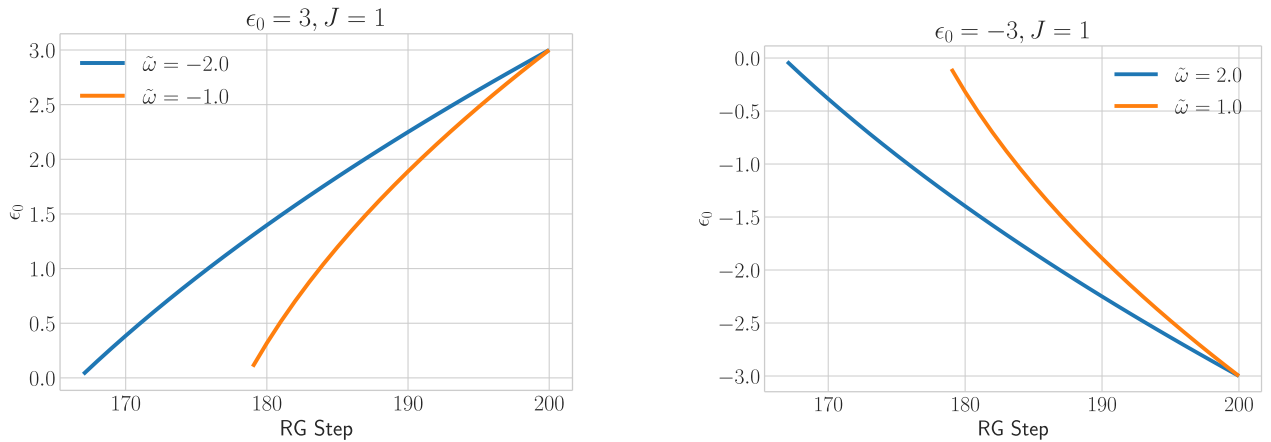


Figure 1.5: Flows where ϵ_0 and $\tilde{\omega}$ have opposite sign. The left and right panels show flows starting from negative and positive values respectively. The two plots in each panel correspond to different values of $\tilde{\omega}$, one greater than the bare ϵ_0 , the other less than that. The fixed point value is 0.

1.3 Example II: URG analysis of the two-channel Kondo problem

The Hamiltonian for the channel-isotropic MCK model is:

$$H = \sum_l \left[\sum_{\substack{k \\ \alpha=\uparrow,\downarrow}} \epsilon_{k,l} \hat{n}_{k\alpha,l} + \frac{J}{2} \sum_{\substack{k,k' \\ \alpha,\alpha'=\uparrow,\downarrow}} \vec{S}_d \cdot \vec{\sigma}_{\alpha\alpha'} c_{k\alpha,l}^\dagger c_{k'\alpha',l} \right]. \quad (3.1)$$

The URG equation for the single-channel Kondo model [?] shows a stable strong coupling fixed point. Ferromagnetic interactions are irrelevant. Strictly speaking, that RG equation already encodes, in principle, the multi-channel behaviour, through a modified $\hat{\omega}$. To extract this information, we consider the strong coupling fixed-point $J \gg D$ as a fixed point and analyze its stability from the star graph perspective. For the exactly-screened case, the star graph decouples from the conduction bath, leaving behind a local Fermi liquid interaction on the first site. Similarly, in the under-screened regime, the ground state is composed of states where the impurity spin is only partially screened by the conduction channels. If a particular configuration of the bath-impurity system has the total conduction bath spin down, the impurity will have a residual up spin. This induces a ferromagnetic super-exchange coupling that is irrelevant under RG, so this fixed point is stable as well.

We now come to the over-screened case, where there is a residual spin on the conduction channel site. The neighbouring electrons will now hop in with spins opposite to that of the impurity, so an antiferromagnetic interaction will be induced, and such an interaction is relevant under the RG. This shows that the over-screened regime cannot have a stable strong coupling fixed point, and we need to search for an intermediate coupling fixed point. We therefore need the generator of the unitary transformation that incorporates third order scattering scatterings explicitly. We should take account of all possible processes that render the set of states $\{|\hat{n}_{q\beta} = 1\rangle, |\hat{n}_{q\beta} = 0\rangle\}$ diagonal. The higher order generator itself has two scattering processes, such that the entire renormalisation term $c_{q\beta}^\dagger T \eta$ has in total three coherent processes. The complete generator up to third order can be written as

$$\eta = \frac{1}{\hat{\omega} - H_D} T^\dagger c \simeq \frac{1}{\omega' - H_D} T^\dagger c + \frac{1}{\omega' - H_D} H_X \frac{1}{\omega' - H_D} T^\dagger c + \frac{1}{\omega' - H_D} T^\dagger c \frac{1}{\omega' - H_D} H_X, \quad (3.2)$$

where $H_X = J \sum_{k,k' < \Lambda_j, \alpha, \alpha'} \vec{S}_d \cdot \vec{\sigma}_{\alpha\alpha'} c_{k\alpha}^\dagger c_{k'\alpha'}$ is scattering between the entangled electrons. There are two third order terms in the above equation corresponding to the two possible sequences in which the processes can occur while keeping the total renormalisation $c_{q\beta}^\dagger T \eta$ diagonal in $q\beta$. The second order processes remain unchanged. The total renormalisation takes the form:

$$\begin{aligned} \Delta H_{(j)} = & \underbrace{c^\dagger T \frac{1}{\omega' - H_D} T^\dagger c + (c^\dagger \leftrightarrow c)}_{\Delta H_{(j)}^{(2)}} \\ & + \underbrace{c^\dagger T \frac{1}{\omega' - H_D} H_X \frac{1}{\omega' - H_D} T^\dagger c + c^\dagger T \frac{1}{\omega' - H_D} T^\dagger c \frac{1}{\omega' - H_D} H_X + (c^\dagger \leftrightarrow c)}_{\Delta H_{(j)}^{(3)}}. \end{aligned} \quad (3.3)$$

$\Delta H_{(j)}^{(2)}$ and $\Delta H_{(j)}^{(3)}$ are the renormalisation arising from the second and third order processes respectively.

It is easier to see the RG flow of the couplings if we write the Hamiltonian in terms of the eigenstates of S_d^z . These eigenstates are defined by $S_d^z |m_d\rangle = m_d |m_d\rangle$, $m_d \in [-S, S]$. In terms of these eigenstates, the Hamiltonian becomes

$$\mathcal{H} = \sum_{k\sigma} \epsilon_k \tau_{k\sigma} + \sum_{m_d=-S}^S \sum_{\substack{kl, \\ \sigma=\uparrow, \downarrow}} J_{m_d}^\sigma |m_d\rangle \langle m_d| c_{k\sigma}^\dagger c_{l\sigma} + \sum_{kl} \sum_{m_d=-S}^{S-1} J_{m_d}^t (|m_d+1\rangle \langle m_d| s_{kl}^- + \text{h.c.}) \quad (3.4)$$

where k, l sum over the momentum states, σ sums over the spin indices, $J_m^\sigma = \frac{1}{2}\sigma m J$ in the UV Hamiltonian, and $J_m^t = J \frac{1}{2} \sqrt{S(S+1) - m(m+1)}$ is the coupling that connects $|m\rangle$ and $|m+1\rangle$. We first calculate $\Delta H_{(j)}^{(2)}$. There will be two types of processes - those processes that start from an occupied state (particle sector) and those that start from a vacant state (hole sector). Due to particle hole symmetry of the Hamiltonian, they will be equal to each other and we will only calculate the particle sector contribution.

In the particle sector, we have ($\hat{n}_{q\beta} = 1$), so we will work at a negative energy shell $\epsilon_q = -D$. The renormalisation can schematically be represented as $H_0^I \frac{1}{\omega - H_{q\beta}^D} H_1^I$. Both H_0^I and H_1^I have all three operators $S_d^z, S_d^\pm$. We first consider specifically the case of spin- $\frac{1}{2}$ impurity. Those terms that have identical operators on both sides can be ignored because $S_d^{z2} = \text{constant}$ and $S^\pm 2 = 0$. All the six terms that *will* renormalise the Hamiltonian have a spin flip operator on at least one side of the Greens function. This means that in the denominator of the Greens function, S_d^z and s_{qq}^z have to be anti-parallel in order to produce a non-zero result for that term. This means we can identically replace $S_d^z s_{qq}^z = -\frac{1}{4}$. Also, in the particle sector, the Greens function always has $c_{q\beta}$ in front of it, so $\epsilon_q \tau_{q\beta} = \frac{D}{2}$. The upshot of all this is that the denominator of all scattering processes for the spin- $\frac{1}{2}$ impurity Hamiltonian will be $\omega - \frac{D}{2} + \frac{J}{4}$.

We now come to the general case of spin- S impurity. The various terms that renormalise the Hamiltonian can be described in terms of the bath spin operators that come into them. For example, the term that has s^z on both sides of the intervening Greens function can be represented as $z|z$. There are 7 such terms: $z|z, \pm|\mp, z|\pm, \pm|z$. Each of these terms occurs both in the particle and the hole sectors. We will demonstrate the calculation of two of these terms. The $z|z + | -$ terms evaluate in the following manner.

$$\begin{aligned} z|z : & \sum_{kk', m, \sigma} c_{q\sigma}^\dagger c_{k'\sigma} |m\rangle \langle m| \frac{J_m^{\sigma 2}}{\omega - \frac{D}{2} + \frac{J}{2}\sigma S_d^z} |m\rangle \langle m| c_{k\sigma}^\dagger c_{q\sigma} = - \sum_{kk', m, \sigma} n_{q\sigma} \frac{J_m^{\sigma 2} c_{k\sigma}^\dagger c_{k'\sigma} |m\rangle \langle m|}{\omega_{m, \sigma} - \frac{D}{2} + \frac{J}{2}\sigma m}, \\ +|- : & \sum_{kk', m} c_{q\uparrow}^\dagger c_{k'\downarrow} |m\rangle \langle m+1| \frac{J_m^2}{\omega - \frac{D}{2} + \frac{J}{2}S_d^z} |m+1\rangle \langle m| c_{k\downarrow}^\dagger c_{q\uparrow} = -n_{q\uparrow} \sum_{kk', m} \frac{J_m^2 c_{k\downarrow}^\dagger c_{k'\downarrow} |m\rangle \langle m|}{\omega_{m+1, \uparrow} - \frac{D}{2} + \frac{J}{2}(m+1)}. \end{aligned} \quad (3.5)$$

We similarly compute the rest of the terms. We again define $\sum_q \hat{n}_{q\sigma} = n(D)$. To compare with the spin- $\frac{1}{2}$ RG equations, we will transform the general spin- S ω to the spin- $\frac{1}{2}$ ω , using $\omega_{m, \sigma} \rightarrow$

$\omega - \frac{J}{2} \left(m\sigma - \frac{1}{2} \right)$. The renormalisation in J_m^σ is

$$\Delta J_m^\sigma = -n(D) \frac{(J_m^\sigma)^2 + \left(J_{m-\frac{1+\sigma}{2}}^t \right)^2}{\omega - \frac{D}{2} + \frac{J}{4}}. \quad (3.6)$$

Here, we have defined $J_m^t = 0$ for $|m| > S$. Two relations can be obtained from this RG equation, the RG equations for the sum and difference of the couplings: $J_m^\pm = \frac{1}{2} (J_m^\uparrow \pm J_m^\downarrow)$. The RG equation for the sum of the couplings is

$$\Delta J_m^+ = -n(D) \frac{\sum_\sigma (J_m^\sigma)^2 + \sum_\sigma \left(J_{m-\frac{1+\sigma}{2}}^t \right)^2}{2 \left(\omega - \frac{D}{2} + \frac{J}{4} \right)} = -n(D) \frac{J^2}{4} \frac{S(S+1)}{\omega - \frac{D}{2} + \frac{J}{4}}. \quad (3.7)$$

This is an m -independent piece, so it can be summed over to produce an impurity-independent potential scattering term, which we ignore.

The second is the RG equation for the difference of the couplings:

$$\Delta J_m^- = -n(D) \frac{1}{2} \frac{\left(J_{m-1}^t \right)^2 - \left(J_m^t \right)^2}{\omega - \frac{D}{2} + \frac{J}{4}} = -\frac{1}{4} \frac{n(D) m J^2}{\omega - \frac{D}{2} + \frac{J}{4}}. \quad (3.8)$$

The usual J Kondo coupling is recovered through $J = 2J_m^-/m$. Substituting this gives

$$\Delta_{\text{p sector}} J = -\frac{1}{2} n(D) \frac{J^2}{\omega - \frac{D}{2} + \frac{J}{4}}. \quad (3.9)$$

We can also obtain the RG equation for J from the transverse renormalisation:

$$\Delta J_m^t = -\frac{n(D) J_m^t (J_m^\downarrow + J_{m+1}^\uparrow)}{\omega - \frac{D}{2} + \frac{J}{4}} = -\frac{1}{2} \frac{n(D) J_m^t J}{\omega - \frac{D}{2} + \frac{J}{4}}. \quad (3.10)$$

Since $J_m^t \propto J$, we have

$$\Delta J_{\text{p sector}} = -\frac{1}{2} \frac{n(D) J^2}{\omega - \frac{D}{2} + \frac{J}{4}}. \quad (3.11)$$

The total renormalisation from both particle and hole sectors, at this order, is

$$\Delta J^{(2)} = -\frac{n(D) J^2}{\omega - \frac{D}{2} + \frac{J}{4}}. \quad (3.12)$$

We now come to the third-order renormalisation. Following eq. (3.3), the next order renormalisation is

$$\Delta H_j^{(3)} = c^\dagger T \frac{1}{\omega' - H_D} H_X \frac{1}{\omega' - H_D} T^\dagger c + c^\dagger T \frac{1}{\omega' - H_D} T^\dagger c \frac{1}{\omega' - H_D} H_X. \quad (3.13)$$

The first term will be of the form

$$\sum_{q,k,l_1} c_{q\beta,l_1}^\dagger c_{k\alpha,l_1} |m_1\rangle \langle m_2| \frac{1}{\omega - \frac{D}{2} - \frac{\epsilon_k}{2} + \frac{\beta J}{4} S_d^z} |m_2\rangle c_{k_1\sigma_1,l_2}^\dagger c_{k_2\sigma_2,l_2} \langle m_3| \frac{1}{\omega - \frac{D}{2} - \frac{\epsilon_k}{2} + \frac{\beta J}{4} S_d^z} |m_3\rangle \langle m_4| c_{k\alpha,l_1}^\dagger c_{q\beta,l_1}. \quad (3.14)$$

We have not bothered to write all the summations and the couplings correctly, because we will only simplify the denominator here. Evaluating the inner products gives

$$\sum_{q,k,l_1} \frac{|m_1\rangle \langle m_4| c_{q\beta}^\dagger c_{k\alpha} c_{k_1\sigma_1,l_2}^\dagger}{\omega_{m_2,\beta} - \frac{D}{2} - \frac{\epsilon_k}{2} + \frac{\beta J m_2}{4}} \frac{c_{k_2\sigma_2,l_2} c_{k\alpha}^\dagger c_{q\beta}}{\omega_{m_3,\beta} - \frac{D}{2} - \frac{\epsilon_k}{2} + \frac{\beta J m_3}{4}}. \quad (3.15)$$

We again use $\omega_{m,\sigma} \rightarrow \omega - \frac{J}{2} \left(m\sigma - \frac{1}{2} \right)$.

$$|m_1\rangle \langle m_4| c_{k_1\sigma_1,l_2}^\dagger c_{k_2\sigma_2,l_2} \sum_{q,k,l_1} \frac{\hat{n}_{q\beta} (1 - \hat{n}_{k\alpha})}{\left(\omega - \frac{D}{2} - \frac{\epsilon_k}{2} + \frac{J}{4} \right)^2}. \quad (3.16)$$

We define $\sum_q \hat{n}_{q\beta} = n(D)$. Performing the sums over k and l_1 gives

$$-\frac{1}{2} |m_1\rangle \langle m_4| c_{k_1\sigma_1,l_2}^\dagger c_{k_2\sigma_2,l_2} \frac{\rho n(D) K}{\omega - \frac{D}{2} + \frac{J}{4}}. \quad (3.17)$$

ρ is the density of states which we have taken to be constant. Reinstating the complete summation and the couplings gives

$$-\frac{1}{2} \frac{\rho n(D) K}{\omega - \frac{D}{2} + \frac{J}{4}} \sum_{\substack{m_1, m_4, k_1, k_2, \\ l_2, \sigma_1 \sigma_2}} \lambda_1 \lambda_2 \lambda_3 |m_1\rangle \langle m_4| c_{k_1\sigma_1,l_2}^\dagger c_{k_2\sigma_2,l_2}. \quad (3.18)$$

There is no sum over m_2 and m_3 because they are constrained by m_1 and m_4 respectively. λ_i represent the couplings present at the three interaction vertices. $k_{1,2}$ sum over the momenta, $\sigma_{1,2}$ sum over the spin indices and l_2 sums over the channels.

The second term in eq. (3.13) can be evaluated in an almost identical fashion. The integral here will be negative of the first term, because of an exchange in the scattering processes.

$$\frac{1}{2} \frac{\rho n(D) K}{\omega - \frac{D}{2} + \frac{J}{4}} \sum_{\substack{m_1, m_4, k_1, k_2, \\ l_2, \sigma_1 \sigma_2}} \lambda_1 \lambda_3 \lambda_2 |m_1\rangle \langle m_4| c_{k_1\sigma_1,l_2}^\dagger c_{k_2\sigma_2,l_2}. \quad (3.19)$$

The first group of terms (those that appear in (3.18)) in the particle sector can be represented as $a|b'|c$, where $a, b, c \in \{z, +, -\}$ and represent the operator for the conduction electrons in the three connected processes. The $\prime$ on b indicates that it is the state of the electrons *not being decoupled*. The second group of terms (those that appear in (3.19)) are therefore represented as $a|b|c'$, because in this group, the interaction H_X between the electrons that are not being decoupled occur at the

very end. We will only calculate the terms in the particle sector, the ones in hole sector will be equal to these because of particle-hole symmetry. The full list of terms is:

$$\begin{array}{cccccccccccc}
 z|z'|z & z|z|z' & -|z'|+ & -|+|z' & +|z'|- & +|-|z' & z|+|z & z|-'|z & +|+|- & -|+|+ \\
 z|z|+ & z|z|- & +|-|+ & +|-|- & -|+|+ & z|z'|z & z|z|z' & -|z'|+ & -|+|z' & +|z'|- \\
 +|-|z' & z|+|z & z|-'|z & +|+|- & -|+|+ & z|z|+ & z|z|- & +|-|+ & +|-|- & -|+|+
 \end{array}$$

The total renormalisation in J_m^σ is

$$\begin{aligned}
 \Delta J_m^\sigma &= \frac{1}{2} \frac{\rho n(D)K}{\omega - \frac{D}{2} + \frac{J}{4}} \left[(J_{m-1}^t)^2 J_m^\sigma + (J_m^t)^2 J_m^\sigma - (J_{m-1}^t)^2 J_{m-1}^\sigma - (J_m^t)^2 J_{m+1}^\sigma \right] \\
 &= \frac{1}{2} \frac{\rho n(D)K}{\omega - \frac{D}{2} + \frac{J}{4}} J^3 m \sigma.
 \end{aligned} \tag{3.20}$$

Since we had defined $J_m^\sigma \equiv \frac{1}{2} J m \sigma$, we have $\Delta J = \frac{2}{m \sigma} \Delta J_m^\sigma$, and we get $\Delta_{\text{p sector}} J = \frac{1}{4} \frac{\rho n(D)K}{\omega - \frac{D}{2} + \frac{J}{4}} J^3$. Combining with the hole sector renormalisation, we get

$$\Delta J^{(3)} = \frac{1}{2} \frac{\rho n(D)K}{\omega - \frac{D}{2} + \frac{J}{4}} J^3. \tag{3.21}$$

The total renormalisation in J after combining all orders is

$$\Delta J = -\frac{n(D)J^2}{\omega - \frac{D}{2} + \frac{J}{4}} + \frac{1}{2} \frac{\rho n(D)K}{\omega - \frac{D}{2} + \frac{J}{4}} J^3. \tag{3.22}$$

1.4 Example III: Particle in a periodic potential

We consider electrons moving in one dimension in the presence of a periodic potential $V(x)$ satisfying $V(x+a) = V(x)$. The full Hamiltonian can be represented as

$$\mathcal{H} = \int_{-\infty}^{\infty} dx \, c^\dagger(x) \left[\hat{p}^2/2m + V(x) \right] c(x), \tag{4.1}$$

where $c(x)$ destroys a fermion at the position x .

Physically, the model represents a set of N_{well} number of wells (located at the minima of the potential) linked to each other through hopping (brought about by the kinetic energy term). Each well is described a potential $V(x)$, whose spectrum consists of N_{levs} single particle levels. These levels are localised in real space (around each minima of width a), so the corresponding momentum states are widely separated; each of these momentum states eventually give rise to an energy band, we therefore represent them using the band index: $k_n = nG$ ($n = 0, 1, \dots, N_{\text{levs}} - 1$), where G is the spacing between the momentum states arising from a single potential well, and is equal to the reciprocal lattice vector $2\pi/a$.

These are of course not the only momentum states in the problem; since each potential well is spaced far apart (at distances greater than a), the multiplicity of the wells leads to closely spaced

momentum states, N_{well} in number (the number of potential wells). Since these are closely spaced, they lead to the dense set of states within each band: $k_{n,j} = nG + \frac{2\pi j}{N_{\text{well}}}$ ($j = 0, 1, \dots, N_{\text{well}} - 1$). The spacing $2\pi/N_{\text{well}}$ between the states within the band is imposed by applying periodic boundary conditions on the system: $\psi(x=0) = \psi(x=aN_{\text{well}})$. We take the thermodynamic limit by allowing $N_{\text{well}} \rightarrow \infty$, keeping all other independent parameters unchanged. Because of the periodicity of the real space potential, it can only scatter states that are separated by a reciprocal lattice vector G ; this means that states within a band (that are within a reciprocal lattice vector of each other) do not scatter off each other: $\langle k_{n+m,j+l} | \hat{V} | k_{n,j} \rangle = \delta_{l,0} \langle k_{n+m,j} | \hat{V} | k_{n,j} \rangle$.

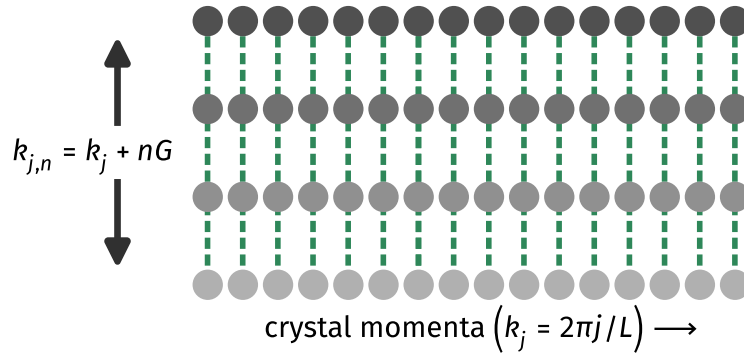


Figure 1.6: The complete set of momentum states in the system. Each row represents a band of states. Each column contains the set of momenta that are connected by a reciprocal lattice vector nG . The states in the lowest band give rise to the crystal momenta. All the momentum states in a column are connected to each other through the potential term (green dotted line). Any two states within a band, on the other hand, are decoupled from each other.

Decomposition into sum of zero-dimensional problems

We begin by writing the Hamiltonian completely in momentum space. A general momentum is represented as $k + nG$, where n is the band index goes from 0 to $N - 1$, and k is the crystal momentum that spans through the lowest band. With this notation, we have

$$\begin{aligned}
 \mathcal{H} &= \frac{\hbar^2}{2m} \int dk \sum_n c^\dagger(k + nG) (k + nG)^2 c(k + nG) \\
 &\quad + \int dk \sum_{m,n} [\mathcal{V}(nG) c^\dagger(k + nG + mG) c(k + mG) + \text{h.c.}] \\
 &= \frac{2\pi}{a} \int dk \mathcal{H}(k)
 \end{aligned} \tag{4.2}$$

where

$$\begin{aligned}
 \mathcal{H}(k) &= \frac{a}{2\pi} \frac{\hbar^2}{2m} \sum_n c^\dagger(k+nG) (k+nG)^2 c(k+nG) \\
 &\quad + \frac{a}{2\pi} \sum_{m,n} [v(nG) c^\dagger(k+nG+mG) c(k+mG) + \text{h.c.}] \\
 &= \frac{a}{2\pi} \int dx' c^\dagger(x') \left[\frac{1}{2m} (-i\hbar\partial_{x'} + \hbar k)^2 + V(x') \right] c(x').
 \end{aligned} \tag{4.3}$$

We have used x' to symbolise the position variable conjugate to the new set of momenta $\{nG\} =$

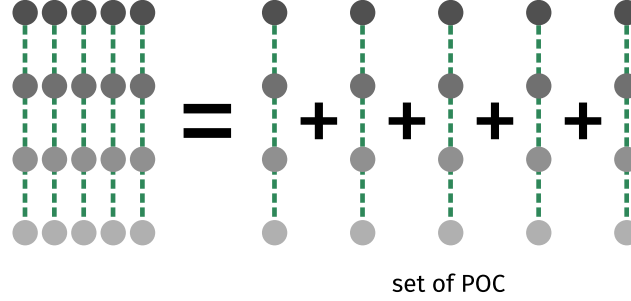


Figure 1.7: Because of the periodicity of the potential, the problem splits into a sum of particle on a circle (POC) problems, each formed by a specific crystal momentum and other momentum states that are related to it by a reciprocal vector.

$-i\partial_{x'}$ that span the restricted Hilbert space of $\mathcal{H}(k)$. Also, $\mathcal{V}(k)$ is the Fourier transform of the real space potential $V(x)$.

This decoupling of the full Hamiltonian $\mathcal{H}$ into a block-diagonal form (each block indexed by the crystal momentum k) can be attributed to the discrete translation symmetry present in the problem and could have been anticipated very generally, as we now show. The Hamiltonian satisfies $[\mathcal{H}, T(a)] = 0$, where $T(x) = e^{i\hat{p}x}$ is the translation operator. By choosing a starting location $x = 0$, applying the translation operator cycles through the set of states $|x = na\rangle$; $n = 0, 1, \dots, N_{\text{well}}$; this set is closed under the application of $T(a)$. It is easy enough to guess the eigenstates of $T(a)$. Since the application of $T(a)$ increases the position monotonically, the only way to create an eigenstate is to involve all states. With that in mind, the simplest ansatz is of the form $|\Phi\rangle = \sum_n e^{-i\Phi n} |an\rangle$. Applying the translation operator on this state gives

$$T(a) |\Phi\rangle = \sum_{n=0}^{N_{\text{well}}-2} e^{-i\Phi n} |a(n+1)\rangle + e^{-i\Phi(N_{\text{well}}-1)} |aN_{\text{well}}\rangle = e^{i\Phi} \left[e^{-i\Phi N_{\text{well}}} |aN_{\text{well}}\rangle + \sum_{n=1}^{N_{\text{well}}-1} e^{-i\Phi n} |an\rangle \right] \tag{4.4}$$

Since we have applied PBC on the system, we used $|aN_{\text{well}}\rangle$ for $|0\rangle$. If the values of Φ are chosen to be some integer multiple of $2\pi/N_{\text{well}}$, we would have $e^{-i\Phi N_{\text{well}}} = 1$, and the entire state can be resummed into the form

$$T(a) |\Phi\rangle = e^{i\Phi} |\Phi\rangle \tag{4.5}$$

This shows that $|\Phi\rangle$ is an eigenstate of the translation operator. Since the set of states has a dimension of N_{well} , the number of eigenstates and the number of possible values of Φ will also be N_{well} . This

fixes the eigenvalues to $\Phi = 2\pi m/N_{\text{well}}, m = 0, 1, \dots, N_{\text{well}} - 1$. These will also be eigenstates of the Hamiltonian, because they are indexed by different values of Φ and the Hamiltonian cannot change Φ (because it commutes with T). As will be shown later, the eigenvalues Φ are actually related to the crystal momentum, through $\Phi = ka$. The set of allowed values of Φ form the first Brillouin zone.

However, $|\Phi\rangle$ as defined above is not the only eigenstate with eigenvalue $e^{i\Phi}$. One can simply shift the starting location $x = 0$ by any distance smaller than a , and repeat the entire procedure. The eigenstates hence obtained will have the same eigenvalues as above:

$$|\Phi, \delta\rangle = \sum_n e^{-i\Phi n} |an + \delta\rangle, \quad 0 \leq \delta < a; \quad (4.6)$$

$$T(a) |\Phi, \delta\rangle = \sum_{n=0}^{N_{\text{well}}-2} e^{-i\Phi n} |a(n+1) + \delta\rangle + e^{-i\Phi(N_{\text{well}}-1)+\delta} |aN_{\text{well}} + \delta\rangle = e^{i\Phi} |\Phi, \delta\rangle \quad (4.7)$$

The translation operator is therefore able to at most classify the complete set of states into N_{well} number of blocks, each labelled by a specific value of Φ . Each block has a large number of states (N_{levs}), and the Hamiltonian will general scatter these states into each other. The Hamiltonian for each of these blocks can be represented as $\mathcal{H}(\Phi) = \mathcal{P}_\Phi \mathcal{H} \mathcal{P}_\Phi$ (where $\mathcal{P}_\Phi$ projects onto the block of states with eigenvalue $e^{i\Phi}$), such that the total Hamiltonian can be represented as $\mathcal{H} = \sum_\Phi \mathcal{H}(\Phi)$. Upon rescaling by $2\pi/a$ and taking the limit of $N_{\text{well}} \rightarrow \infty$, this is the same Hamiltonian decoupling relation as that obtained in eq. 4.2.

Since the spacing between two adjacent momentum is G , the maximum value of x' has to be $2\pi/G = a$, with the ends identified with each other: $\psi(x' = 0) = \psi(x' = a)$. We can then define a dimensionless variable $\theta = 2\pi x'/a$ that goes from 0 to 2π , and is conjugate to the angular wave number $Q = -i\partial_\theta$. This definition ensures $[\theta, Q] = i$. For reasons that will become clear very soon, we will also define a dimensionless crystal momentum $\Phi = ka$. With these modifications, the Hamiltonian takes the simpler form

$$\mathcal{H}(k) = \int_0^{2\pi} d\theta \, c^\dagger(\theta) \left[\frac{\hbar^2}{2ma^2} \left(\hat{Q} + \Phi/2\pi \right)^2 + U(\theta) \right] c(\theta). \quad (4.8)$$

By noting that we have the boundary condition $\psi(\theta = 0) = \psi(\theta = 2\pi)$, we can recognise this Hamiltonian as that describing the dynamics of a particle moving on a circle (POC) with a potential $U(\theta) = \frac{a}{2\pi} V(a\theta/2\pi)$, in the presence of an Aharonov-Bohm flux Φ . Note that $U(\theta)$ has a period of 2π .

Even when reduced to the simpler POC, the problem cannot be simply integrated to obtain the eigenstates. This is because the complete Hamiltonian does not commute with either the position or the momentum operator; it contains quantum fluctuations that need to be resolved in order to obtain the ground state energy. One way of obtaining the energy is to calculate the propagator for transition from $\theta = 0$ to $\theta = 2\pi$, using the method of path integrals [?, ?]. This is shown in Appendix ???. We, however, will attack the problem using a renormalisation group method, called the unitary renormalisation group in order to track the emergence of the tight-binding metal in the limit when the effective amplitude of the single-particle periodic potential at low-energies has become much larger than the typical kinetic energy of the electron.

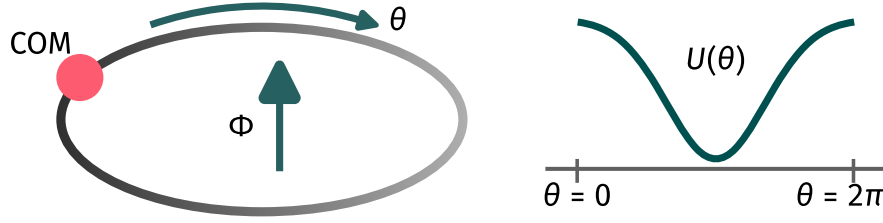


Figure 1.8: The full problem reduces to that of a particle on a circle, in the presence of potential with period 2π , and a flux Φ .

URG analysis of the particle on a circle problem

A summary of the unitary RG method is provided in Appendix ?? . For the purposes of studying the model under renormalisation group, we write the particle on a circle Hamiltonian in momentum(Q -)space. For that, we introduce the Fourier modes

$$c(\theta) = \frac{1}{\sqrt{2\pi}} \int dQ e^{iQ\theta} c(Q), \text{ and } U(\theta) = \int dQ e^{iQ\theta} \mathcal{U}(Q), \quad (4.9)$$

so that the Hamiltonian becomes

$$\mathcal{H}(k) = \int dQ \varepsilon(Q + \Phi/2\pi) \hat{n}(Q) + \int dQ_1 dQ_2 c^\dagger(Q_1) c(Q_2) \mathcal{U}(Q_1 - Q_2). \quad (4.10)$$

where we have defined the dispersion $\varepsilon(Q) = \hbar^2 Q^2 / (2ma^2)$. The second term involves scattering between momentum states Q_1 and Q_2 . The quantum fluctuations that connect the low-energy states to the high-energy states can be integrated out using the URG method [?]. This proceeds through a reduction in the momentum cutoff, ultimately leading to an IR theory. As the large momenta are integrated out, the potential term for scattering between the lower momenta undergo renormalisation. Let, at the l^{th} RG step, the momentum cutoff be Q_l . At this step, fluctuations between all momentum states more energetic than Q_l have been integrated out. The RG equations for the renormalisation of the potential $\mathcal{U}_{ij}^{(l)}$ that scatters any two lower momenta Q_i and Q_j ($Q_i, Q_j < Q_l$) is given by [?]

$$\Delta \mathcal{U}_{ij}^{(l)}(\omega) = \frac{\mathcal{U}_{il} \mathcal{U}_{lj}}{\omega - \varepsilon(Q_l + \Phi/2\pi)}, \quad (4.11)$$

where ω is an energy scale that tracks the amount of quantum fluctuations present in the system, and is characteristic of the URG method. The dispersion also undergoes renormalisation::

$$\Delta \varepsilon^{(l)}(Q_i, \omega) = \frac{|\mathcal{U}_{il}|^2}{\omega - \varepsilon(Q_l + \Phi/2\pi)}, \quad (4.12)$$

Effective description at low energies

Appearance of bands and band gaps

As the URG proceeds, the quantum fluctuations between the momentum states are resolved, leading to a simpler theory for the IR. We focus on the final two states $|Q_0 + \Phi/2\pi\rangle$, $|Q_1 + \Phi/2\pi\rangle$, for $\Phi = 0$

(Brillouin zone edge). The Hamiltonian for their dynamics, just prior to the decoupling of Q_j , is

$$H_{01}^* = \varepsilon^*(Q_0) |Q_0\rangle \langle Q_0| + \varepsilon^*(Q_1) |Q_1\rangle \langle Q_1| + (\mathcal{U}_{01}^* |Q_1\rangle \langle Q_0| + \text{h.c.}) \quad (4.13)$$

Since both the states are at the edge of the band, we can assume $\varepsilon^*(Q_0) \simeq \varepsilon^*(Q_1)$, in which case the problem becomes that of a symmetric double well with tunnelling. The eigenstates are $|Q_0\rangle \pm |Q_1\rangle$, with energies $\varepsilon^*(Q_0) \pm |\mathcal{U}_{01}^*|$. We can substitute this estimate of the energy eigenvalues into the RG equations in the form of the quantum fluctuation scale ω . At the last decoupling, the shifts in the kinetic energy are therefore

$$\Delta\varepsilon^*(Q_i, \omega^\pm) \simeq \frac{|\mathcal{U}_{01}^*|^2}{\varepsilon^*(Q_0) \pm |\mathcal{U}_{01}^*| - \varepsilon(Q_0)} = \pm |\mathcal{U}_{01}^*| = \pm |\mathcal{U}^*(G)|, \quad (4.14)$$

where we used $\mathcal{U}_{01} = \mathcal{U}(Q_1 - Q_0) = \mathcal{U}(G)$. The difference between the two shifts gives the energy difference between the two lowest states of the POC. In this way, the above RG equation shows the emergence of the single-particle band gap and the formation of the band insulator (if the chemical potential is placed within the gap). If we now reintroduce the flux degree of freedom, the lowest energy eigenstates of all the POCs can be combined to form a band of closely spaced energy levels $\{\varepsilon^*(Q_0 + \Phi_j/2\pi)\}$, $\Phi = k_{0,j}a$ (see Fig. 1.9). These levels are indexed by $k_{0,j}$ with j ranging through the first Brillouin zone; $k_{0,j}$ are the crystal momenta of the system.

The second band of states is formed similarly by considering the first excited of the POCs: $\{\varepsilon^*(Q_1 + \Phi_j/2\pi)\}$, $\Phi = k_{0,j}a$. The gap between the first and second band is set by the energy difference between the ground state and first excited state of the POC at $\Phi = 0$. This difference, and hence the gap, is equal to $2|\mathcal{U}_{01}^*|$.

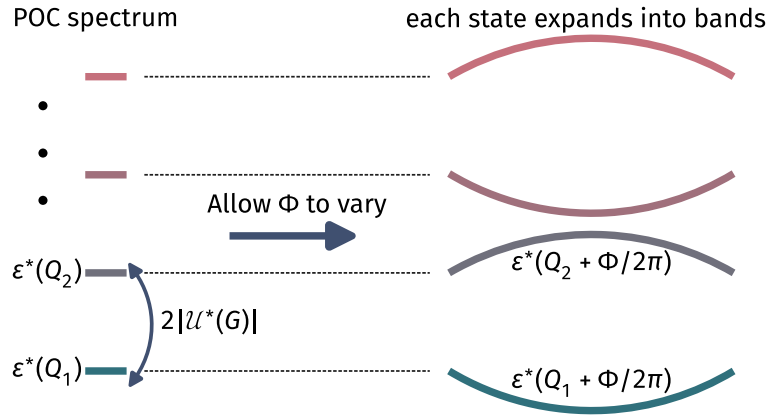


Figure 1.9: Spectral flow transformations on the POC by tuning the flux Φ generates the full band of delocalised single-particle electronic states whose crystal momenta correspond to $\hbar\Phi/a$.

Dispersion relation of the lowest band

In order to derive the dispersion for the lowest band, we note that the final fixed point Hamiltonian (obtained upon decoupling all higher momentum states) is described by the single momentum state

$|Q_0\rangle$. In terms of real space hopping, this means that all short wavelength processes have been removed, and the only process that remains is the longest hopping, that from 0 to 2π . The fixed point Hamiltonian can be written in terms of this process:

$$\begin{aligned} \varepsilon^*(Q_0) |Q_0\rangle \langle Q_0| &= \frac{1}{2} \varepsilon^*(2\pi) (|\theta = 0\rangle + |\theta = 2\pi\rangle) (\langle\theta = 0| + \langle\theta = 2\pi|) = \frac{1}{2} \varepsilon^*(2\pi) (\hat{n}(0) + \hat{n}(2\pi)) \\ &\quad + \frac{1}{2} \varepsilon^*(2\pi) (c^\dagger(0)c(2\pi) + \text{h.c.}) \end{aligned} \quad (4.15)$$

We now recall that we had mapped the spatial point $x = 0$ ($x = a$) to the angular point $\theta = 0$ ($\theta = 2\pi$) point. We now transform back to spatial coordinates to get

$$\frac{1}{2} \varepsilon^*(2\pi) (\hat{n}(0) + \hat{n}(a)) + \frac{1}{2} \varepsilon^*(a) (c^\dagger(0)c(a) + \text{h.c.}) . \quad (4.16)$$

This is computed for a specific value $\Phi = 0$ for the flux (or the crystal momentum). The POC corresponding to a particular flux is built from the states within a particular well. Changing the flux is then equivalent to studying the POC corresponding to the states within a different well. What this means is that tuning the flux by one interval amounts to translating the fixed point Hamiltonian by one period a of the potential. Adding the contributions from all the values of the flux leads to the following tight-binding Hamiltonian:

$$H_{TB} = \varepsilon^*(2\pi) \sum_{j=0}^{N_{\text{well}}-1} \hat{n}(ja) + \frac{1}{2} \varepsilon^*(a) \sum_{j=0}^{N_{\text{well}}-1} (c^\dagger(ja)c((j+1)a) + \text{h.c.}) , \quad (4.17)$$

where the first term corresponds to a constant chemical potential applied to the tight-binding metal in one dimension with a single-particle dispersion $\varepsilon(k) = -\varepsilon^*(a) \cos(ka)$ where k corresponds to the set of crystal momenta $k \in [-\frac{\pi}{a}, \frac{\pi}{a}]$. The exercise, therefore, shows the emergence of nearest-neighbour hopping in the lowest band by starting from a more microscopic model of a particle in a periodic potential. (An alternate derivation of the tight-binding metal has also been shown from path-integral methods in Appendix ??).

1.5 Alternative Approach Towards The URG: Starting from Anderson's Poor Man's Scaling Approach

Recasting Poor Man's scaling as a block-diagonalisation algorithm

We first describe the formalism of the poor man's scaling (PMS) method, first formulated by Anderson [6]. The problem is defined as

$$\mathcal{H} |\Psi\rangle = E |\Psi\rangle \quad (5.1)$$

$\mathcal{H}$ is the total Hamiltonian and $|\Psi\rangle$ and E are the exact eigenstate and eigenvalue of $\mathcal{H}$. We imagine a separation of the total Hilbert space into two set of states, and we call these two states $|0\rangle$ and

$|1\rangle$. This separation depends on which scattering term we want to kill by this transformation. For example, in the URG, we typically select a particular electron $q\beta$ and then kill the scattering terms that change the number of this state. In that case, $|0\rangle$ will refer to the set of states $\{|\hat{n}_{q\beta} = 0\rangle \otimes |\phi_0\rangle\}$ and $|1\rangle$ will refer to the set of states $\{|\hat{n}_{q\beta} = 1\rangle \otimes |\phi_1\rangle\}$. $|\phi_{0,1}\rangle$ refer to the states of all the other electrons. As another example, if we wanted to separate the charge-Kondo and the spin-Kondo from the SIAM, we would want to kill the terms that scatter between the spin-full subspace $\hat{n}_d = 1$ to the spin-less subspace $\hat{n}_d = 0, 2$. These two will then be the $|0\rangle$ and $|1\rangle$ sets.

Keeping this separation in mind, the exact eigenstate $|\Psi\rangle$ can be split as

$$|\Psi\rangle = \sum_i |\phi_0^i\rangle + \sum_i |\phi_1^i\rangle \quad (5.2)$$

The Hamiltonian can also be split as

$$\mathcal{H} = H_0 + V_+ + V_- \quad (5.3)$$

H_0 does not scatter between $\{|0\rangle\}$ and $\{|1\rangle\}$. It contains the diagonal parts as well as scatterings inside the subspaces. $V_{\pm}$ scatter between the subspaces:

$$\begin{aligned} V_+ \{|0\rangle\} &\mapsto \{|1\rangle\}, & V_+ |1\rangle &\rightarrow 0 \\ V_- \{|1\rangle\} &\mapsto \{|0\rangle\}, & V_- |0\rangle &\rightarrow 0 \end{aligned} \quad (5.4)$$

The Schrodinger equation can thus be split into

$$\begin{aligned} H_0 \sum_i |\phi_0^i\rangle + V_- \sum_i |\phi_1^i\rangle &= E \sum_i |\phi_0^i\rangle \\ H_0 \sum_i |\phi_1^i\rangle + V_+ \sum_i |\phi_0^i\rangle &= E \sum_i |\phi_1^i\rangle \end{aligned} \quad (5.5)$$

Eliminating $\sum_i |\phi_1^i\rangle$ gives

$$H_0 \sum_i |\phi_0^i\rangle + V_- \frac{1}{E_1 - H_0} V_+ \sum_i |\phi_0^i\rangle = E \sum_i |\phi_0^i\rangle \quad (5.6)$$

The effective Hamiltonian in this subspace is therefore

$$\tilde{\mathcal{H}}_0 = H_0 + V_- \frac{1}{E - H_0} V_+ \quad (5.7)$$

Similarly, eliminating $\sum_i |\phi_0^i\rangle$ gives the effective Hamiltonian in the other subspace,

$$\tilde{\mathcal{H}}_1 = H_0 + V_+ \frac{1}{E - H_0} V_- \quad (5.8)$$

The total effective Hamiltonian that does not scatter between the two subspaces is

$$\tilde{\mathcal{H}}(E) = H_0 + \underbrace{V_- \frac{1}{E - H_0} V_+ + V_+ \frac{1}{E - H_0} V_-}_{\text{renormalization}} \quad (5.9)$$

This is of course a function of whatever exact energy eigenvalue we chose, E . Different choices will give different effective Hamiltonians. The renormalization will now be written in terms of the matrix elements. Since the entire $\mathcal{H}_X$ must be Hermitian, we must have $V_- = V_+^\dagger \equiv V$.

$$\Delta\mathcal{H}(E) = V \frac{1}{E - H_0} V^\dagger + V^\dagger \frac{1}{E - H_0} V \quad (5.10)$$

Now take the first term and insert complete bases on both sides of V and $V^\dagger$.

$$\begin{aligned} V \frac{1}{E - H_0} V^\dagger &= \sum_{ijk} |\phi_0^i\rangle \langle \phi_0^i| V |\phi_1^j\rangle \langle \phi_1^j| \frac{1}{E - H_0} |\phi_1^j\rangle \langle \phi_1^j| V^\dagger |\phi_0^k\rangle \langle \phi_0^k| \\ &= \sum_{ijk} |\phi_0^i\rangle V_{ij} \langle \phi_1^j| \frac{1}{E - H_0} |\phi_1^j\rangle V_{kj}^\dagger \langle \phi_0^k|, \end{aligned} \quad (5.11)$$

where we defined $\langle \phi_0^i| V |\phi_1^j\rangle = V_{ij}$. We now approximate H_0 by keeping just the diagonal part, and allowing the balance to redefine E into ω . Then, $(E - H_0) |\phi_{0,1}^j\rangle \equiv (\omega_{0,1} - E_{0,1}^j) |\phi_{0,1}^j\rangle$. That gives

$$V \frac{1}{E - H_0} V^\dagger = \sum_{ijk} |\phi_0^i\rangle \langle \phi_0^k| \frac{V_{ij} V_{kj}^\dagger}{\omega_1 - E_1^j} \quad (5.12)$$

The second term similarly gives

$$V^\dagger \frac{1}{E - H_0} V = \sum_{ijk} |\phi_1^i\rangle \langle \phi_1^k| \frac{V_{ji}^\dagger V_{jk}}{\omega_0 - E_0^j} \quad (5.13)$$

The total renormalization becomes

$$\Delta\mathcal{H}(E) = \sum_{ijk} \left(\frac{1}{\omega_1 - E_1^j} |\phi_0^i\rangle \langle \phi_0^k| V_{ij} V_{kj}^\dagger + \frac{1}{\omega_0 - E_0^j} |\phi_1^i\rangle \langle \phi_1^k| V_{ji}^\dagger V_{jk} \right) \quad (5.14)$$

This is a general expression that would work irrespective of whether you are decoupling multiple electrons or a single electron. However, the ω are unknown and we need some prescription for replacing them. Since the E is the eigenstate of the initial state on which the scattering terms act, it makes sense to replace them with the initial state energy.

$$\Delta\mathcal{H}(E) = \sum_{ijk} \left(\frac{1}{E_0^k - E_1^j} |\phi_0^i\rangle \langle \phi_0^k| V_{ij} V_{kj}^\dagger + \frac{1}{E_1^k - E_0^j} |\phi_1^i\rangle \langle \phi_1^k| V_{ji}^\dagger V_{jk} \right) \quad (5.15)$$

However, closer inspection reveals that this choice makes the renormalization non-Hermitian. So

the correct choice is to keep both the initial and final energies.

$$\begin{aligned}
 \Delta\mathcal{H} &= \frac{1}{2} \sum_{ijk} \frac{1}{\omega_1 - E_1^j} \left(|\phi_0^i\rangle \langle \phi_0^k| V_{ij} V_{kj}^\dagger + |\phi_0^k\rangle \langle \phi_0^i| V_{kj} V_{ij}^\dagger \right) \\
 &\quad + \frac{1}{2} \sum_{ijk} \frac{1}{\omega_0 - E_0^j} \left(|\phi_1^i\rangle \langle \phi_1^k| V_{ji}^\dagger V_{jk} + |\phi_1^k\rangle \langle \phi_1^i| V_{jk}^\dagger V_{ji} \right) \\
 &= \frac{1}{2} \sum_{ijk} \left(\frac{1}{E_0^k - E_1^j} |\phi_0^i\rangle \langle \phi_0^k| V_{ij} V_{kj}^\dagger + \frac{1}{E_0^k - E_1^j} |\phi_0^k\rangle \langle \phi_0^i| V_{kj} V_{ij}^\dagger \right) \\
 &\quad + \frac{1}{2} \sum_{ijk} \left(\frac{1}{E_1^k - E_0^j} |\phi_1^i\rangle \langle \phi_1^k| V_{ji}^\dagger V_{jk} + \frac{1}{E_1^k - E_0^j} |\phi_1^k\rangle \langle \phi_1^i| V_{jk}^\dagger V_{ji} \right)
 \end{aligned} \tag{5.16}$$

Therefore,

$$\Delta\mathcal{H} = \frac{1}{2} \sum_{ijk} \left(\frac{1}{E_0^k - E_1^j} + \frac{1}{E_0^i - E_1^j} \right) |\phi_0^i\rangle \langle \phi_0^k| V_{ij} V_{kj}^\dagger + \frac{1}{2} \sum_{ijk} \left(\frac{1}{E_1^k - E_0^j} + \frac{1}{E_1^i - E_0^j} \right) |\phi_1^i\rangle \langle \phi_1^k| V_{ji}^\dagger V_{jk} \tag{5.17}$$

In summary, the prescription of replacing all ω with the initial state energy will be correct only if the initial and final states are the same. This happens when we are decoupling a single-electron state - then the total renormalization is of the form $c^\dagger T^\dagger c$ such that we start from an initial state, scatter to an intermediate state and then go back to the initial state so that the final state is the same as the initial state. However, if we are using PMS to decouple states in one-shot, each subspace will have multiple states and there might be terms where we do not end up at the initial state we started with. Then the correct prescription would be to use the mean of the initial and final state denominators.

One might wonder how we can generate higher order terms in this method. Eq. 5.17, as it stands, has only $O(V^2)$ terms. The higher order terms were actually dropped when we replaced H_0 with its diagonal part in eq. 5.12. To see the higher order term, we do not drop the off-diagonal part in the denominator, but split the total H_0 into a diagonal and an off-diagonal part: $H_0 = H_d + X$. Note that X is off-diagonal in the subspace of the states that have not been decoupled yet but will be decoupled later. They represent scattering between the lower energy states. X is still diagonal with respect to the states that we are decoupling presently. The total effective Hamiltonian becomes

$$\tilde{\mathcal{H}}(E) = H_0 + V \frac{1}{G_0(E)^{-1} - X} V^\dagger + V^\dagger \frac{1}{G_0(E)^{-1} - X} V \tag{5.18}$$

where $G_0(E)^{-1} = E - H_d$ is the inverse of the non-interacting Greens function. To allow computation, we can expand the denominator in powers of $XG_0(E)^{-1}$:

$$\begin{aligned}
 \Delta H \equiv \tilde{\mathcal{H}}(E) - H_0 &= V G_0(E) [1 + X G_0(E) + X G_0(E) X G_0(E) + \dots] V^\dagger + V^\dagger \frac{1}{G_0(E)^{-1} - X} V \\
 &= \underbrace{V G_0(E) V^\dagger + V^\dagger G_0(E) V}_{\text{two vertex or one loop correction}} + \underbrace{V G_0(E) X G_0(E) V^\dagger + V^\dagger G_0(E) X G_0(E) V}_{\text{three vertex or two loop correction}} \\
 &\quad + \text{higher loop corrections}
 \end{aligned} \tag{5.19}$$

PMS in the language of the URG - obtaining the η operators

To make a better connection with URG, we next show how the PMS formalism works out for a single-electron decoupling. The corresponding problem can be phrased in the following manner. We want to decouple one electron at momentum q from the full Hamiltonian. We can split the exact wavefunction as

$$|\Psi\rangle = |\Psi_0\rangle + |\Psi_1\rangle \quad (5.20)$$

where $|\Psi_0\rangle = (1 - \hat{n}_q) |\Psi^N\rangle$ is that part of the wavefunction where the state q is occupied. $|\Psi_1^N\rangle = \hat{n}_q |\Psi\rangle$ is that part of the wavefunction where the state q is occupied. We can also split the Hamiltonian as

$$\mathcal{H} = \mathcal{H}^d + V_0 + V_+ + V_- \quad (5.21)$$

$\mathcal{H}^d$ is the diagonal part; it has the purely energy terms as well as self-energies that may arise from the diagonal parts of interactions; V_0 is the purely off-diagonal term that does not change $\hat{n}_q$; it is the scattering *inside* the low energy subspace. V_+ and V_- are the purely off-diagonal terms that *do* change $\hat{n}_q$; V_+ takes you from $\hat{n}_q = 0$ to $\hat{n}_q = 1$ and V_- does the opposite.

Substituting eqs. 5.21 and 5.20 in eq. 5.1 gives

$$(\mathcal{H}^d + V_0 + V_+ + V_-) (|\Psi_0\rangle + |\Psi_1\rangle) = E (|\Psi_0\rangle + |\Psi_1\rangle) \quad (5.22)$$

Gathering the kets with $\hat{n}_q = 0, 1$ gives

$$\begin{aligned} (\mathcal{H}_0^d + V_0) |\Psi_0\rangle + V_- |\Psi_1\rangle &= E |\Psi_0\rangle \\ (\mathcal{H}_1^d + V_0) |\Psi_1\rangle + V_+ |\Psi_0\rangle &= E |\Psi_1\rangle \end{aligned} \quad (5.23)$$

The second equation can be written as

$$|\Psi_1\rangle = \eta^\dagger |\Psi_0\rangle \quad (5.24)$$

where

$$(\eta^\dagger)_{\text{PMS}} = \frac{1}{E - \mathcal{H}_1^d - V_0} V_+ \quad (5.25)$$

Substituting this in the first equation gives

$$(\mathcal{H}_0^d + V_0 + V_- \eta^\dagger) |\Psi_0\rangle = E |\Psi_0\rangle \quad (5.26)$$

This new Hamiltonian,

$$\tilde{\mathcal{H}}_0 = \mathcal{H}_0^d + V_0 + V_- \eta^\dagger \quad (5.27)$$

has the high energy mode removed; the scattering terms start from the low energy subspace and end at the low energy subspace as well. The renormalization in the low energy subspace scatterings is

$$\Delta V_0 = V_- \eta^\dagger \quad (5.28)$$

If we eliminate $|\Psi_0\rangle$ instead of $|\Psi_1\rangle$, we get the renormalized equation in the high energy subspace:

$$|\Psi_0\rangle = \eta |\Psi_1\rangle \quad (5.29)$$

where

$$(\eta)_{\text{PMS}} = \frac{1}{E - \mathcal{H}_0^d - V_0} V_- \quad (5.30)$$

,so

$$\left(\mathcal{H}_1^d + V_0 + V_+ \eta \right) |\Psi_1\rangle = E |\Psi_1\rangle \quad (5.31)$$

The renormalized Hamiltonian in the high energy subspace is thus

$$\tilde{\mathcal{H}}_1 = \mathcal{H}_1^d + V_0 + V_+ \eta \quad (5.32)$$

If we want to keep both the high energy and low energy parts of the Hamiltonian, the new Hamiltonian is

$$\begin{aligned} \tilde{\mathcal{H}} &= \tilde{\mathcal{H}}_1 \hat{n} + \tilde{\mathcal{H}}_0 (1 - \hat{n}) \\ &= \mathcal{H}_0^d + \mathcal{H}_1^d + V_0 + V_+ \eta + V_- \eta^\dagger \end{aligned} \quad (5.33)$$

The total renormalization is

$$(\Delta \mathcal{H})_{\text{PMS}} = V_+ (\eta)_{\text{PMS}} + V_- (\eta^\dagger)_{\text{PMS}} \quad (5.34)$$

It can be shown that if we define a unitary operator $U = 1 - \eta + \eta^\dagger$, the transformed Hamiltonian $U \mathcal{H} U^\dagger$ is the same as eq. 5.33. This, along with the properties of η , have been shown in section ???. The important feature of eq. 5.33 is that there is no term in the transformed Hamiltonian which scatters between $|\Psi_0\rangle$ and $|\Psi_0\rangle$ - the two subspaces have been truly decoupled.

$$[U \mathcal{H} U^\dagger, n_q] = 0 \quad (5.35)$$

We can write down the renormalized Schrodinger equation in the low energy subspace, from eq. 5.26,

$$\tilde{\mathcal{H}}_0 |\Psi_0\rangle = E |\Psi_0\rangle \quad (5.36)$$

and again repeat the entire process. $\tilde{\mathcal{H}}_0$ now takes the place of $\mathcal{H}$ and $|\Psi_0\rangle$ takes the place of $|\Psi\rangle$ in eq. 5.1.

The expression for URG is obtained in an almost identical way. The only difference is that instead of starting with the exact eigenpair $(E, |\Psi\rangle)$, we start with a more general pair $(\tilde{\mathcal{H}}, |\Phi\rangle)$ where $|\Phi\rangle$ is not necessarily an exact eigenstate of $\mathcal{H}$. It is defined by $\mathcal{H}'$, which is in turn defined as $\hat{n}_q \mathcal{H}' (1 - \hat{n}_q) = 0$. $|\Phi\rangle$ is then defined by

$$\mathcal{H} |\Phi\rangle = \mathcal{H}' |\Phi\rangle \quad (5.37)$$

This definition of $\mathcal{H}'$ is the very minimum that we must have in order to fulfill our goal (decouple q).

The operators η and its conjugate change accordingly:

$$\begin{aligned} (\eta)_{\text{URG}} &= \frac{1}{\tilde{\mathcal{H}} - \mathcal{H}_0^d - V_0} V_- \\ &= \frac{1}{\hat{\omega} - \mathcal{H}_0^d} V_- \end{aligned} \quad (5.38)$$

where $\hat{\omega} \equiv \mathcal{H}' - V_0$ now embodies the quantum fluctuations inherent in the Hamiltonian through the scattering term V_0 . Similarly,

$$(\eta^\dagger)_{\text{URG}} = \frac{1}{\hat{\omega} - \mathcal{H}_1^d} V_+ \quad (5.39)$$

The renormalization is again

$$(\Delta\mathcal{H})_{\text{URG}} = V_+ (\eta)_{\text{URG}} + V_- (\eta^\dagger)_{\text{URG}} \quad (5.40)$$

This again allows us to write down a unitary operator that decouples the entangled state:

$$U = 1 - \eta + \eta^\dagger, [\hat{n}_q, U\mathcal{H}U^\dagger] = 0 \quad (5.41)$$

where $\tilde{\mathcal{H}} = U^\dagger \mathcal{H} U$. We can now write down a new problem in this decoupled space with the rotated items and attempt to decouple another electron q' . We will again choose some general eigenpair $(\mathcal{H}', |\Phi\rangle)$ such that $\tilde{\mathcal{H}} |\Phi\rangle = \mathcal{H}' |\Phi\rangle$ and $[\mathcal{H}', \hat{n}_{q'}] = 0$.

Summarizing, the general Hamiltonian is not diagonal in the Fock space basis. URG, in order to proceed, selects one non-Fock basis of states $|\Phi\rangle$ such that q is decoupled in that Hamiltonian. Since there can be lots of such basis, there is a freedom in this choice. With this basis in mind, URG then finds a unitary operator which when operated on the Hamiltonian takes us to the form in which it is diagonal in the Fock space basis. Note that this form is a function of the chosen $|\Phi\rangle$. We then select the second degree of freedom and repeat the process. What PMS does is, it exploits the freedom of choice and selects the exact eigenstate $|\Psi\rangle$ of the Hamiltonian as the non-Fock basis $|\Phi\rangle$. Doing that returns a rotated Hamiltonian which is diagonal in q , and is a function of the chosen state, same as URG. The conclusion is that depending on which state we choose as our diagonal non-Fock basis, URG and PMS will cause flows along different lines in general.

As the couplings flow, V_0 will also flow, leading to a flow of $\hat{\omega}$. Just at the fixed point, the denominator of URG vanishes, giving the equation

$$(\hat{\omega} - \mathcal{H}_1^d) V_+ |\Psi_0\rangle \text{ or } (\hat{\omega} - \mathcal{H}_1^d) V_- |\Psi_1\rangle \quad (5.42)$$

This means that one of the eigenvalues of $\hat{\omega}$ matches with the eigenvalue of the diagonal part $\mathcal{H}^d$, either in the occupied sector ($\mathcal{H}_1^d$) or unoccupied sector ($\mathcal{H}_0^d$). Since the eigenvalues are unchanged during the unitary renormalization, this implies that ω takes up one of the eigenvalues of the whole Hamiltonian $\mathcal{H}$. This will correspond to the fixed point obtained from PMS if we had started PMS with that eigenvalue.

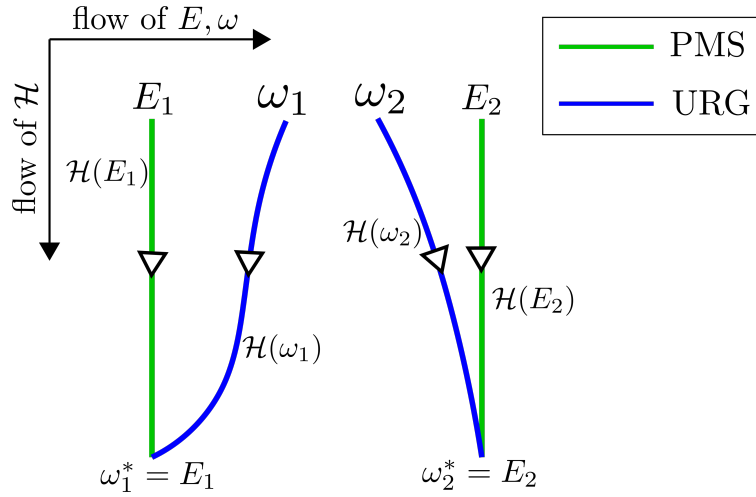


Figure 1.10: Flows of PMS (green) and URG (blue)

In short, while the PMS flow is parametrised by one of the exact energy eigenvalues E , the URG flow is parametrised by a non-trivial operator $\hat{\omega}$ which incorporates both a diagonal part and an off-diagonal part and itself flows under the URG. At the fixed point, the off-diagonal part cancels out and the $\hat{\omega}$ finally flows to one of the energy eigenvalues and the URG fixed point matches with one of the PMS fixed points.

- The *only* difference in the formalism of PMS and URG is that while PMS uses the exact energy eigenvalue E to parameterise the flow, URG uses a general intermediate decoupled Hamiltonian to do the same. Since the E is also, technically, an intermediate decoupled Hamiltonian (it is the final Hamiltonian), PMS can be seen as an URG but with a specific choice for the paramter.
- In practise, PMS replaces $E - V_0$ with the diagonal part of the initial state at the current step of the RG. We are talking about the energy of the initial state, not the intermediate state. This is because, from eq. 5.1, E is the energy of the initial state on which $V_{\pm}$ act.
- The ideal solution would have been to substitute the exact energy and the total scattering term V , but since we do not know E and keeping the V would make the thing untractable, we use our current best guess (renormalised diagonal part). As the RG flows, both E_j and V flow, such that at the fixed point, V becomes zero (scattering terms get removed) and E_j morphs into the exact E .
- In practise, URG replaces the $\hat{\omega}$ with a guess for the final energy E . This however ignores the renormalization of $\hat{\omega}$. A better approach would be to replace it with E_j , following PMS. That would act like the one-particle renormalization of $\hat{\omega}$.
- PMS usually drops any diagonal component of the scattering from the denominator. For example, in the PMS of the Kondo model by Anderson [6] or that of the anisotropic power

law Kondo model by Chenge et.al [8], they do not keep the term $J_z S_d^z s^z$ in the denominator although it is number (spin) conserving. Such terms are kept in the denominator of the URG though. It must be mentioned however that ref. [9] *does* bring a diagonal charge-charge interaction in the denominator in the PMS of the extended Anderson model.

1.6 The Unitary RG as a non-perturbative Schrieffer-Wolff transformation

Formalism

We have a general Hamiltonian

$$\mathcal{H} = \mathcal{H}_0 + \mathcal{H}_X \quad (6.1)$$

$\mathcal{H}_0$ is diagonal w.r.t a particular degree of freedom. V is off-diagonal w.r.t that same degree of freedom. Let S be an *anti-Hermitian* and *off-diagonal* operator. $U = e^S$ is then a unitary transformation.

$$\begin{aligned} U\mathcal{H}U^\dagger &= e^S (\mathcal{H}_0 + \mathcal{H}_X) e^{-S} \\ &= (\cosh(S) + \sinh(S)) (\mathcal{H}_0 + \mathcal{H}_X) (\cosh(S) - \sinh(S)) \\ &= H_1 + H_2 \end{aligned} \quad (6.2)$$

where H_1 is diagonal and H_2 is off-diagonal.

$$\begin{aligned} H_1 &= \cosh(S) \mathcal{H}_0 \cosh(S) - \sinh(S) \mathcal{H}_0 \sinh(S) - \cosh(S) \mathcal{H}_X \sinh(S) \\ &\quad + \sinh(S) \mathcal{H}_X \cosh(S) \\ H_2 &= -\cosh(S) \mathcal{H}_0 \sinh(S) + \sinh(S) \mathcal{H}_0 \cosh(S) + \cosh(S) \mathcal{H}_X \cosh(S) \\ &\quad - \sinh(S) \mathcal{H}_X \sinh(S) \end{aligned} \quad (6.3)$$

The decoupling condition is $H_2 = 0$.

For small S , we have $\sinh S \sim S$ and $\cosh S \sim 1 + \frac{1}{2}S^2$. Therefore, the off-diagonal part, up to second order, is

$$H_2 = -\mathcal{H}_0 S + S \mathcal{H}_0 + \mathcal{H}_X + O(S^3) = [S, \mathcal{H}_0] + \mathcal{H}_X \quad (6.4)$$

The second order decoupling condition is thus

$$[S, \mathcal{H}_0] = -\mathcal{H}_X \quad (6.5)$$

The effective Hamiltonian is what remains, H_1 . That becomes, at second order,

$$\begin{aligned}
 H_1 &= \left(1 + \frac{1}{2}S^2\right) \mathcal{H}_0 \left(1 + \frac{1}{2}S^2\right) - S\mathcal{H}_0S - \left(1 + \frac{1}{2}S^2\right) \mathcal{H}_XS + S\mathcal{H}_X \left(1 + \frac{1}{2}S^2\right) \\
 &= \mathcal{H}_0 + \frac{1}{2} \{S^2, \mathcal{H}_0\} - S\mathcal{H}_0S - \mathcal{H}_XS + S\mathcal{H}_X + O(S^3) \\
 &= \mathcal{H}_0 + \frac{1}{2}S [S, \mathcal{H}_0] - \frac{1}{2} [S, \mathcal{H}_0] S + [S, \mathcal{H}_X] + O(S^3) \\
 &= \mathcal{H}_0 + \frac{1}{2} [S, [S, \mathcal{H}_0]] + [S, \mathcal{H}_X] + O(S^3) \\
 &= \mathcal{H}_0 + \frac{1}{2} [S, -\mathcal{H}_X] + [S, \mathcal{H}_X] + O(S^3) \\
 &= \mathcal{H}_0 + \frac{1}{2} [S, \mathcal{H}_X] + O(S^3)
 \end{aligned} \tag{6.6}$$

Avoiding the perturbative route, we can take $S = \frac{\pi}{4} (\eta^\dagger - \eta)$, where η and its conjugate are non-perturbative and Fermionic - they satisfy $\eta^2 = \eta^{\dagger 2} = 0$ and $\{\eta, \eta^\dagger\} = 1$. We can then write

$$\begin{aligned}
 e^S &= \exp \left\{ \frac{\pi}{4} (\eta^\dagger - \eta) \right\} \\
 &= 1 + (\eta^\dagger - \eta) \frac{\pi}{4} + \frac{1}{2!} (\eta^\dagger - \eta)^2 \left(\frac{\pi}{4}\right)^2 + \frac{1}{3!} (\eta^\dagger - \eta)^3 \left(\frac{\pi}{4}\right)^3 + \dots \\
 &= 1 + (\eta^\dagger - \eta) \frac{\pi}{4} - \frac{1}{2!} \left(\frac{\pi}{4}\right)^2 - \frac{1}{3!} (\eta^\dagger - \eta) \left(\frac{\pi}{4}\right)^3 + \frac{1}{4!} \left(\frac{\pi}{4}\right)^4 + \dots \\
 &= \cos \frac{\pi}{4} + (\eta^\dagger - \eta) \sin \frac{\pi}{4} \\
 &= \frac{1}{\sqrt{2}} (1 + \eta^\dagger - \eta)
 \end{aligned} \tag{6.7}$$

There we used

$$(\eta^\dagger - \eta)^2 = \eta^{\dagger 2} + \eta^2 - \{\eta^\dagger, \eta\} = -1 \quad \left[\because \eta^2 = \eta^{\dagger 2} = 0 \right] \tag{6.8}$$

and hence

$$(\eta^\dagger - \eta)^3 = -1 (\eta^\dagger - \eta) \tag{6.9}$$

and so on. This simplification allows us to write

$$\cosh S = \frac{1}{2} [e^S + e^{-S}] = \frac{1}{2\sqrt{2}} (1 + \eta^\dagger - \eta + 1 - \eta^\dagger + \eta) = \frac{1}{\sqrt{2}} \tag{6.10}$$

and

$$\sinh S = \frac{1}{2} [e^S - e^{-S}] = \frac{1}{2\sqrt{2}} (1 + \eta^\dagger - \eta - 1 + \eta^\dagger - \eta) = \frac{1}{\sqrt{2}} (\eta^\dagger - \eta) \tag{6.11}$$

The off-diagonal part now becomes

$$H_2 = \frac{1}{2} (\mathcal{H}_X - \eta^\dagger \mathcal{H}_X \eta^\dagger - \eta \mathcal{H}_X \eta + [\eta^\dagger - \eta, \mathcal{H}_0]) \tag{6.12}$$

	renormalization	decoupling condition
SWT	$\frac{1}{2} [S, \mathcal{H}_X]$	$[S, \mathcal{H}_0] = -\mathcal{H}_X$
URG	$\frac{2}{\pi} [S, \mathcal{H}_X]$	$[S, \mathcal{H}_0] = -\frac{\pi}{4} \mathcal{H}_X + \frac{4}{\pi} S \mathcal{H}_X S$

Table 1.1: Comparison of perturbative and non-perturbative canonical transformations

The vanishing of this quantity is now the decoupling condition, and is also given in eq 16 of ref. [1].

To look for a decoupling condition similar to eq. 6.5, we can re-express the cosh and sinh in eq. 6.10 in terms of S , by substituting $\eta^\dagger - \eta = \frac{4}{\pi} S$:

$$\cosh S = \frac{1}{\sqrt{2}}, \text{ and } \sinh S = \frac{4}{\sqrt{2}\pi} S \quad (6.13)$$

That gives

$$H_2 = \frac{1}{2} \left(\frac{4}{\pi} [S, \mathcal{H}_0] + \mathcal{H}_X - \frac{16}{\pi^2} S \mathcal{H}_X S \right) \quad (6.14)$$

The decoupling condition becomes

$$[S, \mathcal{H}_0] = -\frac{\pi}{4} \mathcal{H}_X + \frac{4}{\pi} S \mathcal{H}_X S \quad (6.15)$$

This can be compared to the second order condition: $[S, \mathcal{H}_0] = -\mathcal{H}_X$. We can also write the effective Hamiltonian for this non-perturbative case.

$$U \mathcal{H} U^\dagger = H_1 = \frac{1}{2} \mathcal{H}_0 - \frac{4}{\pi^2} S \mathcal{H}_0 S + \frac{2}{\pi} [S, \mathcal{H}_X] \quad (6.16)$$

The differences between the perturbative and non-perturbative ways are summarized in table 1.1. There appear to be two differences between these decoupling conditions: (a) a pre-factor of $\frac{\pi}{4}$ for the first term on the right hand side, and (b) the altogether new second term on the right hand side. Both are outcomes of the non-perturbative nature of URG. This offers evidence that the physics captured by the effective Hamiltonian (and its associated low-energy many-particle Hilbert space) obtained from URG lies well beyond that obtained from SWT. Further, it shows that the SWT can only be justified as an expansion in a small parameter (say, $\frac{1}{U}$) in the Anderson impurity problem), followed by a truncation of the BCH expansion and a projection onto a particular low-energy subspace. The truncation and projection are adopted simultaneously, and appear to impose the limit of $U = \infty$ by hand. The URG flow never attains such a limit, thus suggesting that there exists a lot of interesting physics that could potentially be lost in the SWT procedure. Further, the projection finally applied within SWT means that we can never recover what is thrown away. This is again not the case with URG.

Obtaining renormalization via Schrieffer-Wolff transformation - Comparison with Poor Man's scaling and URG

Similar to the situation in Poor Man's scaling, one can visualize two set of states and let $\mathcal{H}_X = V_+ + V_-$ be the scattering that connects them and hence the one we want to kill. Let S be of the form

$$S = \sum_{ij} \left[s |\phi_1^i\rangle \langle \phi_0^j| - s^\dagger |\phi_0^j\rangle \langle \phi_1^i| \right] \quad (6.17)$$

This form is of course chosen to make S anti-Hermitian and off-diagonal. The part s can be determined from the decoupling condition:

$$-\mathcal{H}_X = [S, H_0] = SH_0 - H_0S \quad (6.18)$$

Multiplying with $\langle \phi_0^a|$ and $|\phi_1^b\rangle$ from the left and right respectively gives

$$-\langle \phi_0^a| V + V^\dagger |\phi_1^b\rangle = \langle \phi_0^a| SH_0 - H_0S |\phi_1^b\rangle \quad (6.19)$$

Since $V^\dagger$ acts on $|0\rangle$, it will not affect the LHS. Also, $\langle \phi_0^a| V |\phi_1^b\rangle = V_{ab}$. If we now consider only the diagonal part of H_0 , we can write $H_0(|\phi_0^a\rangle, |\phi_1^b\rangle) = (E_{0,a} |\phi_0^a\rangle, E_{1,b} |\phi_1^b\rangle)$. We then get

$$\begin{aligned} -V_{ab} &= \langle \phi_0^a| \sum_i [S |\phi_1^i\rangle \langle \phi_1^i| H_0 - H_0 |\phi_0^i\rangle \langle \phi_0^i| S] |\phi_1^b\rangle = \sum_i [S_{ai} E_1^i \delta_{bi} - E_0^i \delta_{ai} S_{ib}] = S_{ab} E_1^b - E_0^a S_{ab} \\ \implies S_{ab} &= \frac{V_{ab}}{E_0^a - E_1^b} \end{aligned} \quad (6.20)$$

where we defined $\langle \phi_0^x| S |\phi_1^y\rangle = S_{xy}$. The total generator is

$$S = \sum_{ij} \left[S_{ij} |\phi_0^i\rangle \langle \phi_1^j| - S_{ij}^\dagger |\phi_1^j\rangle \langle \phi_0^i| \right] = \sum_{ij} \frac{1}{E_0^i - E_1^j} \left[V_{ij} |\phi_0^i\rangle \langle \phi_1^j| - V_{ij}^\dagger |\phi_1^j\rangle \langle \phi_0^i| \right] \quad (6.21)$$

The renormalization is thus

$$\begin{aligned} \Delta\mathcal{H} &= \frac{1}{2} [S, \mathcal{H}_X] \\ &= \frac{1}{2} \sum_{ij,kl} \left[\frac{1}{E_0^i - E_1^j} \left(V_{ij} |\phi_0^i\rangle \langle \phi_1^j| - V_{ij}^\dagger |\phi_1^j\rangle \langle \phi_0^i| \right), V_{kl} |\phi_0^k\rangle \langle \phi_1^l| + V_{kl}^\dagger |\phi_1^l\rangle \langle \phi_0^k| \right] \\ &= \frac{1}{2} \sum_{ij,kl} \left[\frac{1}{E_0^i - E_1^j} \left(V_{ij} V_{kl}^\dagger |\phi_0^i\rangle \langle \phi_0^k| \delta_{jl} - V_{ij}^\dagger V_{kl} |\phi_1^j\rangle \langle \phi_1^l| \delta_{ik} \right. \right. \\ &\quad \left. \left. - V_{kl}^\dagger V_{ij} |\phi_1^l\rangle \langle \phi_1^j| \delta_{ki} + V_{kl} V_{ij}^\dagger |\phi_0^k\rangle \langle \phi_0^i| \delta_{lj} \right) \right] \\ &= \frac{1}{2} \sum_{ijk} \left[\frac{1}{E_0^i - E_1^j} \left(V_{ij} V_{kj}^\dagger |\phi_0^i\rangle \langle \phi_0^k| - V_{ij}^\dagger V_{ik} |\phi_1^j\rangle \langle \phi_1^k| - V_{ik}^\dagger V_{ij} |\phi_1^k\rangle \langle \phi_1^j| + V_{kj} V_{ij}^\dagger |\phi_0^k\rangle \langle \phi_0^i| \right) \right] \\ &= \frac{1}{2} \sum_{ijk} \left[\left(\frac{1}{E_0^i - E_1^j} + \frac{1}{E_0^k - E_1^j} \right) V_{ij} V_{kj}^\dagger |\phi_0^i\rangle \langle \phi_0^k| - \left(\frac{1}{E_0^i - E_1^j} + \frac{1}{E_0^i - E_1^k} \right) V_{ij}^\dagger V_{ik} |\phi_1^j\rangle \langle \phi_1^k| \right] \end{aligned} \quad (6.22)$$

This is the same as the PMS result eq. 5.17. It is easy to see that since this transformation is unitary, it has zero trace so as to preserve the trace of the Hamiltonian:

$$\text{Tr}[\mathcal{H}] = \sum_l \left(\langle \phi_0^l | + \langle \phi_1^l | \right) \Delta \mathcal{H} \left(| \phi_0^l \rangle + | \phi_1^l \rangle \right) = \frac{1}{2} \sum_{jl} \frac{2}{E_0^l - E_1^j} V_{lj} V_{lj}^\dagger - \frac{1}{2} \sum_{ji} \frac{2}{E_0^i - E_1^j} V_{il}^\dagger V_{il} = 0 \quad (6.23)$$

We can also make a comparison to the renormalization obtained from URG.

$$\Delta \mathcal{H} = \frac{1}{2} [\eta^\dagger - \eta, \mathcal{H}] \quad (6.24)$$

where

$$\begin{aligned} \eta &= \frac{1}{\omega - \mathcal{H}^d} \sum_{ij} V_{ij} | \phi_0^i \rangle \langle \phi_1^j | = \sum_{ij} \frac{1}{\omega_1^j - E_0^i} V_{ij} | \phi_0^i \rangle \langle \phi_1^j | \\ \Rightarrow \eta^\dagger &= \sum_{ij} \frac{1}{\omega_1^j - E_0^i} V_{ij}^\dagger | \phi_1^j \rangle \langle \phi_0^i | \\ \Rightarrow \eta^\dagger - \eta &= \sum_{ij} \frac{1}{\omega_1^j - E_0^i} \left(V_{ij}^\dagger | \phi_1^j \rangle \langle \phi_0^i | - V_{ij} | \phi_0^i \rangle \langle \phi_1^j | \right) \end{aligned} \quad (6.25)$$

This can be thought of as the generator for the unitary transformations of URG. Comparing with the generator S of eq. 6.21, the prescription to go from URG to SWT is to replace $\omega_1^j \rightarrow E_1^j$. Doing a similar calculation gives

$$\Delta \mathcal{H}_{URG} = \frac{1}{2} \sum_{ijk} \left[\left(\frac{1}{E_0^i - \omega_1^j} + \frac{1}{E_0^k - \omega_1^j} \right) V_{ij} V_{kj}^\dagger | \phi_0^i \rangle \langle \phi_0^k | - \left(\frac{1}{E_0^i - \omega_1^j} + \frac{1}{E_0^i - \omega_1^k} \right) V_{ij}^\dagger V_{ik} | \phi_1^j \rangle \langle \phi_1^k | \right] \quad (6.26)$$

1.7 A comparison of URG, SWT and PMS on the Anderson model

The SWT for the single-impurity Anderson model is briefly sketched below. In order to decouple a state $q\beta$ from the SIAM ($\epsilon_q > 0$), we take an ansatz $S = (A + B \hat{n}_{d\bar{\beta}})(c_{q\beta}^\dagger c_{d\beta} - \text{h.c.})$. Plugging this into the decoupling condition gives

$$-\epsilon_q (A + B \hat{n}_{d\bar{\beta}}) + \epsilon_d (A + B \hat{n}_{d\bar{\beta}}) + U (A + B) \hat{n}_{d\bar{\beta}} = -V \quad (7.1)$$

which gives

$$S = V \left[\frac{1 - \hat{n}_{d\bar{\beta}}}{\epsilon_q - \epsilon_d} + \frac{\hat{n}_{d\bar{\beta}}}{\epsilon_q - \epsilon_d - U} \right] (c_{q\beta}^\dagger c_{d\beta} - \text{h.c.}) \quad (7.2)$$

The remaining diagonal part constitutes the effective Hamiltonian.

$$\begin{aligned}
 U\mathcal{H}U^\dagger &= H_1 = \mathcal{H}_0 + \frac{1}{2} \{ \mathcal{H}_0, S^2 \} - S\mathcal{H}_0S + [S, \mathcal{H}_X] \\
 &= \mathcal{H}_0 + \frac{1}{2} [[\mathcal{H}_0, S], S] + [S, \mathcal{H}_X] \\
 &= \mathcal{H}_0 + \frac{1}{2} [\mathcal{H}_X, S] + [S, \mathcal{H}_X] \\
 &= \mathcal{H}_0 + \frac{1}{2} [S, \mathcal{H}_X]
 \end{aligned} \tag{7.3}$$

For the SIAM (and noting that we are decoupling $q\beta$), the two parts are

$$\begin{aligned}
 \mathcal{H}_0 &= \sum_{k\sigma} \epsilon_k \hat{n}_{k\sigma} + \epsilon_d \hat{n}_d + U \hat{n}_{d\uparrow} \hat{n}_{d\downarrow} + \sum_{k\sigma \neq q\beta} \left(c_{k\sigma}^\dagger c_{d\sigma} + \text{h.c.} \right) \\
 \mathcal{H}_X &= c_{q\beta}^\dagger c_{d\beta} + \text{h.c.}
 \end{aligned} \tag{7.4}$$

The renormalization in the effective Hamiltonian from decoupling a high energy particle state is thus

$$\begin{aligned}
 \frac{1}{2} [S, \mathcal{H}_X] \Big|_{\hat{n}_{q\beta}=0} &= |V|^2 \left[\frac{1 - \hat{n}_{d\bar{\beta}}}{\epsilon_q - \epsilon_d} + \frac{\hat{n}_{d\bar{\beta}}}{\epsilon_q - \epsilon_d - U} \right] \left[\hat{n}_{q\beta} (1 - \hat{n}_{d\beta}) - \hat{n}_{d\beta} (1 - \hat{n}_{q\beta}) \right] \Big|_{\hat{n}_{q\beta}=0} \\
 &= -\hat{n}_{d\beta} |V|^2 \left[\frac{1 - \hat{n}_{d\bar{\beta}}}{\epsilon_q - \epsilon_d} + \frac{\hat{n}_{d\bar{\beta}}}{\epsilon_q - \epsilon_d - U} \right]
 \end{aligned} \tag{7.5}$$

In the last step, we put $\hat{n}_{q\beta} = 0$ because previously we assumed $\epsilon_q > 0$ and high energy virtual excitations above the Fermi surface must necessarily be vacant in the initial state (at $T = 0$). We can obtain the renormalization from decoupling a high energy *hole* state directly from this expression, just by choosing $\hat{n}_{q\beta} = 1$ and setting $\epsilon_q \rightarrow -\epsilon_q$.

$$\frac{1}{2} [S, \mathcal{H}_X] \Big|_{\hat{n}_{q\beta}=1} = -(1 - \hat{n}_{d\beta}) |V|^2 \left[\frac{1 - \hat{n}_{d\bar{\beta}}}{\epsilon_q + \epsilon_d} + \frac{\hat{n}_{d\bar{\beta}}}{\epsilon_q + \epsilon_d + U} \right] \tag{7.6}$$

These two results - the renormalization in the particle and hole sectors - is identical to the result obtained from PMS of the SIAM [7, 10]. The renormalizations in the various energy levels of the impurity can be read off now, after summing over all states in the interval we are decoupling.

$$\begin{aligned}
 \Delta E_2 &= -2 \sum_q \frac{|V_q|^2}{\epsilon_q - \epsilon_d - U} \\
 \Delta E_1 &= - \sum_q \frac{|V_q|^2}{\epsilon_q - \epsilon_d} - \sum_q \frac{|V_q|^2}{\epsilon_q + \epsilon_d + U} \\
 \Delta E_0 &= -2 \sum_q \frac{|V_q|^2}{\epsilon_q + \epsilon_d}
 \end{aligned} \tag{7.7}$$

This can be compared with the URG result, eq. ??,

$$\begin{aligned}
 \Delta E_2 &= 2 \sum_q \frac{|V_q|^2}{\omega - \frac{1}{2}\epsilon_q + \epsilon_d + U} \\
 \Delta E_1 &= \sum_q \frac{|V_q|^2}{\omega - \frac{1}{2}\epsilon_q - \epsilon_d - U} + \sum_q \frac{|V_q|^2}{\omega - \frac{1}{2}\epsilon_q + \epsilon_d} \\
 \Delta E_0 &= 2 \sum_q \frac{|V_q|^2}{\omega - \frac{1}{2}\epsilon_q - \epsilon_d}
 \end{aligned} \tag{7.8}$$

We can transform the URG result to the SWT result if we ignore the effect of the quantum fluctuations in ω (arising from the presence of the off-diagonal term $\mathcal{H}^i$) and replace it with the renormalised diagonal value of $-\frac{1}{2}\epsilon_q$. This means that SWT tracks the effect of the off-diagonal terms only in the numerator. Of course, all this assumes we are doing an iterative SWT instead of a one-shot SWT; the latter is the conventional way. A second difference is that URG has a Green's function like structure in the renormalization such that a fixed point is reached when the diagonal part $\mathcal{H}^d$ matches one of the eigenvalues of ω (see 1.1). SWT does not have such a fixed point structure.

Another point to note is that decoupling a single electron does not generate all the charge-charge or spin-spin interactions that come out when one performs a one-shot SWT. This implies that such terms are a result of decoupling the non-local interactions of the impurity (it is talking to all the mobile electrons), and cannot be generated when we remove just the local interactions of the mobile electrons. Instead, if one performs a URG in which we non-perturbatively kill the 2-point vertices in the SIAM, such 4-point vertices are generated. This is shown in the next subsection.

1.8 Deriving the Kondo model from the Anderson model via a one-shot URG

Here we will show how we can obtain the spin-spin interaction of the Kondo model by performing a one-shot URG on the SIAM. This should justify that the action of performing an SWT is analogous to decoupling the whole band via URG. There are three departures from the conventional way of doing URG (or PMS).

- We will be severing the connections of the impurity with all the mobile electrons in one-shot, and not iteratively.
- We will have to trivialize the quantum fluctuation operator $\hat{\omega}$ by replacing it with the diagonal part of the initial state energy.

Since we are decoupling the whole band, the off-diagonal part that we want to remove is

$$\mathcal{H}^I = \sum_{k\sigma} \left[V_k c_{k\sigma}^\dagger c_{d\sigma} + \text{h.c.} \right] \tag{8.1}$$

The diagonal part is the rest of the Hamiltonian.

$$\mathcal{H}^d = \sum_{k\sigma} \epsilon_k \hat{n}_{k\sigma} + \epsilon_d \hat{n}_d + U \hat{n}_{d\uparrow} \hat{n}_{d\downarrow} = \sum_{k\sigma} \epsilon_k \tau_{k\sigma} + \epsilon_d \hat{n}_d + U \hat{n}_{d\uparrow} \hat{n}_{d\downarrow} \quad (8.2)$$

Following eq. 1.47, the renormalization is

$$\Delta\mathcal{H} = \frac{1}{2} [\eta^\dagger - \eta, \mathcal{H}_X] \quad (8.3)$$

The transition operator η is

$$\begin{aligned} \eta &= \frac{1}{\omega - \mathcal{H}^d} \sum_{k\sigma} V_k^* c_{d\sigma}^\dagger c_{k\sigma} = \sum_{k\sigma} \frac{1}{\omega + \frac{1}{2}\epsilon_k - \epsilon_d - (\epsilon_d + U) \hat{n}_{d\bar{\sigma}}} V_k^* c_{d\sigma}^\dagger c_{k\sigma} \\ &= \sum_{k\sigma} \left[\frac{\hat{n}_{d\bar{\sigma}}}{\omega_1 + \frac{1}{2}\epsilon_k - 2\epsilon_d - U} + \frac{1 - \hat{n}_{d\bar{\sigma}}}{\omega_0 + \frac{1}{2}\epsilon_k - \epsilon_d} \right] V_k^* c_{d\sigma}^\dagger c_{k\sigma} = \sum_{k\sigma} \left[\frac{\hat{n}_{d\bar{\sigma}}}{E_k^1} + \frac{1 - \hat{n}_{d\bar{\sigma}}}{E_k^0} \right] V_k^* c_{d\sigma}^\dagger c_{k\sigma} \end{aligned} \quad (8.4)$$

where $E_k^1 = \omega_1 + \frac{1}{2}\epsilon_k - 2\epsilon_d - U$ and $E_k^0 = \omega_0 + \frac{1}{2}\epsilon_k - \epsilon_d$. The total generator is therefore

$$\eta^\dagger - \eta = \sum_{k\sigma} \left[\frac{\hat{n}_{d\bar{\sigma}}}{E_k^1} + \frac{1 - \hat{n}_{d\bar{\sigma}}}{E_k^0} \right] \left(V_k c_{k\sigma}^\dagger c_{d\sigma} - V_k^* c_{d\sigma}^\dagger c_{k\sigma} \right) \quad (8.5)$$

The renormalization is

$$\Delta\mathcal{H}(\omega_1, \omega_0) = \frac{1}{2} \sum_{kq\sigma\alpha} \left[\left(\frac{\hat{n}_{d\bar{\sigma}}}{E_k^1} + \frac{1 - \hat{n}_{d\bar{\sigma}}}{E_k^0} \right) \left(V_k c_{k\sigma}^\dagger c_{d\sigma} - V_k^* c_{d\sigma}^\dagger c_{k\sigma} \right), V_q c_{q\alpha}^\dagger c_{d\alpha} + V_q^* c_{d\alpha}^\dagger c_{q\alpha} \right] \quad (8.6)$$

The summation has two parts, $\Delta_{1,2}$ - one where $\sigma = \alpha$ and another where $\sigma = \bar{\alpha}$. The first part Δ_1 gives

$$\begin{aligned}
 \Delta_1 &= \frac{1}{2} \sum_{kq\sigma=\alpha} \left[\left(\frac{\hat{n}_{d\bar{\sigma}}}{E_k^1} + \frac{1 - \hat{n}_{d\bar{\sigma}}}{E_k^0} \right) \left(V_k c_{k\sigma}^\dagger c_{d\sigma} - V_k^* c_{d\sigma}^\dagger c_{k\sigma} \right), V_q c_{q\sigma}^\dagger c_{d\sigma} + V_q^* c_{d\sigma}^\dagger c_{q\sigma} \right] \\
 &= \frac{1}{2} \sum_{kq\sigma} \left(\frac{\hat{n}_{d\bar{\sigma}}}{E_k^1} + \frac{1 - \hat{n}_{d\bar{\sigma}}}{E_k^0} \right) \left[V_k c_{k\sigma}^\dagger c_{d\sigma} - V_k^* c_{d\sigma}^\dagger c_{k\sigma}, V_q c_{q\sigma}^\dagger c_{d\sigma} + V_q^* c_{d\sigma}^\dagger c_{q\sigma} \right] \\
 &= \frac{1}{2} \sum_{kq\sigma} \left(\frac{\hat{n}_{d\bar{\sigma}}}{E_k^1} + \frac{1 - \hat{n}_{d\bar{\sigma}}}{E_k^0} \right) \left\{ V_k V_q \left[c_{k\sigma}^\dagger c_{d\sigma}, V_q^* c_{d\sigma}^\dagger c_{q\sigma} \right] - V_k^* V_q \left[c_{d\sigma}^\dagger c_{k\sigma}, c_{q\sigma}^\dagger c_{d\sigma} \right] \right\} \\
 &= \frac{1}{2} \sum_{kq\sigma} \left(\frac{\hat{n}_{d\bar{\sigma}}}{E_k^1} + \frac{1 - \hat{n}_{d\bar{\sigma}}}{E_k^0} \right) \left\{ V_k V_q \left[c_{k\sigma}^\dagger c_{d\sigma}, V_q^* c_{d\sigma}^\dagger c_{q\sigma} \right] + V_k^* V_q \left[c_{q\sigma}^\dagger c_{d\sigma}, c_{d\sigma}^\dagger c_{k\sigma} \right] \right\} \\
 &= \frac{1}{2} \sum_{kq\sigma} \left[\hat{n}_{d\bar{\sigma}} \left(\frac{1}{E_k^1} + \frac{1}{E_q^1} \right) + (1 - \hat{n}_{d\bar{\sigma}}) \left(\frac{1}{E_k^0} + \frac{1}{E_q^0} \right) \right] V_k V_q^* \left[c_{k\sigma}^\dagger c_{d\sigma}, c_{d\sigma}^\dagger c_{q\sigma} \right] \\
 &= \sum_{kq\sigma} \left[\frac{1}{2} V_k V_q^* \left(\frac{1}{E_k^0} + \frac{1}{E_q^0} \right) + \hat{n}_{d\bar{\sigma}} \frac{1}{2} V_k V_q^* \left(\frac{1}{E_k^1} + \frac{1}{E_q^1} - \frac{1}{E_k^0} - \frac{1}{E_q^0} \right) \right] \left(c_{k\sigma}^\dagger c_{q\sigma} - c_{d\sigma}^\dagger c_{d\sigma} \delta_{kq} \right)
 \end{aligned} \tag{8.7}$$

We can now define two new energy scales:

$$W_{kq} = \frac{1}{2} V_k V_q^* \left(\frac{1}{E_k^0} + \frac{1}{E_q^0} \right), \quad J_{kq} = \frac{1}{2} V_k V_q^* \left(\frac{1}{E_k^1} + \frac{1}{E_q^1} - \frac{1}{E_k^0} - \frac{1}{E_q^0} \right) \tag{8.8}$$

The renormalization Δ_1 becomes

$$\begin{aligned}
 \Delta_1 &= \sum_{kq\sigma} [W_{kq} + \hat{n}_{d\bar{\sigma}} J_{kq}] \left(c_{k\sigma}^\dagger c_{q\sigma} - c_{d\sigma}^\dagger c_{d\sigma} \delta_{kq} \right) \\
 &= \sum_{kq\sigma} [W_{kq} + \hat{n}_{d\bar{\sigma}} J_{kq}] c_{k\sigma}^\dagger c_{q\sigma} - \sum_{k\sigma} [W_{kk} + \hat{n}_{d\bar{\sigma}} J_{kk}] \hat{n}_{d\sigma} \\
 &= \sum_{kq\sigma} \left[W_{kq} + \frac{1}{2} \hat{n}_d J_{kq} \right] c_{k\sigma}^\dagger c_{q\sigma} - \sum_{kq\sigma} \sigma J_{kq} S_d^z c_{k\sigma}^\dagger c_{q\sigma} - \sum_{k\sigma} [W_{kk} + \hat{n}_{d\bar{\sigma}} J_{kk}] \hat{n}_{d\sigma}
 \end{aligned} \tag{8.9}$$

There we exchanged $\hat{n}_{d\bar{\sigma}}$ for S_d^z and $\hat{n}_d$, in the first term, by using the definitions $\hat{n}_{d\sigma} + \hat{n}_{d\bar{\sigma}} = \hat{n}_{d\sigma}$ and $\hat{n}_{d\sigma} - \hat{n}_{d\bar{\sigma}} = 2\sigma S_d^z$.

The second term in the summation comes from the choice $\sigma = \bar{\alpha}$.

$$\begin{aligned}
 \Delta_2 &= \frac{1}{2} \sum_{kq\bar{\sigma}=\alpha} \left[\left(\frac{\hat{n}_{d\bar{\sigma}}}{E_k^1} + \frac{1 - \hat{n}_{d\bar{\sigma}}}{E_k^0} \right) \left(V_k c_{k\sigma}^\dagger c_{d\sigma} - V_k^* c_{d\sigma}^\dagger c_{k\sigma} \right), V_q c_{q\bar{\sigma}}^\dagger c_{d\bar{\sigma}} + V_q^* c_{d\bar{\sigma}}^\dagger c_{q\bar{\sigma}} \right] \\
 &= \frac{1}{2} \sum_{kq\sigma} \left(V_k c_{k\sigma}^\dagger c_{d\sigma} - V_k^* c_{d\sigma}^\dagger c_{k\sigma} \right) \left[\frac{\hat{n}_{d\bar{\sigma}}}{E_k^1} + \frac{1 - \hat{n}_{d\bar{\sigma}}}{E_k^0}, V_q c_{q\bar{\sigma}}^\dagger c_{d\bar{\sigma}} + V_q^* c_{d\bar{\sigma}}^\dagger c_{q\bar{\sigma}} \right] \\
 &= \frac{1}{2} \sum_{kq\sigma} \left(V_k c_{k\sigma}^\dagger c_{d\sigma} - V_k^* c_{d\sigma}^\dagger c_{k\sigma} \right) \left(V_q^* c_{d\bar{\sigma}}^\dagger c_{q\bar{\sigma}} - V_q c_{q\bar{\sigma}}^\dagger c_{d\bar{\sigma}} \right) \left(\frac{1}{E_k^1} - \frac{1}{E_k^0} \right) \\
 &= \frac{1}{2} \sum_{kq\sigma} \left(V_k V_q^* c_{k\sigma}^\dagger c_{d\sigma} c_{d\bar{\sigma}}^\dagger c_{q\bar{\sigma}} - V_k V_q c_{k\sigma}^\dagger c_{d\sigma} c_{q\bar{\sigma}}^\dagger c_{d\bar{\sigma}} - V_k^* V_q^* c_{d\sigma}^\dagger c_{k\sigma} c_{d\bar{\sigma}}^\dagger c_{q\bar{\sigma}} + V_k^* V_q c_{d\sigma}^\dagger c_{k\sigma} c_{q\bar{\sigma}}^\dagger c_{d\bar{\sigma}} \right) \\
 &\quad \times \left(\frac{1}{E_k^1} - \frac{1}{E_k^0} \right)
 \end{aligned} \tag{8.10}$$

We now use the following trick to combine the first and fourth terms:

$$\begin{aligned}
 &\frac{1}{2} \sum_{kq\sigma} \left(V_k V_q^* c_{k\sigma}^\dagger c_{d\sigma} c_{d\bar{\sigma}}^\dagger c_{q\bar{\sigma}} + V_k^* V_q c_{d\sigma}^\dagger c_{k\sigma} c_{q\bar{\sigma}}^\dagger c_{d\bar{\sigma}} \right) \times \left(\frac{1}{E_k^1} - \frac{1}{E_k^0} \right) \\
 &= \frac{1}{2} \sum_{kq\sigma} V_k V_q^* c_{k\sigma}^\dagger c_{d\sigma} c_{d\bar{\sigma}}^\dagger c_{q\bar{\sigma}} \left(\frac{1}{E_k^1} - \frac{1}{E_k^0} \right) + \frac{1}{2} \sum_{kq\sigma} V_k^* V_q c_{d\sigma}^\dagger c_{k\sigma} c_{q\bar{\sigma}}^\dagger c_{d\bar{\sigma}} \left(\frac{1}{E_k^1} - \frac{1}{E_k^0} \right) \\
 &= \frac{1}{2} \sum_{kq\sigma} V_k V_q^* c_{k\sigma}^\dagger c_{d\sigma} c_{d\bar{\sigma}}^\dagger c_{q\bar{\sigma}} \left(\frac{1}{E_k^1} - \frac{1}{E_k^0} \right) + \frac{1}{2} \sum_{qk\sigma} V_q^* V_k c_{d\bar{\sigma}}^\dagger c_{q\bar{\sigma}} c_{k\sigma}^\dagger c_{d\sigma} \left(\frac{1}{E_q^1} - \frac{1}{E_q^0} \right) \\
 &= - \sum_{kq\sigma} J_{kq} c_{k\sigma}^\dagger c_{q\bar{\sigma}} c_{d\bar{\sigma}}^\dagger c_{d\sigma}
 \end{aligned} \tag{8.11}$$

In the penultimate step, we interchanged the dummy indices k and q and changed $\sigma \leftrightarrow \bar{\sigma}$ in the second term.

Similarly, for the second term, we get

$$\begin{aligned}
 & \frac{1}{2} \sum_{kq\sigma} V_k V_q c_{k\sigma}^\dagger c_{d\sigma} c_{q\bar{\sigma}}^\dagger c_{d\bar{\sigma}} \left(\frac{1}{E_k^1} - \frac{1}{E_k^0} \right) \\
 &= \frac{1}{4} \sum_{kq\sigma} \left[V_k V_q \left(\frac{1}{E_k^1} - \frac{1}{E_k^0} \right) c_{k\sigma}^\dagger c_{d\sigma} c_{q\bar{\sigma}}^\dagger c_{d\bar{\sigma}} + \underbrace{V_k V_q \left(\frac{1}{E_k^1} - \frac{1}{E_k^0} \right) c_{k\bar{\sigma}}^\dagger c_{d\bar{\sigma}} c_{q\sigma}^\dagger c_{d\sigma}}_{\sigma \leftrightarrow \bar{\sigma}} \right] \\
 &= \frac{1}{4} \sum_{kq\sigma} \left[V_k V_q \left(\frac{1}{E_k^1} - \frac{1}{E_k^0} \right) c_{k\sigma}^\dagger c_{d\sigma} c_{q\bar{\sigma}}^\dagger c_{d\bar{\sigma}} + \underbrace{V_q V_k \left(\frac{1}{E_q^1} - \frac{1}{E_q^0} \right) c_{q\bar{\sigma}}^\dagger c_{d\bar{\sigma}} c_{k\sigma}^\dagger c_{d\sigma}}_{k \leftrightarrow q} \right] \\
 &= \sum_{kq\sigma} V_k V_q \frac{1}{4} \left(\frac{1}{E_k^1} - \frac{1}{E_k^0} + \frac{1}{E_q^1} - \frac{1}{E_q^0} \right) c_{k\sigma}^\dagger c_{d\sigma} c_{q\bar{\sigma}}^\dagger c_{d\bar{\sigma}} \\
 &= \frac{1}{2} \sum_{kq\sigma} K_{kq} c_{k\sigma}^\dagger c_{d\sigma} c_{q\bar{\sigma}}^\dagger c_{d\bar{\sigma}}
 \end{aligned}$$

where K_{kq} is yet another energy scale.

$$K_{kq} = \frac{1}{2} V_k V_q \left(\frac{1}{E_k^1} - \frac{1}{E_k^0} + \frac{1}{E_q^1} - \frac{1}{E_q^0} \right) \quad (8.12)$$

The third term gives

$$\frac{1}{2} \sum_{kq\sigma} V_k^* V_q^* c_{d\sigma}^\dagger c_{k\sigma} c_{d\bar{\sigma}}^\dagger c_{q\bar{\sigma}} \left(\frac{1}{E_k^1} - \frac{1}{E_k^0} \right) = \sum_{kq\sigma} K_{kq} c_{d\sigma}^\dagger c_{k\sigma} c_{d\bar{\sigma}}^\dagger c_{q\bar{\sigma}} \quad (8.13)$$

The total renormalization can thus be written as

$$\begin{aligned}
 \Delta \mathcal{H}(\omega_1, \omega_0) &= - \sum_{k\sigma} [W_{kk} + \hat{n}_{d\bar{\sigma}} J_{kk}] \hat{n}_{d\sigma} && \text{[renormalization in } \epsilon_d, U] \\
 &+ \sum_{kq\sigma} \left[W_{kq} + \frac{1}{2} \hat{n}_d J_{kq} \right] c_{k\sigma}^\dagger c_{q\sigma} && \text{[potential scattering]} \\
 &- \sum_{kq\sigma} J_{kq} \left[S_d^z \sigma c_{k\sigma}^\dagger c_{q\sigma} + \sum_{kq\sigma} J_{kq} c_{k\sigma}^\dagger c_{q\bar{\sigma}} c_{d\bar{\sigma}}^\dagger c_{d\sigma} \right] && \text{[spin Kondo]} \\
 &+ \sum_{kq\sigma} K_{kq} c_{k\sigma}^\dagger c_{d\sigma} c_{q\bar{\sigma}}^\dagger c_{d\bar{\sigma}} + \text{h.c.} && \text{[charge Kondo]}
 \end{aligned} \quad (8.14)$$

Note that this renormalization is in a particular eigendirection of the total quantum fluctuation operator $\hat{\omega}$. In other words, the single perturbative J_{kq} has been replaced with 2^N scales, each with its own value of ω . This is where the complexity has been transferred in going from the second-order SWT to the non-perturbative URG. The new energy scales are thus the non-perturbative variants of the ones generated in SWT.

$$\begin{aligned}
 W_{kq}^{SWT} &= \frac{1}{2} V_k V_q^* \left(\frac{1}{\epsilon_k - \epsilon_d} + \frac{1}{\epsilon_q - \epsilon_d} \right) \\
 J_{kq}^{SWT} &= \frac{1}{2} V_k V_q^* \left(\frac{1}{\epsilon_k - \epsilon_d - U} + \frac{1}{\epsilon_q - \epsilon_d - U} - \frac{1}{\epsilon_k - \epsilon_d} - \frac{1}{\epsilon_q - \epsilon_d} \right) \\
 W_{kq}^{URG}(\omega) &= \frac{1}{2} V_k V_q^* \left(\frac{1}{\omega_0 + \frac{1}{2}\epsilon_k - \epsilon_d} + \frac{1}{\omega_0 + \frac{1}{2}\epsilon_q - \epsilon_d} \right) \\
 J_{kq}^{URG}(\omega) &= \frac{1}{2} V_k V_q^* \left(\frac{1}{\omega_1 + \epsilon_k - \epsilon_d - U} + \frac{1}{\omega_1 + \epsilon_q - \epsilon_d - U} - \frac{1}{\omega_0 + \epsilon_k - \epsilon_d} - \frac{1}{\omega_0 + \epsilon_q - \epsilon_d} \right)
 \end{aligned} \tag{8.15}$$

To recover the SWT scales from the URG ones, we have to substitute each ω_i by the energy of the initial state to which it corresponds. From eq. 8.4, we note that ω_1 refers to the initial state in which $\hat{n}_{k\sigma} = \hat{n}_{d\bar{\sigma}} = 1 - \hat{n}_{d\sigma} = 1$. Therefore, $\omega_1 = \frac{1}{2}\epsilon_k + \epsilon_d$. Similarly, ω_0 refers to the initial state in which $\hat{n}_{k\sigma} = 1 - \hat{n}_{d\bar{\sigma}} = 1 - \hat{n}_{d\sigma} = 1$. Therefore, $\omega_0 = \frac{1}{2}\epsilon_k$. Substituting these into the URG energy scales gives back the SWT scales.

1.9 Continuous unitary transformation RG

Formalism

The following equation generates a family of unitary Hamiltonians.

$$\frac{d\mathcal{H}(l)}{dl} = [\mathcal{H}, \eta(l)] \tag{9.1}$$

To prove the unitarity [11], we construct the unitary operator $U(l)$ that connects the Hamiltonians $\mathcal{H}(l)$ and $\mathcal{H}(l=0)$. Let $\mathcal{H}(l) = U(l)\mathcal{H}(l=0)U^\dagger(l)$, where $U(l)$ is defined by

$$\eta(l) = \frac{dU}{dl}U^\dagger = -U\frac{dU^\dagger}{dl} \quad \left[UU^\dagger = 1 \implies \frac{d(UU^\dagger)}{dl} = 0 \right] \tag{9.2}$$

This will give

$$\begin{aligned}
 \frac{d\mathcal{H}(l)}{dl} &= \frac{dU}{dl}\mathcal{H}(0)U^\dagger(l) + U\mathcal{H}(0)\frac{dU^\dagger}{dl} \\
 &= \frac{dU}{dl}U^\dagger\mathcal{H}(l) + \mathcal{H}(l)U\frac{dU(l)^\dagger}{dl} \\
 &= \eta(l)\mathcal{H}(l) - \mathcal{H}(l)\eta(l) \\
 &= [\eta(l), \mathcal{H}(l)]
 \end{aligned} \tag{9.3}$$

This proves that the family of Hamiltonians $\mathcal{H}(l)$ satisfy the flow equation eq. 9.1. $\eta(l)$ is referred to as the generator of the flow equation. It is chosen so as to reduce the off-diagonal part of the Hamiltonian, either progressively or in one shot. In Schrieffer-Wolff transformation, the transformation is one-shot, and the η there is the S that sits on top of the exponential in the unitary transformation. In URG, the generator to decouple one electron $q\beta$ is $\eta_{q\beta}^\dagger - \eta_{q\beta}$.

In Continuous unitary transformation (CUT) RG [12], we progressively block-diagonalize the Hamiltonian by removing off-diagonal terms that are farthest from the diagonal, through infinitesimal unitary transformations. The change is described as a flow against the parameter l . The canonical choice of the generator is $\eta(l) = [\mathcal{H}_d, \mathcal{H}_X]$, where $\mathcal{H}_d$ is the diagonal part of the Hamiltonian and $\mathcal{H}_X = \mathcal{H} - \mathcal{H}_d$ is the off-diagonal part of the Hamiltonian. Therefore,

$$\frac{d\mathcal{H}}{dl} = [[\mathcal{H}_d(l), \mathcal{H}_X(l)], \mathcal{H}(l)] \quad (9.4)$$

To see how this choice of the generator results in a decay of the off-diagonal terms, we can consider a simple 2-particle Hamiltonian:

$$\mathcal{H} = \sum_i \epsilon_i \hat{n}_i + \sum_{i \neq j} g_{ij} c_i^\dagger c_j \quad (9.5)$$

where $g_{ij}^* = g_{ji}$. The canonical generator then turns out to be

$$\eta = \left[\sum_i \epsilon_i \hat{n}_i, \sum_{j \neq k} g_{jk} c_j^\dagger c_k \right] = \sum_{k \neq i} \epsilon_i \left[g_{ik} c_i^\dagger c_k - g_{ki} c_k^\dagger c_i \right] = \sum_{k \neq i} g_{ik} c_i^\dagger c_k (\epsilon_i - \epsilon_k) \quad (9.6)$$

and the renormalization in the Hamiltonian is

$$\frac{d\mathcal{H}}{dl} = [\eta, \mathcal{H}] = \left[\sum_{k \neq i} g_{ik} c_i^\dagger c_k (\epsilon_i - \epsilon_k), \sum_i \epsilon_i \hat{n}_i + \sum_{i \neq j} g_{ij} c_i^\dagger c_j \right] \quad (9.7)$$

The first commutator gives

$$- \sum_{i \neq k} g_{ik} (\epsilon_i - \epsilon_k)^2 c_i^\dagger c_k \quad (9.8)$$

The second commutator gives

$$\sum_{\substack{k \neq i \\ j}} \left[g_{kj} g_{ik} (\epsilon_i - \epsilon_k) c_i^\dagger c_j + g_{ji} g_{ik} (\epsilon_k - \epsilon_i) c_j^\dagger c_k \right] = \sum_{\substack{k \neq i \\ j}} g_{ik} g_{kj} (\epsilon_i + \epsilon_j - 2\epsilon_k) c_i^\dagger c_j \quad (9.9)$$

The total renormalization is

$$\frac{d\mathcal{H}}{dl} = - \sum_{i \neq j} g_{ij} (\epsilon_i - \epsilon_j)^2 c_i^\dagger c_j + \sum_{\substack{k \neq i \\ j}} g_{ik} g_{kj} (\epsilon_i + \epsilon_j - 2\epsilon_k) c_i^\dagger c_j \quad (9.10)$$

The couplings renormalize as

$$\begin{aligned}\frac{d\epsilon_i}{dl} &= \sum_{k \neq i} 2|g_{ik}|^2 (\epsilon_i - \epsilon_k) \\ \frac{dg_{ij}}{dl} &= -g_{ij} (\epsilon_i - \epsilon_j)^2 + \sum_{k \neq i} g_{ik}g_{kj} (\epsilon_i + \epsilon_j - 2\epsilon_k)\end{aligned}\tag{9.11}$$

To see the decay of the off-diagonal terms, first we will relate the off-diagonal flow to the diagonal flow using the invariance of the trace under a unitary transformation:

$$0 = \frac{d\text{Tr}(\mathcal{H})^2}{dl} = \frac{d\text{Tr}(\mathcal{H})^2}{dl} = \sum_i \frac{d\epsilon_i^2}{dl} + \sum_{i \neq j} \frac{d|g_{ij}|^2}{dl} \implies \sum_{i \neq j} \frac{d|g_{ij}|^2}{dl} = - \sum_i \frac{d\epsilon_i^2}{dl}\tag{9.12}$$

From the flow equation, we can see that

$$\sum_i \frac{d\epsilon_i^2}{dl} = 2 \sum_{i \neq k} \epsilon_i \frac{d\epsilon_i}{dl} = 2 \sum_i |g_{ik}|^2 (\epsilon_i - \epsilon_k)^2 \geq 0\tag{9.13}$$

Therefore,

$$\sum_{i \neq j} \frac{d|g_{ij}|^2}{dl} \leq 0\tag{9.14}$$

This implies that at $l \rightarrow \infty$, the only off-diagonal terms that survive are those with g_{ij} that scatter between degenerate states, that is, those with $\epsilon_i - \epsilon_j = 0$.

Deriving CUT RG from URG

We will now see that the renormalization in URG can also be put into a similar form. From eq. 1.47, we can write the URG renormalization in the diagonal part as

$$\Delta\mathcal{H}^0 = \frac{1}{2} [\eta^\dagger - \eta, \mathcal{H}]\tag{9.15}$$

where $\mathcal{H}^0 = \mathcal{H}^d + \mathcal{H}^i$. The URG generator can be recast (starting from the definitions of η , eqs. 1.14) as

$$\begin{aligned}\eta^\dagger - \eta &= G_1 c^\dagger T - G_0 T^\dagger c \\ &= \frac{1}{\omega_1 - \omega_0} [G_1 (\omega_1 - \omega_0) c^\dagger T - G_0 (\omega_1 - \omega_0) T^\dagger c] \\ &= \frac{1}{\omega_1 - \omega_0} [G_1 \omega_1 c^\dagger T - c^\dagger T \omega_0 G_0 - T^\dagger c \omega_1 G_1 + G_0 \omega_0 T^\dagger c]\end{aligned}\tag{9.16}$$

In the last step, we changed the second and fourth terms using the constraints $G_1 c^\dagger T = c^\dagger T G_0$ and $G_0 T^\dagger c = T^\dagger c G_1$, eq. 1.16. We now add and subtract $G_1 G_1^{-1} c^\dagger T = c^\dagger T$ and $G_0 G_0^{-1} T^\dagger c = T^\dagger c$ for

each term.

$$\begin{aligned}
 \eta^\dagger - \eta &= \frac{1}{\omega_1 - \omega_0} \left[G_1 \left(\omega_1 - G_1^{-1} \right) c^\dagger T + c^\dagger T - c^\dagger T \left(\omega_0 - G_0^{-1} \right) G_0 - c^\dagger T \right. \\
 &\quad \left. - T^\dagger c \left(\omega_1 - G_1^{-1} \right) G_1 - T^\dagger c + G_0 \left(\omega_0 - G_0^{-1} \right) T^\dagger c + T^\dagger c \right] \\
 &= \frac{1}{\omega_1 - \omega_0} \left[G_1 \left(\omega_1 - G_1^{-1} \right) c^\dagger T - c^\dagger T \left(\omega_0 - G_0^{-1} \right) G_0 \right. \\
 &\quad \left. - T^\dagger c \left(\omega_1 - G_1^{-1} \right) G_1 + G_0 \left(\omega_0 - G_0^{-1} \right) T^\dagger c \right]
 \end{aligned} \tag{9.17}$$

In the last step, the extra $c^\dagger T$ and $T^\dagger c$ terms canceled out. In the second and third terms, we can bring the Greens function closer to the operators $c^\dagger T$ and $T^\dagger c$ because $(\omega_j - G_j^{-1}) G_j = G_j (\omega_j - G_j^{-1})$:

$$\begin{aligned}
 \eta^\dagger - \eta &= \frac{1}{\omega_1 - \omega_0} \left[G_1 \left(\omega_1 - G_1^{-1} \right) c^\dagger T - c^\dagger T G_0 \left(\omega_0 - G_0^{-1} \right) \right. \\
 &\quad \left. - T^\dagger c G_1 \left(\omega_1 - G_1^{-1} \right) + G_0 \left(\omega_0 - G_0^{-1} \right) T^\dagger c \right] \\
 &= \frac{1}{\omega_1 - \omega_0} \left[G_1 \left(\omega_1 - G_1^{-1} \right) c^\dagger T - G_1 c^\dagger T \left(\omega_0 - G_0^{-1} \right) \right. \\
 &\quad \left. - G_0 T^\dagger c \left(\omega_1 - G_1^{-1} \right) + G_0 \left(\omega_0 - G_0^{-1} \right) T^\dagger c \right]
 \end{aligned} \tag{9.18}$$

In the last step, we again used the constraint $G_1 c^\dagger T = c^\dagger T G_0$ and its partner. From the definition of the Green's function $G = (\omega - \mathcal{H}^d)^{-1}$, we can write $\omega_j - G_j^{-1} = \mathcal{H}_j^d$. Therefore,

$$\begin{aligned}
 \eta^\dagger - \eta &= \frac{1}{\omega_1 - \omega_0} \left[G_1 \mathcal{H}_1^d c^\dagger T - G_1 c^\dagger T \mathcal{H}_0^d - G_0 T^\dagger c \mathcal{H}_1^d + G_0 \mathcal{H}_0^d T^\dagger c \right] \\
 &= \frac{1}{\omega_1 - \omega_0} \left[G \mathcal{H}^d c^\dagger T - G c^\dagger T \mathcal{H}^d - G T^\dagger c \mathcal{H}^d + G \mathcal{H}^d T^\dagger c \right] \\
 &= \frac{1}{\omega_1 - \omega_0} G \left[\mathcal{H}^d, c^\dagger T + T^\dagger c \right]
 \end{aligned} \tag{9.19}$$

where $\mathcal{H}^d = \mathcal{H}_1^d \hat{n} + \mathcal{H}_0^d (1 - \hat{n})$ and $G = G_1 \hat{n} + G_0 (1 - \hat{n}) = (\hat{\omega} - \mathcal{H}^d)^{-1}$. For URG, the relevant off-diagonal part of the Hamiltonian for the current node is $\mathcal{H}^I = c^\dagger T + T^\dagger c$. Therefore,

$$\eta^\dagger - \eta = \frac{1}{\omega_1 - \omega_0} G \left[\mathcal{H}^d, \mathcal{H}^I \right] = \left[\mathcal{H}^d, \frac{1}{\omega_1 - \omega_0} G \mathcal{H}^I \right] \tag{9.20}$$

The last equality comes about because both G and $\mathcal{H}^d$ are completely diagonal and hence commute. The renormalization in the Hamiltonian under URG, which is a function of both the quantum fluctuation scale ω and the running bandwidth D , can thus be written as

$$\Delta \mathcal{H}(\omega, D) = \left[\left[\mathcal{H}^d, \tilde{\mathcal{H}}^I \right], \mathcal{H} \right] - \mathcal{H}_X \tag{9.21}$$

The most obvious difference with the CUT version is the presence of the off-diagonal piece $-\mathcal{H}_X$. This is a result of the philosophical difference between URG and CUT-RG - while CUT-RG gradually suppresses the off-diagonal matrix elements, URG makes the off-diagonal components in each 2×2 block vanish completely. We can instead look at the renormalization in the diagonal part of the Hamiltonian under URG:

$$\Delta\mathcal{H}^0(\omega, D) = [[\mathcal{H}^d, \tilde{\mathcal{H}}^I], \mathcal{H}] \quad (9.22)$$

where $\tilde{\mathcal{H}}^I = \frac{1}{\omega_1 - \omega_0} G \mathcal{H}^I$. This can be compared to the analogous equation for CUT (eq. 9.4),

$$\Delta\mathcal{H}(\lambda) = \Delta\lambda [[\mathcal{H}_d, \mathcal{H}_X], \mathcal{H}] \quad (9.23)$$

Leaving aside the obvious differences in the philosophies (the presence of ω in URG or the fact that while URG decouples as a flow in the bandwidth, CUT uses a general parameter λ or the algorithmic difference that while URG decouples one specific node, CUT tries to make the Hamiltonian band-diagonal), the major physical difference is the presence of the total Green's function in the URG equation. It must be noted that while CUT employs the entire off-diagonal part in $\mathcal{H}_X$, the $\mathcal{H}^I$ of URG contains only those parts that are off-diagonal with respect to the node being decoupled at this step.

To bring the URG form closer to CUT, we can make some approximations.

$$G = (\hat{\omega} - \mathcal{H}^d)^{-1} \approx -(\mathcal{H}^d)^{-1} \quad (9.24)$$

where we assumed that the quantum fluctuations are small and can be ignored w.r.t the diagonal contribution $\mathcal{H}^d$. This gives

$$\frac{\Delta\mathcal{H}^0(\omega, D)}{[\mathcal{H}_1^d(\omega_0 - \omega_1)]^{-1}} = [[\mathcal{H}^d, \mathcal{H}^I], \mathcal{H}] \quad (9.25)$$

We can thus make the connection,

$$\Delta\lambda \sim [\mathcal{H}_1^d(\omega_0 - \omega_1)]^{-1} \quad (9.26)$$

Note that in going from eq. 9.22 to the simplified form eq. 9.25, we had to drop all quantum fluctuations in the denominator and we lose the fixed point structure in the process. This results in the distinction that while URG can reach a fixed point theory in a finite number of steps, CUT cannot do so.

1.10 Comparison of renormalisation group approaches based on canonical transformations

We have considered three canonical transformations in this section: the Poor Man's scaling (PMS), the Schrieffer-Wolff transformation (SWT) and the continuous unitary transformation renormalization group (CUT-RG). PMS and SWT are more or less identical; they differ in the context in which

they are used. PMS is used when there is an entire spectrum of energies in the model, ranging from a low energy limit to a high energy; it is then employed to decouple the highest energy modes in an iterative fashion. SWT is used when the Hamiltonian can be split into one high and one low energy part, and we need to decouple these two modes. It is clear when seen from this perspective that SWT is like a one-shot PMS; it decouples the UV from the IR in one step, compared to the iterative approach of PMS. However, as has been shown in the previous subsections, both PMS and SWT can be formalized in an identical fashion, so that one can be switched for the other in both contexts.

Now that we have established that PMS and SWT are formally identical, we can relate them to URG. URG is similar in philosophy to PMS in the fact that URG also successively decouples high energy modes from the low energy ones. The difference is that URG takes care of the quantum fluctuations, at least in principle, by introducing a new set of energy scales ω . These ω then parameterise the RG flows of URG, compared to the single RG flow of PMS. Since SWT is formally the same as PMS, there is also a comparison between SWT and URG. Both PMS and SWT trivialize the quantum fluctuation scales of URG by replacing it with the bare energy values.

CUT-RG is philosophically different from the other transformations. Its goal is to gradually reduce the contributions of the off-diagonal terms by making them decay along a certain scale l . Off-diagonal terms that connect states with large energy differences decay faster. In this sense, there is no sequential dropping out of off-diagonal terms; all off-diagonal terms disappear at $l = \infty$. In this sense, it is like a continuous version of SWT. While SWT strives to remove an entire off-diagonal term in one-shot, CUT RG does this gradually by introducing a scale l . This separation of scales is absent in SWT. It does exist in URG and PMS, albeit in a different fashion. There, the separation comes in when we decouple single electron states starting from the Brillouin zone boundary Λ_N and come down to the Fermi surface Λ_0 . Each Λ provides a natural energy scale for separating the high and low energy physics.

If one integrates the continuum generator η over all the scales, one should recover something analogous to the SWT generator. This generator is however necessarily perturbative in the off-diagonal term, since by definition η will only be of first order in $\mathcal{H}_X$. This is in contrast to the non-perturbative generators in URG and, at least in principle, PMS. This non-perturbative form is encapsulated in the presence of a second completely different set of energy scales ω (or E in PMS). This second energy scale is absent in CUT RG because it takes care of at most the second order term.

Another point to note is that since SWT keeps the entire off-diagonal piece in the generator, new terms will almost certainly be generated at every step, and they have to be truncated. This is not the case with URG or PMS, because in those methods, we decouple just a single-electron at each step, and so those electrons become integrals of motion at that step, leading to their removal from the off-diagonal piece, and very often the Hamiltonian retains the same form as the bare model. This makes the philosophy of URG and PMS easier to work with. Tied to this is the fact that CUT RG often takes a certain type of interaction in the generator part, and not the entire off-diagonal piece. Hence, at the limit of the flow parameter going to ∞ , the chosen off-diagonal piece goes to zero but the remaining off-diagonal pieces still remain. As a result, the Hamiltonian is at most block diagonal at this stage. On the other hand, URG progressively decouples single electrons, meaning all scattering terms corresponding to that electron vanish at each step.

One last thing to note is that URG, being unitarily implemented with a well-defined generator,

does not accomodate for spontaneous symmetry breaking (SSB). In order to see SSB, the symmetry-breaking term has to be added to the bare model explicitly; if this term grows under the RG, then the ground state will be symmetry-broken. CUT RG, however, does allow for SSB through the idea that the generator is not uniquely defined. If the generator commutes with a particular symmetry, then the family of Hamiltonians will also have the symmetry [11]. However, if the generator is replaced by something that is normal ordered w.r.t a particular expectation value, then the CUT RG flow will usually be towards either the symmetry-preserved state or the symmetry-broken state, depending on whichever is stable [13].

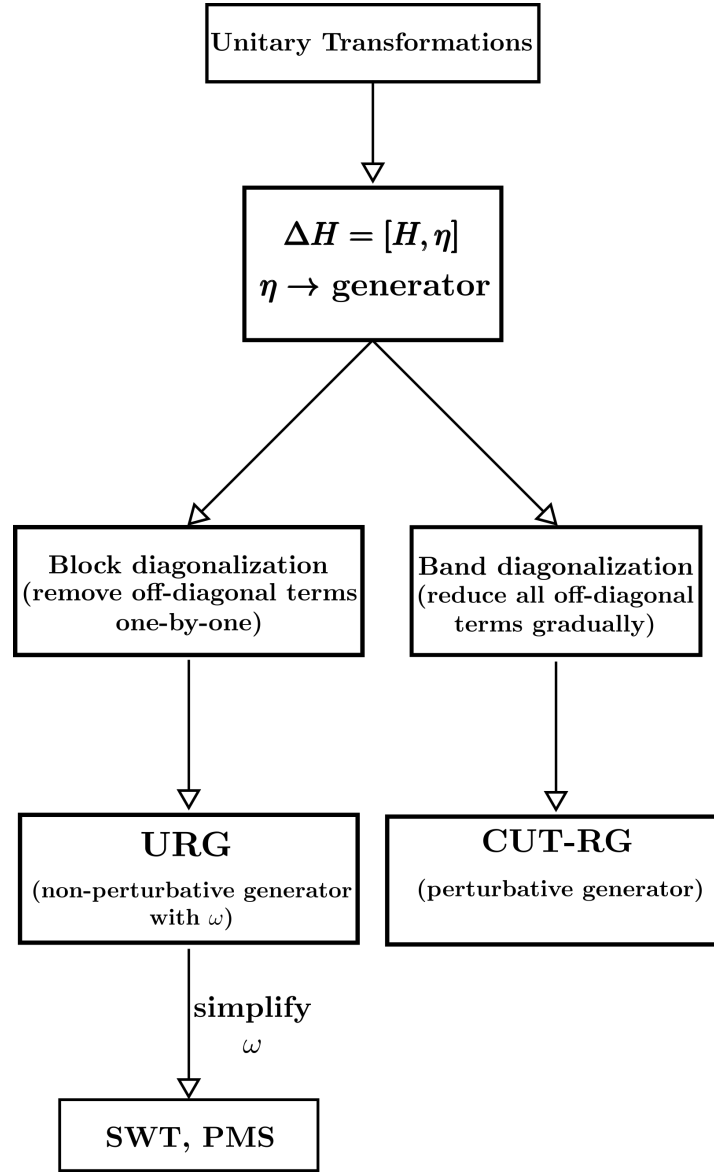


Figure 1.11: Comparison of the various unitary transformations and their relationships to each other.

1.11 Iterative Diagonalisation of Quantum Impurity Models

The broad idea is the following. We have a Hamiltonian describing an interacting set of states (in real or momentum space). We can express the total Hamiltonian in an incremental fashion: $H = \sum_i H_i$, where H_{i+1} involves more number of states than H_i . For example, a tight-binding model can be written in that form, with the definition $H_i = c_{i+1}^\dagger c_i + \text{h.c.}$. The iterative diagonalisation method obtains the low-energy spectrum of this problem in the following manner: We first diagonalise the Hamiltonian H for a smaller value of i , small enough such that this can be done exactly. We then truncate the spectrum to a predefined size, and rotate all existing operators to this truncated basis, including the Hamiltonian H . We then consider the "bonding Hamiltonian" ΔH between the existing sites and the new sites, and rotate the same into the truncated basis. Adding the previous rotated Hamiltonian H and the rotated increment Hamiltonian ΔH gives us a truncated but effective Hamiltonian for the increased number of sites. We again diagonalise this, and again retain only a fixed number of states in the spectrum. We keep repeating this until we reach the required number of sites.

Jordan-Wigner Matrix Representation of Fermionic Operators

For a single qubit, the creation/annihilation matrices are

$$c = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, c^\dagger = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad (11.1)$$

given the convention that $|1\rangle = (0 \ 1)$ is the occupied state. For a many-body system, these must be replaced with field operators that have the canonical fermionic algebra. For a system of N 1-particle levels, this can be accomplished through a *Jordan-Wigner-like* transformation

$$c_j = \left(\otimes_1^j \sigma_z \right) \otimes c \otimes \left(\otimes_1^{N-j} \mathbb{I} \right), j \in [0, N-1], \quad (11.2)$$

where $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ and $\mathbb{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. For example, for spinless electrons on a three-site lattice, we can have three fermionic operators: c_1, c_2 , and c_3 . Following the above expression, their matrices are

$$c_1 = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, c_2 = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, c_3 = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix},$$

For later use, we define the *antisymmetriser matrix* $S(j)$ and identity matrix $\mathbb{I}(j)$,

$$\begin{aligned} S(j) &= \otimes_1^j \sigma_z = \sigma_z \otimes \dots j \text{ times } \dots \otimes \sigma_z, \\ \mathbb{I}(j) &= \otimes_1^{N-j} \mathbb{I} = \mathbb{I} \otimes \dots j \text{ times } \dots \otimes \mathbb{I}, \end{aligned} \quad (11.3)$$

to express the fermionic representation compactly:

$$c_j = S(j) \otimes c \otimes \mathbb{I}(N - j) . \quad (11.4)$$

Structure of Hamiltonian

We consider a general impurity model, of L number of sites (excluding the impurity site):

$$H_{\text{sys}}(L) = H_{\text{imp}} + H_{\text{imp-bath}} + -t \sum_{\sigma, j=1}^{L-1} \left(c_{j,\sigma}^\dagger c_{j+1,\sigma} + \text{h.c.} \right) , \quad (11.5)$$

where H_{imp} is the Hamiltonian for the decoupled impurity site and $H_{\text{imp-bath}}$ is the impurity-bath coupling. Such a system has $2L + 2$ single-particle levels in the full system (2 spin levels for each of the $L + 1$ sites).

Iteration scheme (the initial number of states is L_0):

$$\begin{aligned} H_0(L_0) &= H_{\text{sys}}(L_0) , \\ H_{r+1}(L_0) &= H_r(L_0) + \Delta H_r(L_0), \quad r \geq 0 , \\ \Delta H_r(L_0) &= -t \sum_{\sigma} \left(c_{L_0+r,\sigma}^\dagger c_{L_0+r+1,\sigma} + \text{h.c.} \right) . \end{aligned} \quad (11.6)$$

H_0 is the initial Hamiltonian, consisting of L_0 lattice sites in the bath, $H_{r+1}(L_0)$ is the Hamiltonian after $r + 1$ iterations having started with L_0 sites, and $\Delta H_r(L_0)$ is the increment term that gets added to $H_r(L_0)$ during the $(r + 1)^{\text{th}}$ iteration to give rise to $H_{r+1}(L_0)$.

Iterative Diagonalization

Let the starting Hamiltonian be $H_0(L_0)$, consisting of L_0 single-particle levels (hence a Hilbert space dimension of 2^{L_0}). We start with a single-particle computational basis and construct the L_0 fermionic operators $c_1, c_2, \dots, c_{L_0}$ in this basis, using eq. 11.4. We also keep track of a large antisymmetriser matrix $S(L_0)$ that will be used to attach new sites when we expand the system. Let M_s be the maximum number of eigenstates we retain in the spectrum at any given step. The value of M_s should be chosen so that a $M_s \times M_s$ matrix can be diagonalised in reasonable time.

- S1. Construct the complete Hamiltonian matrix H_0 in our present basis using the field operators $\{c_j\}$. Diagonalise the Hamiltonian (of size $D_0 \times D_0$) and obtain the eigenvalues E_n and eigenstates X_n . Each X_n is a column vector of size D_0 .
- S2. Retain at most M_s number of eigenstates, preferring the ones with lower energy. The reduced basis for this rotated truncated subspace is constructed by stacking the column vectors X_n horizontally:

$$R = [X_1 X_2 \dots X_{M_s}]_{D_0 \times M_s} . \quad (11.7)$$

This matrix also acts as the transformation to rotates and truncate all operators from the old basis into the new one.

- S3. We rotate our Hamiltonian H_0 , our fermionic operators $\{c_j\}$ and the large antisymmetrizer matrix $S(L_0)$ into the new reduced basis, using the transformation $O \rightarrow R^\dagger O R$.
- S4. We now need to expand our system by adding the increment Hamiltonian ΔH_0 . Let the number of new 1-particle levels in ΔH_0 be L_Δ . These new levels will be indexed as $L_0+1, \dots, L_0+L_\Delta$. We need to define antisymmetrized fermionic operators for the new sites:

$$\begin{aligned} c_{L_0+1} &= c \otimes \mathbb{I}(2^{L_\Delta-1}), \\ c_{L_0+2} &= \sigma_z \otimes c \otimes \mathbb{I}(2^{L_\Delta-2}), \\ &\dots \\ c_{L_0+L_\Delta} &= (\otimes_1^{L_\Delta-1} \sigma) \otimes c. \end{aligned} \tag{11.8}$$

These have to be calculated in the local computational basis (of size 2^{L_Δ}) of the new levels.

- S5. Combining the new sites with the old sites leads to a combined Hilbert space dimension of $(M_s + 2^{L_\Delta}) \times (M_s + 2^{L_\Delta})$. To allow all operators to act on the enlarged Hilbert space, we expand both sets of operators:

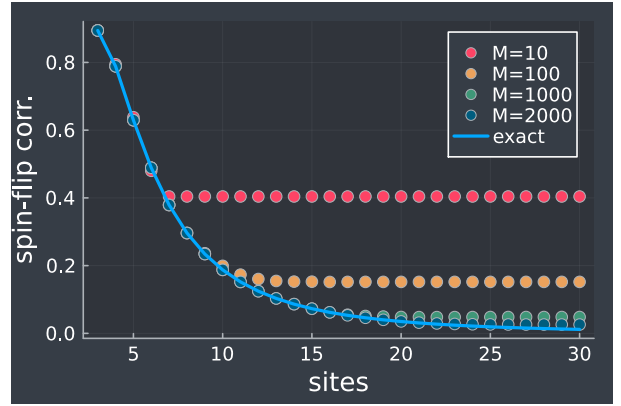
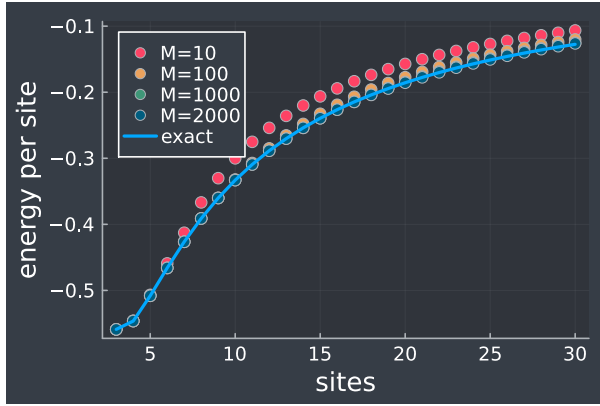
$$\begin{aligned} c_j &= S(L_0) \otimes c_j; \quad j = L_0 + 1, \dots, L_\Delta, \\ c_j &= c_j \otimes \mathbb{I}(2^{L_\Delta}); \quad j = 1, 2, \dots, L_0, \\ H_0 &\rightarrow H_0 \otimes \mathbb{I}(2^{L_\Delta}), \\ S(L_1) &= S(L_0) \otimes \mathbb{I}(2^{L_\Delta}). \end{aligned} \tag{11.9}$$

where $\mathbb{I}(2^{L_\Delta})$ is an identity matrix of dimension $2^{L_\Delta} \times 2^{L_\Delta}$. Note that the operators in the last three equations are the rotated ones (following Step 3).

- S6. Using the transformed operators $c_{L_0+1}, \dots, c_{L_0+L_\Delta}$ for the new sites, construct the difference Hamiltonian matrix ΔH_0 and hence the updated Hamiltonian $H_1 = H_0 + \Delta H_0$ for the next step. Repeat the process starting from step 2 with the new Hamiltonian H_1 , the new operators c_j and the new matrix $S(L_1)$ replacing the old counterparts.

Static correlations

We are in general interested in n -point correlations of the form $\langle O_1 O_2 \dots O_n \rangle$, where O_i are operators that act on 1-particle Hilbert spaces. Let the earliest step of the iterative diagonalisation procedure at which all these operators have entered the system be r . At this step r , construct the correlation operator $O_1 O_2 \dots O_n$ using the fermionic matrices c_j at that step (recall that these matrices will be highly rotated versions of the matrices we started with). Having constructed the operator O , we expand and rotate it after the completion of every future step: $O_{n+1} = (R_n^\dagger O_n R_n) \otimes \mathbb{I}(2^{L_\Delta})$, where R_n is the rotation matrix for the n^{th} step. The last step n^* of the iterative diagonalisation consists of only a diagonalisation and no expansion, resulting in a final set of eigenstates $\{X_i\}$. The form of the correlation operator in this basis is $O_{n^*} = R_{n^*-1}^\dagger O_{n^*-1} R_{n^*}$. The expectation value can now be calculated using the matrix O_{n^*} and the ground state of $\{X_i\}$.



Reduced density matrices

Examples and Benchmarks: Single-Impurity Anderson Model

The single-impurity Anderson model at half-filling is obtained by setting $H_{\text{imp}} = -\frac{U}{2} (n_{d\uparrow} - n_{d\downarrow})^2$ and $H_{\text{imp-bath}} = -V \sum_{\sigma} (c_{d\sigma}^{\dagger} c_{0\sigma} + \text{h.c.})$ in eq. 11.5. We studied this model using the above approach to benchmark the ground state energy and spin-flip correlation $\langle \frac{1}{2} S_d^+ S_0^- + \text{h.c.} \rangle$ against exact diagonalization (ED). In order to extend the ED to a larger number of sites, we restricted ourselves to just the $N = 2$ sector, N being the total occupancy. We find very good agreement for $M_s \sim 1000$ and above. These results are shown in Fig. ??.

1.12 Optimising Fermi Surface Calculations

Rotated basis for simplified representation

At the renormalisation group fixed point, we have a renormalised theory for the interaction of the impurity spin S_f with the f -layer Fermi surface:

$$H^* = \sum_{\alpha, \beta} \mathbf{S}_f \cdot \boldsymbol{\sigma}_{\alpha, \beta} \sum_{\mathbf{k}, \mathbf{q}} J_f^*(\mathbf{k}, \mathbf{q}) f_{\mathbf{k}, \alpha}^{\dagger} f_{\mathbf{q}, \beta} . \quad (12.1)$$

We will now obtain a more minimal representation of this interaction. Each momentum label is summed over all four quadrants Q_1 through Q_4 ; eq. ?? relates Q_1 with Q_3 and Q_2 with Q_4 :

$$\begin{aligned}
 \sum_{\mathbf{k}, \mathbf{q}} J_f^*(\mathbf{k}, \mathbf{q}) f_{\mathbf{k}, \alpha}^\dagger f_{\mathbf{q}, \beta} &= \sum_{\mathbf{q}} \left[\sum_{\mathbf{k} \in Q_1, Q_2} J_f^*(\mathbf{k}, \mathbf{q}) f_{\mathbf{k}, \alpha}^\dagger + \sum_{\mathbf{k} \in Q_3, Q_4} J_f^*(\mathbf{k}, \mathbf{q}) f_{\mathbf{k}, \alpha}^\dagger \right] f_{\mathbf{q}, \beta} \\
 &= \sum_{\mathbf{q}} \left[\sum_{\mathbf{k} \in Q_1, Q_2} J_f^*(\mathbf{k}, \mathbf{q}) f_{\mathbf{k}, \alpha}^\dagger + \sum_{\mathbf{k} \in Q_1} J_f^*(\mathbf{k} - \mathbf{Q}_1, \mathbf{q}) f_{\mathbf{k} - \mathbf{Q}_1, \alpha}^\dagger + \sum_{\mathbf{k} \in Q_2} J_f^*(\mathbf{k} + \mathbf{Q}_2, \mathbf{q}) f_{\mathbf{k} + \mathbf{Q}_2, \alpha}^\dagger \right] f_{\mathbf{q}, \beta} \\
 &= \sum_{\mathbf{q}} \left[\sum_{\mathbf{k} \in Q_1} J_f^*(\mathbf{k}, \mathbf{q}) (f_{\mathbf{k}, \alpha}^\dagger - f_{\mathbf{k} - \mathbf{Q}_1, \alpha}^\dagger) + \sum_{\mathbf{k} \in Q_2} J_f^*(\mathbf{k}, \mathbf{q}) (f_{\mathbf{k}, \alpha}^\dagger - f_{\mathbf{k} + \mathbf{Q}_2, \alpha}^\dagger) \right] f_{\mathbf{q}, \beta} .
 \end{aligned} \tag{12.2}$$

For ease of notation, we define new fermionic operators $A_{\mathbf{k}, \sigma}$:

$$A_{\mathbf{k}, \sigma, \pm} = \begin{cases} \frac{1}{\sqrt{2}} (f_{\mathbf{k}, \sigma} \pm f_{\mathbf{k} - \mathbf{Q}_1, \sigma}) , & \mathbf{k} \in Q_1 , \\ \frac{1}{\sqrt{2}} (f_{\mathbf{k}, \sigma} \pm f_{\mathbf{k} + \mathbf{Q}_2, \sigma}) , & \mathbf{k} \in Q_2 , \end{cases} \tag{12.3}$$

which satisfy the appropriate algebra: $\{A_{\mathbf{k}, \sigma, p}, A_{\mathbf{k}', \sigma', p'}\} = \delta_{\mathbf{k}, \mathbf{k}'} \delta_{\sigma, \sigma'} \delta_{p, p'}$, with $p = \pm$ denoting the flavours of the A fields. Note that only the $p = -1$ flavour enters the Hamiltonian. Henceforth, we drop the label $\pm$ and it is implied that A refers to the $p = -1$ variant.

Decomposing the sum over $\mathbf{q}$ in a similar fashion, we get

$$\sum_{\mathbf{k}, \mathbf{q}} J_f^*(\mathbf{k}, \mathbf{q}) f_{\mathbf{k}, \alpha}^\dagger f_{\mathbf{q}, \beta} = 2 \sum_{\mathbf{k}, \mathbf{q} \in Q_1, Q_2} J_f^*(\mathbf{k}, \mathbf{q}) A_{\mathbf{k}, \alpha}^\dagger A_{\mathbf{q}, \beta} . \tag{12.4}$$

Correlations in truncated representation

In order to calculate equal-time correlations (such as $\langle S_d^+ f_{\mathbf{k}\downarrow}^\dagger f_{\mathbf{k}'\uparrow} \rangle$), we use eq. 12.3 to express the bare fields $f_{\mathbf{k}, \sigma}$ in terms of the new fields $A_{\mathbf{k}, \sigma, \pm}$ and $B_{\mathbf{k}, \sigma, \pm}$:

$$\begin{aligned}
 f_{\mathbf{k}, \sigma} &= \frac{1}{\sqrt{2}} (A_{\mathbf{k}, \sigma, +} + A_{\mathbf{k}, \sigma, -}) , & \mathbf{k} \in Q_1, Q_2 \\
 f_{\mathbf{k} - \mathbf{Q}_1, \sigma} &= \frac{1}{\sqrt{2}} (A_{\mathbf{k}, \sigma, +} - A_{\mathbf{k}, \sigma, -}) , & \mathbf{k} \in Q_1 \\
 f_{\mathbf{k} + \mathbf{Q}_2, \sigma} &= \frac{1}{\sqrt{2}} (A_{\mathbf{k}, \sigma, +} - A_{\mathbf{k}, \sigma, -}) , & \mathbf{k} \in Q_2
 \end{aligned} \tag{12.5}$$

where the four relations act on operators in four quadrants. These relations allow us to express correlations $F_{k_i, k_j} = \langle S_d^+ c_{k_i, \uparrow}^\dagger c_{k_j, \downarrow} \rangle$ for the bare fields f in terms of the rotated fields. Using the fact

that the Hamiltonian does not couple the fields B_+ and A_+ , we have

$$\begin{aligned} F_{k_1,k_1} &= F_{k_3,k_3} = -F_{k_1,k_3} = \frac{1}{2} \langle S_d^+ A_{k_1,\downarrow,-}^\dagger A_{k_1,\uparrow,-} \rangle , \\ F_{k_2,k_2} &= F_{k_4,k_4} = -F_{k_2,k_4} = \frac{1}{2} \langle S_d^+ A_{k_2,\downarrow,-}^\dagger A_{k_2,\uparrow,-} \rangle , \\ F_{k_1,k_2} &= F_{k_3,k_4} = -F_{k_1,k_4} = -F_{k_2,k_3} = \frac{1}{2} \langle S_d^+ A_{k_1,\downarrow,-}^\dagger A_{k_1,\uparrow,-} \rangle , \end{aligned} \quad (12.6)$$

where the momentum state k_i belongs to the quadrant Q_i .

Effective Star Graph Representation of Kondo Interaction

To make computation less intensive, we want to make the Hamiltonian in eq. 12.4 sparser, by going to a star graph representation where each node of the graph couples only with the impurity spin (central node) and not to the other nodes of the graph. We therefore consider a rotated basis of excitations

$$\gamma_{i,\sigma}^\dagger = \sum_k \Lambda_{i,k} A_{k,\sigma}^\dagger, \text{ where } \sum_k \Lambda_{i,k}^* \Lambda_{j,k} = \delta_{i,j}. \quad (12.7)$$

These fermions of course satisfy the appropriate algebra:

$$\{\gamma_{i,\sigma}, \gamma_{j,\sigma'}^\dagger\} = \sum_{k,q} \Lambda_{i,k}^* \Lambda_{j,q} \{A_{k,\sigma}, A_{q,\sigma'}^\dagger\} = \delta_{i,j} \delta_{\sigma,\sigma'}. \quad (12.8)$$

In order to calculate the coefficients $\Lambda_{i,k}$, we impose the star graph simplification condition described above:

$$\sum_{\mathbf{k},\mathbf{q}} J_f^*(\mathbf{k},\mathbf{q}) A_{\mathbf{k},\alpha}^\dagger A_{\mathbf{q},\beta} = \sum_i K_i \gamma_{i,\alpha}^\dagger \gamma_{i,\beta}, \quad (12.9)$$

where K_i are the eigenvalues of the Kondo coupling matrix $J_f^*(\mathbf{k},\mathbf{q})$. Substituting the definition of the γ fields (eq. 12.7) leads to the equation

$$J_f^*(\mathbf{k},\mathbf{q}) = \sum_i \Lambda_{k,i} K_i \Lambda_{i,q}^\dagger \implies \mathcal{J} = \Lambda K \Lambda^\dagger, \quad (12.10)$$

where $\mathcal{J}$ is the Kondo coupling matrix with matrix elements $J_f^*(\mathbf{k},\mathbf{q})$, K is a diagonal matrix with the eigenvalues (K_i) of $\mathcal{J}$ along its diagonal, and Λ is the matrix $\Lambda_{i,k}$ (the rotated indices i along the row, and the earlier momentum indices along the column). From eq. 12.10, it is clear that since $\mathcal{J}$ is Hermitian, the matrix Λ is simply the unitary matrix which diagonalises the Kondo matrix $\mathcal{J}$:

$$\Lambda^\dagger \mathcal{J} \Lambda = K. \quad (12.11)$$

The prescription is therefore as follows. We compute the unitary transformation Λ that diagonalises $\mathcal{J}$ and obtain the eigenvalues $\{K_i\}$. Using the eigenvalues, we work with the sparser Hamiltonian

$$\vec{S}_f \cdot \sum_{\alpha,\beta} \vec{\sigma}_{\alpha,\beta} \sum_i K_i \gamma_{i,\alpha}^\dagger \gamma_{i,\beta}. \quad (12.12)$$

If we have to calculate a certain correlation $C_{k,q}$, we instead calculate the correlation matrix $\tilde{C}_{i,j}$, and then transform back using the unitary transformation:

$$C = \Lambda \tilde{C} \Lambda^\dagger . \quad (12.13)$$

1.13 Measures of entanglement and their computation

The present millennium has seen a rapid surge in the use of quantum entanglement towards the study of quantum and condensed matter systems [14–19]. By tracking quantum correlations among particles across both small and large distances, entanglement measures offer valuable insights into the nature of the ground state and low-energy excitations. This, for example, allows us to distinguish strongly-correlated topologically ordered states characterised by long-ranged entanglement (in the form of a sub-leading topological term in their entanglement entropy [20–23]) from weakly-interacting topologically trivial phases that have short-ranged area-law entanglement [16, 24, 25].

1.13.1 Reduced density matrix elements as equal-time correlation functions

A density matrix plays the role of a probability density function (PDF) for a quantum system. A classical variable X that can take values $x \in [-X^*, X^*]$ has a PDF $F(X = x)$, which is used, for example, for calculating the average value (expectation value) of the variable:

$$\langle X \rangle = \int_{-X^*}^{X^*} x F(x) dx . \quad (13.1)$$

In quantum mechanics, the variables we are interested in are operators O , and a quantum PDF ρ will then give the expectation value of the operator, in a fashion similar to its classical counterpart:

$$\langle O \rangle = \sum_{\phi} \langle \phi | \rho O | \phi \rangle ; \quad (13.2)$$

here, the integral over the classical values x is replaced by a sum over the all possible quantum states $|\phi\rangle$ (complete orthonormal basis) of the system. However, we already know that the expectation value of a quantum operator is given by

$$\langle O \rangle = \langle \Psi | O | \Psi \rangle . \quad (13.3)$$

To bring this form closer to the one in terms of ρ (eq. 2), we insert an identity resolution: $1 = \sum_{\phi} |\phi\rangle \langle \phi|$:

$$\langle O \rangle = \sum_{\phi} \langle \Psi | O | \phi \rangle \langle \phi | \Psi \rangle = \sum_{\phi} \langle \phi | \Psi \rangle \langle \Psi | O | \phi \rangle . \quad (13.4)$$

A comparison of this equation to eq. (2) gives

$$\rho = |\Psi\rangle \langle \Psi| ; \quad (13.5)$$

a density matrix (also called density operator) is therefore formally defined as the outer product of the wavefunction with its conjugate. It encodes all information about the state; for example, the probability of measuring the system to be at a position x (hence in a state $|x\rangle$) is given by $P(x) = \langle x|\rho|x\rangle$. From its definition in eq. (5), the DM has the very important property that it has unity trace:

$$\text{Tr}[\rho] = 1 , \quad (13.6)$$

where $\text{Tr}[\cdot]$ is the trace of an operator. This is seen very easily by taking the trace in an orthonormal basis in which the state $|\Psi\rangle$ is one of the basis states. For example, consider the DM

$$\rho = \frac{1}{\sqrt{2}} [|\Psi_1\rangle \langle\Psi_1| + |\Psi_2\rangle \langle\Psi_2|] , \quad (13.7)$$

where $|\Psi_{1,2}\rangle$ are orthogonal normalised states. The trace of ρ^2 for such a state is only $1/2$; such states satisfying $\text{Tr}[\rho^2] < 1$ are termed *mixed states*; the DM describing such states cannot be written in the form $\rho = |\Psi\rangle \langle\Psi|$ for any wavefunction $|\Psi\rangle$. On the other hand, the DMs that *can* be expressed like that are termed *pure states*, and have $\text{Tr}[\rho^2] = 1$.

One way in which a pure state can be realistically converted into a mixed state is through a measurement process. Consider the zero total spin state formed by two spins:

$$|\Psi\rangle = \frac{1}{\sqrt{2}} [|\uparrow, \downarrow\rangle + |\downarrow, \uparrow\rangle] , \rho = |\Psi\rangle \langle\Psi| . \quad (13.8)$$

The state $|\uparrow, \downarrow\rangle$ has the first spin pointing up and the second one down, and this is in a superposition with its flipped counterpart $|\downarrow, \uparrow\rangle$. This is clearly a pure state.

If we now measure the state of the second spin, it will collapse into one of its two states; the other spin will collapse into the opposite state (given the antiparallel configurations of the spins). What this means is that until one checks what the result of the measurement is, the first spin remains in a statistical superposition of both outcomes (up and down). This is formally expressed through a partial trace operation: the state of the first spin after a measurement of the second spin is obtained by taking the full density matrix ρ and computing its trace over the states $|\sigma_B\rangle$ of the second spin, leaving out the states of the first spin from the trace:

$$\rho_1 = \text{Tr}_2[\rho] = \sum_{\sigma_2} \langle\sigma_2|\rho|\sigma_2\rangle ; \quad (13.9)$$

where $|\sigma_2\rangle$ sums over the states $|\uparrow\rangle$ and $|\downarrow\rangle$ of the second spin. this partial tracing operation Tr_2 codifies the measurement process, and the partially traced density matrix ρ_1 is termed a reduced density matrix (RDM) of the first spin, because it only captures information about the first spin (the second spin has been traced out).

To see the connection with mixedness, let us compute the RDM for the first spin explicitly:

$$\rho_1 = \langle\uparrow_2|\rho|\uparrow_2\rangle + \langle\downarrow_2|\rho|\downarrow_2\rangle = \frac{1}{2} [|\uparrow_1\rangle \langle\uparrow_1| + |\downarrow_1\rangle \langle\downarrow_1|] , \quad (13.10)$$

where the numbered subscripts on the kets and bras indicate that they only act on the respective Hilbert spaces. To see that ρ_1 describes a mixed state, we adopt a specific representation:

$$\begin{aligned} |\uparrow_1\rangle &= \begin{pmatrix} 1 \\ 0 \end{pmatrix}, |\downarrow_1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \\ \Rightarrow \rho_1 &= \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \rho_1^2 = \frac{1}{4} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \end{aligned} \quad (13.11)$$

The trace of ρ_1^2 is half, hence it is a mixed state.

The deviation of trace of ρ^2 from its maximum value of unity can be thought of as a measure of the lost information. In the example above, when we traced out the states of the second spin, we lost some information about the state of the first spin as well. This is because the two spins were *entangled*; they had quantum correlations between them. The entanglement arises from the fact that the states of the two spins are anti-correlated.

If instead we had chosen a non-entangled parent state to start from, the reduced density matrix would not have been mixed. For example, consider the state

$$|\Psi\rangle = \frac{1}{2} (|\uparrow_1\rangle + |\downarrow_1\rangle) \otimes (|\uparrow_2\rangle + |\downarrow_2\rangle). \quad (13.12)$$

It is easy to see that this state does not contain any correlations between the two spins - no matter the configuration of the second spin, the first spin is always in the state $|\uparrow_1\rangle + |\downarrow_1\rangle$. The RDM for the first spin comes out to be

$$\rho_1 = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \rho_1^2 = \frac{1}{4} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad (13.13)$$

revealing that ρ_1 is not mixed.

The above method of calculating an RDM works for simple cases but can fail if (i) we have fermion signs to account for, or (ii) the full basis cannot be expressed as a direct product of local basis sets. The former arises if we are dealing with fermions, and the latter arises if we are working in a rotated basis. We now derive an alternative method for calculating the RDM, one that automatically takes care of the fermion signs and works even if the basis is rotated and complicated to write down. We start by defining the RDM more precisely. Given a state $|\Psi\rangle$ and a bipartition into subsystems A and its complement A' , the density matrix for the state is $\rho = |\Psi\rangle\langle\Psi|$. A basis state $|B_{i,j}\rangle$ of the full system can be written as $|B_{i,j}\rangle = |A_i\rangle \otimes |A'_j\rangle$, where $\{A_i\}$ and $\{A'_j\}$ are complete sets of basis states describing A and A' respectively. The RDM for A is then defined as

$$\rho_A = \sum_{i,j,k} \langle B_{i,j} | \rho | B_{i,k} \rangle |A_j\rangle \langle A_k| = \sum_{i,j,k} \langle B_{i,j} | \Psi \rangle \langle \Psi | B_{i,k} \rangle |A_j\rangle \langle A_k|. \quad (13.14)$$

The matrix elements of ρ_A in the basis $\{A_i\}$ can be expressed as

$$\langle A_l | \rho_A | A_k \rangle = \sum_i \langle A'_j | \langle A_i | \Psi \rangle \langle \Psi | A_i \rangle | A'_k \rangle = \sum_i \langle A'_j | \langle \Psi | A_i \rangle | \Psi \rangle \langle \Psi | A_i \rangle | A'_k \rangle. \quad (13.15)$$

By bringing the summation over

Quantum Fisher information: a witness of multipartite entanglement

The quantum Fisher information (QFI) F_Q [26, 27] quantifies how sensitive a state ρ is to unitary transformations generated by an observable $\hat{O}$. For a pure state $\rho = |\psi\rangle\langle\psi|$, $F_Q(\psi, \hat{O})$ is defined in terms of the variance of the operator $\hat{O}$ (i.e., the connected correlation function):

$$F_Q(\psi, \hat{O}) = 4\Delta(\hat{O})^2 = 4 \left(\langle\psi|\hat{O}^2|\psi\rangle - \langle\psi|\hat{O}|\psi\rangle^2 \right), \quad (13.16)$$

and it provides an upper bound on the precision that can be achieved in measuring the parameter θ that is dual to the observable $\hat{O}$: $\mathcal{N}(\Delta\theta)^2 \geq F_Q^{-1}$, $\mathcal{N}$ being the number of independent measurements [26, 27]. $F_Q(\psi, \hat{O})$ is thus a measure of the entanglement arising from quantum fluctuation content in $|\psi\rangle$ (considered with respect to an eigenstate of $\hat{O}$). This can be made more manifest by considering a traceless operator $\hat{M}$ with eigenstates $\{|m\rangle\}$. Without loss of generality, we can rescale the operator such that $\sum m^2 = 1/2$. If $\varepsilon_m(\psi) \equiv 1 - |\langle m|\psi\rangle|^2$ is the geometric entanglement between a given state $|\psi\rangle$ and the eigenstates of $\hat{M}$, the QFI corresponding to $\hat{M}$ in the state ψ can be written as

$$F_Q(\psi, \hat{M}) = 4 \left[\frac{1}{2} - \sum_m m^2 \varepsilon_m - \left(\sum_m m \varepsilon_m \right)^2 \right]. \quad (13.17)$$

The total geometric entanglement $\sum_m \varepsilon_m$ is constrained to be $d_M - 1$, where d_M is the dimension of the operator $\hat{M}$. Because each eigenvalue m has a magnitude of at most $1/\sqrt{2}$ (all m^2 must add up to $1/2$), the maximum magnitude of the last two terms in eq. (13.17) (and hence the minimum value of the QFI) is attained for the case when some of the ε_m are zero. However, because the eigenstates are orthogonal, only one of them can have perfect overlap with the state $|\psi\rangle$ and only one ε_m can be zero at a time. The case of minimum QFI therefore corresponds to $\varepsilon_{m^*} = 0, \varepsilon_{m \neq m^*} = 1$, leading to $F_Q = 0$. The opposite situation arises when all the geometric entanglement measures are non-zero and equal, $\varepsilon_m = 1/d_M$, leading to a maximum QFI value of $F_Q = 2/d_M$. A uniformly spread geometric entanglement distribution, therefore, leads to a larger QFI, while a more focused distribution leads to a smaller QFI. This sums up the relation between the quantum Fisher information and geometric entanglement.

If the operator $\hat{M}$ is such that its eigenstates are separable states from the perspective of a certain party, the vanishing QFI arises when $|\psi\rangle$ has zero geometric entanglement with respect to one of the separable states, and the QFI saturates when $|\psi\rangle$ has a finite geometric entanglement with all of the states. A large QFI can thus be associated with more entanglement. Moreover, if the eigenstates of $\hat{M}$ are symmetry-broken states, a vanishing QFI indicates that $|\psi\rangle$ is very close to such a symmetry-broken state, while a large QFI indicates that $|\psi\rangle$ is a uniform superposition of such symmetry-broken states and therefore preserves the symmetry as a whole. A larger QFI points towards the presence of more quantum fluctuations and a reduced susceptibility towards symmetry-breaking.

We will now relate the QFI to some other measures of correlation. The QFI corresponding to an observable can be directly related to the static lesser Greens functions associated with that observable:

$$F_Q(\psi, \hat{O}_2) = 4 \left(\left\langle \hat{O}_2 \hat{O}_1^\dagger \right\rangle_{\hat{O}_1^\dagger = \hat{O}_2} - \left\langle \hat{O}_2 \hat{O}_1^\dagger \right\rangle_{\hat{O}_1 = 1}^2 \right) = 4 \left[G_{\hat{O}_2, \hat{O}_2}^<(t \rightarrow \infty) - \left(G_{1, \hat{O}_2}^<(t \rightarrow \infty) \right)^2 \right] \quad (13.18)$$

Further, the retarded Greens function defined above in eq. (5.5) can very generally be related to the QFI. By using the simplified forms $\frac{i}{2}G_{O_2, O_2}^R(t \rightarrow \infty) = \langle (O_2)^2 \rangle$ and $\frac{i}{2}G_{1, O_2}^R(t \rightarrow \infty) = \langle O_2 \rangle$, we get

$$F_Q(\psi, \hat{O}_2) = 2iG_{O_2, O_2}^R(t \rightarrow \infty) + \left(G_{1, O_2}^R(t \rightarrow \infty)\right)^2. \quad (13.19)$$

For the sake of completeness, we generalise the above relations to finite temperatures by adopting the approach of Hauke et al. [27]. For a mixed state at inverse temperature β characterised by the density matrix $\rho = \sum_n f_n |n\rangle \langle n|$ where $(E_n, |n\rangle)$ are the eigenvalue-eigenstate pairs and $f_n = e^{-\beta E_n} / \sum_n e^{-\beta E_n}$ are the Boltzmann weights, the FQI can be assigned a spectral representation of the form [27]

$$F_Q(\rho, \hat{O}) = 2 \sum_{m,n} \frac{(f_m - f_n)^2}{f_m + f_n} |\langle m | \hat{O} | n \rangle|^2. \quad (13.20)$$

By using the identity $\frac{f_m - f_n}{f_m + f_n} = \int_{-\infty}^{\infty} d\omega \delta(\omega + E_m - E_n) \tanh(\beta\omega/2)$, the finite temperature FQI can be written in the form

$$F_Q(\rho, \hat{O}) = 2 \int_{-\infty}^{\infty} d\omega \tanh(\beta\omega/2) \sum_{m,n} (f_m - f_n) \delta(\omega + E_m - E_n) |\langle m | \hat{O} | n \rangle|^2. \quad (13.21)$$

In order to relate this with a correlation function, we now point out that the lesser Greens function $G^<$ and the greater Greens function $G^>$ that we will now define admit spectral representations in terms of the eigenstates $|n\rangle$ of the full problem:

$$G_{O^\dagger, O}^<(t) \equiv i \langle \hat{O}^\dagger \hat{O}(t) \rangle \implies G_{O^\dagger, O}^<(\omega) = 2\pi i \sum_{m,n} f_n \delta(\omega + E_m - E_n) |\langle m | \hat{O}^\dagger | n \rangle|^2 \quad (13.22)$$

$$G_{O^\dagger, O}^>(t) \equiv -i \langle \hat{O}(t) \hat{O}^\dagger \rangle \implies G_{O^\dagger, O}^>(\omega) = -2\pi i \sum_{m,n} f_m \delta(\omega + E_m - E_n) |\langle m | \hat{O} | n \rangle|^2 \quad (13.23)$$

If $\hat{O}$ is a Hermitian operator, eqs. (13.22) can be used to rewrite the QFI in terms of $G_{O^\dagger, O}^\lessgtr(\omega)$. Indeed, by combining eqs. (13.22) and eq. (13.21) with the assumption $\hat{O} = \hat{O}^\dagger$, we get

$$F_Q(\rho, \hat{O}) = -\frac{1}{\pi i} \int_{-\infty}^{\infty} d\omega \tanh(\beta\omega/2) \left(G_{O, O}^>(\omega) + G_{O, O}^<(\omega) \right). \quad (13.24)$$

This constitutes the general relation between the finite temperature QFI for a general Hermitian many-particle operator $\hat{O}$ and the corresponding many-particle correlations at zero and finite frequencies.

The Kubo linear response function associated with the operator $\hat{O}(t)$ due to a perturbation from the same operator is associated with a correlation function/susceptibility $C_{\hat{O}}(t, t') = i\theta(t - t') \langle [\hat{O}(t), \hat{O}(t')] \rangle$. By going to the frequency domain, the imaginary part of such a function can

be related to the sum of the lesser and greater Greens functions:

$$\begin{aligned}
 C_{\hat{O}}(\omega) &= i \int_0^\infty d(t-t') e^{i\omega(t-t')} \left(\langle \hat{O}(t) \hat{O}(t') \rangle - \langle \hat{O}(t') \hat{O}(t) \rangle \right) \\
 \Rightarrow C_{\hat{O}}(\omega) - C_{\hat{O}}(\omega)^* &= i \int_0^\infty d(t-t') \left(e^{i\omega(t-t')} - e^{-i\omega(t-t')} \right) \left(\langle \hat{O}(t) \hat{O}(t') \rangle - \langle \hat{O}(t') \hat{O}(t) \rangle \right) \\
 &= i \int_{-\infty}^\infty d(t-t') e^{i\omega(t-t')} \left(\langle \hat{O}(t) \hat{O}(t') \rangle - \langle \hat{O}(t') \hat{O}(t) \rangle \right) \\
 &= - \left(G_{O,O}^>(\omega) + G_{O,O}^<(\omega) \right) .
 \end{aligned}$$

This allows us to connect the QFI with the imaginary part of the Kubo correlation function/susceptibility obtained in Ref. [27]:

$$F_Q(\rho, \hat{O}) = \frac{1}{\pi i} \int_{-\infty}^\infty d\omega \tanh(\beta\omega/2) (C_{\hat{O}}(\omega) - C_{\hat{O}}(\omega)^*) . \quad (13.25)$$

1.13.2 Decomposition of entanglement entropy in terms of multipartite information I_n

We first define a partitioning of the entire system into N disjoint regions, $\mathcal{S}_N = \{\nu_1, \nu_2, \dots, \nu_N\}$. Going beyond bipartite mutual information, one can define the n -partite information I_n between the members of $\mathcal{S}_n = \{\nu_1, \nu_2, \dots, \nu_n\}$, which is itself a subset of the exhaustive set of regions $\mathcal{S}_N$:

$$I_n(\nu_1, \dots, \nu_n) = \sum_{\sigma \in \mathcal{P}(\mathcal{S}_n)} (-1)^{1+|\sigma|} S_{EE}(\sigma) , \quad (13.26)$$

where σ is a subset drawn from the power set of $\mathcal{S}_n$, and $|\sigma|$ is the number of members in the subset σ . The 4-partite information can be written as

$$\begin{aligned}
 I_4(\nu_1, \nu_2, \nu_3, \nu_4) &= S_{EE}(\nu_1) + S_{EE}(\nu_2) + S_{EE}(\nu_3) + S_{EE}(\nu_4) - S_{EE}(\nu_1 \cup \nu_2) - S_{EE}(\nu_1 \cup \nu_3) \\
 &\quad - S_{EE}(\nu_1 \cup \nu_4) - S_{EE}(\nu_2 \cup \nu_3) - S_{EE}(\nu_2 \cup \nu_4) - S_{EE}(\nu_3 \cup \nu_4) + S_{EE}(\nu_1 \cup \nu_2 \cup \nu_3) \\
 &\quad + S_{EE}(\nu_1 \cup \nu_2 \cup \nu_4) + S_{EE}(\nu_1 \cup \nu_3 \cup \nu_4) + S_{EE}(\nu_2 \cup \nu_3 \cup \nu_4) + S_{EE}(\nu_1 \cup \nu_2 \cup \nu_3 \cup \nu_4) .
 \end{aligned} \quad (13.27)$$

One can also define the average n -partite information $\bar{I}_n$ by considering all sets of n regions:

$$\bar{I}_n = \frac{(N-n)!n!}{N!} \sum_{S_n \in C_n(\mathcal{S}_N)} I_n(S_n) = \frac{(N-n)!n!}{N!} \sum_{S_n \in C_n(\mathcal{S}_N)} \sum_{\sigma \in \mathcal{P}(\mathcal{S}_n)} (-1)^{1+|\sigma|} S_{EE}(\sigma) , \quad (13.28)$$

where $C_n(\mathcal{S}_N)$ represents all ways of choosing n distinct elements out of the full set $\mathcal{S}_N$ of subsystems.

The expression for $\bar{I}_n$ can be simplified by splitting the summation over the elements of the power set $\mathcal{P}$ into sets of definite number:

$$\bar{I}_n = \frac{(N-n)!n!}{N!} \sum_{S_n \in C_n(\mathcal{S}_N)} I_n(S_n) = \frac{(N-n)!n!}{N!} \sum_{S_n \in C_n(\mathcal{S}_N)} \sum_{m=1}^n \sum_{\sigma_m \in \mathcal{P}_m(\mathcal{S}_n)} (-1)^{1+|\sigma|} S_{EE}(\sigma_m) , \quad (13.29)$$

where $\mathcal{P}_m$ represents all subsets of the power set that have m elements. With some combinatorial magic, it can be shown that

$$\sum_{S_n \in C_n(S_N)} \sum_{\sigma_m \in \mathcal{P}_m(S_n)} = \binom{N-m}{n-m} \sum_{\sigma_m \in C_m(S_N)} . \quad (13.30)$$

To see how this comes about, note that what this equation says is that summing over combinations of subsets of length n of the super set S_N and further summing over all subsets of length $m \leq n$ for each set S_n is equivalent to summing just once over subsets of length m of the super set S_N but now each subset of length m appears with a frequency $f_m = \binom{N-m}{n-m}$. The value for the frequency f_m is simply the number of subsets S_n in which the smaller subset σ_m appears. This can be calculated by allowing m of the n slots in S_n to be filled by the members of σ_m , and the remaining $n - m$ slots then have to be filled by choosing from the remaining $N - m$ members of the super set S_N ; there's obviously $\binom{N-m}{n-m}$ ways of carrying this out.

Implementing this into the expression for $\bar{I}_n$ gives

$$\bar{I}_n = \sum_{m=1}^n (-1)^{1+m} \frac{\binom{n}{m}}{\binom{N}{m}} \sum_{\sigma_m \in C_m(S_N)} S_{EE}(\sigma_m) = \sum_{m=1}^n (-1)^{1+m} \binom{n}{m} S_{EE}^{(m)} , \quad (13.31)$$

where we have define the average entanglement entropy $S_{EE}^{(m)}$ for all m -member sets of regions:

$$S_{EE}^{(m)} = \frac{1}{\binom{N}{m}} \sum_{\sigma_m \in C_m(S_N)} S_{EE}(\sigma_m) . \quad (13.32)$$

With most of the machinery in place, we would now like to express the entanglement entropy S_{EE} purely in terms multipartite measures I_n , for the reasons stated above. For a system that is invariant under real-space translation (as our lattice model is), the entanglement entropy of any real-space interval should only depend on the extent of the interval. It therefore suffices to calculate the entanglement entropy averaged over all such intervals, and the corresponding measure in our formulation is $S_{EE}^{(1)}$.

We will now express $S_{EE}^{(1)}$ in terms of $\bar{I}_n$, by eliminating $S_{EE}^{(m)}$ ($m > 1$) from the equations. We adopt the ansatz

$$S_{EE}^{(1)} = \sum_{n=2}^N \lambda_n \bar{I}_n = \sum_{n=2}^N \lambda_n \binom{n}{m} \sum_{m=1}^n (-1)^{1+m} S_{EE}^{(m)} . \quad (13.33)$$

In order to separate $S_{EE}^{(1)}$ from the rest of the $S_{EE}^{(m)}$, we exchange the order of summation:

$$S_{EE}^{(1)} = \sum_{m=1}^N (-1)^{1+m} S_{EE}^{(m)} \sum_{n=m}^N \lambda_n \binom{n}{m} - S_{EE}^{(1)} \lambda_1 = S_{EE}^{(1)} \sum_{n=2}^N n \lambda_n - \sum_{m=2}^{N-1} S_{EE}^{(m)} (-1)^m \sum_{n=m}^N \binom{n}{m} \lambda_n , \quad (13.34)$$

where the constraint of $m \leq n$ in the previous form is now enforced by letting n run from m to N . The $N - 1$ coefficients $\{\lambda_n\}$ will be determined by requiring that the $S_{EE}^{(m)}$ ($m > 1$) vanish on the

right-hand side, as well as by matching the coefficients of $S_{\text{EE}}^{(1)}$ on both sides. Note that m runs only upto $N - 1$ in the second term; this is because, $S_{\text{EE}}^{(N)}$ is the entanglement entropy of the full system and hence automatically zero. The equations are

$$\sum_{n=2}^N n \lambda_n = 1, \quad (13.35)$$

$$\sum_{n=m}^N \binom{n}{m} \lambda_n = 0; \quad m = 2, \dots, N - 1. \quad (13.36)$$

We focus on the second equation first, since it is very short for m close to N . For $m = N - 1$ and $m = N - 2$, we get

$$\begin{aligned} m = N - 1 : \lambda_{N-1} + N \lambda_N = 0 &\implies \lambda_{N-1} = -N \lambda_N \\ m = N - 2 : \lambda_{N-2} + (N - 1) \lambda_{N-1} + \frac{N(N - 1)}{2} \lambda_N = 0 &\implies \lambda_{N-2} = \frac{N(N - 1)}{2} \lambda_N. \end{aligned} \quad (13.37)$$

We extrapolate this to the general solution,

$$\lambda_m = (-1)^m \binom{N}{m} \lambda_N, \quad m < N, \quad (13.38)$$

and then prove that if this is true for λ_{N-m} , it also solves the equation for λ_{N-m-1} , thus amounting to a proof by mathematical induction. The general equation for λ_{N-m-1} is

$$\sum_{n=N-m-1}^N \binom{n}{N-m-1} \lambda_n = 0 \implies \lambda_{N-m-1} + \sum_{m'=0}^m \binom{N-m'}{N-m-1} \lambda_{N-m'} = 0. \quad (13.39)$$

Substituting the extrapolated solution gives

$$\begin{aligned} \lambda_{N-m-1} &= - \sum_{m'=0}^m (-1)^{N-m'} \binom{N-m'}{N-m-1} \binom{N}{m'} \lambda_N = \binom{N}{m+1} \lambda_N \sum_{m'=0}^m (-1)^{N-m'-1} \binom{m+1}{m'} \\ &= (-1)^{N-m-1} \binom{N}{m+1} \lambda_N, \end{aligned} \quad (13.40)$$

which is consistent with the general solution eq. 13.38. This determines all coefficients λ_n for $n < N$ in terms of λ_N . To determine λ_N itself, we use eq. 13.35. Substituting eq. 13.38 into it gives

$$\lambda_N \sum_{n=2}^N n (-1)^n \binom{N}{n} = 1 \implies \lambda_N = \frac{(-1)^N}{N}. \quad (13.41)$$

The decomposition of entanglement entropy into multipartite measures is now complete:

$$\begin{aligned} S_{\text{EE}}^{(1)} &= \frac{1}{N} \sum_{n=2}^N (-1)^n \binom{N}{n} \bar{I}_n, \\ \bar{I}_n &= \frac{(N-n)!n!}{N!} \sum_{S_n \in C_n(S_N)} I_n(S_n), \end{aligned} \quad (13.42)$$

where we have assumed N is even in order to drop the global phase factor $(-1)^N$.

1.14 Wavefunction Renormalisation Group

1.14.1 General idea

A reverse URG analysis involves studying the evolution of wavefunctions from the IR to the UV, by implementing inverse renormalisation group transformations. This is made possible by the fact that the IR fixed point Hamiltonians are often tractable and allow the computation of wavefunctions. Such an analysis gives insight into how states change as high-energy quantum fluctuations are resolved during the Hamiltonian renormalisation from the UV to the IR. It also allows studying the evolution of entanglement measures and correlation functions.

We will start with a very simple IR ground state wavefunction, and then go back towards the UV ground state by applying the inverse unitary operator $U^\dagger$:

$$\begin{aligned}
 U : \underbrace{|1, 2, \dots, N\rangle}_{\text{UV ground state}} &\rightarrow |1, 2, \dots, N-1\rangle |N\rangle \rightarrow \dots \rightarrow \underbrace{|1, 2, \dots, N^*\rangle |N^*+1\rangle \dots |N\rangle}_{\text{IR ground state}} \\
 U^\dagger : \underbrace{|1, 2, \dots, N^*\rangle |N^*+1\rangle \dots |N\rangle}_{\text{IR ground state}} &\rightarrow |1, 2, \dots, N^*+1\rangle |N^*+2\rangle \dots |N\rangle \rightarrow \dots \rightarrow \underbrace{|1, 2, \dots, N\rangle}_{\text{UV ground state}}
 \end{aligned}$$

The first process is the forward RG which we used to obtain the scaling equations. The second process is the reverse RG which we will undertake now. The method has already been applied to study the renormalisation of entanglement in various models of strong correlation. In general, we start with an IR wavefunction that consists a certain number of momentum states n_1 still entangled with the impurity, and the remaining momentum states n_2 disentangled from the impurity. The former are said to be part of the emergent Kondo cloud, while the latter are said to be part of the integrals of motion (IOMs) and appear in direct product with the cloud+impurity system in the ground state wavefunction. Each momentum state is tagged with two conduction bath levels, one above the Fermi surface and one below. These will be represented as $k, \pm$. If we represent the emergent cloud momentum states as $\{k_i\}$ and the IOM states as $\{q_i\}$, the ground state wavefunction can be written as

$$|\Psi\rangle_0 = \left(\bigotimes_{i=1}^{n_2} |q_i, -, \uparrow\rangle |q_i, -, \downarrow\rangle \right) |\Phi\rangle_{\text{cloud}} \left(\bigotimes_{i=1}^{n_2} |q_i, +, \uparrow\rangle |q_i, +, \downarrow\rangle \right) \quad (14.1)$$

Throughout, we have assumed that the IOMs below the Fermi surface are occupied while those above are unoccupied. This means:

$$|\Psi\rangle_0 = \left(\bigotimes_{i=1}^{n_2} |\hat{n}_{q_i, -, \uparrow} = 1\rangle |\hat{n}_{q_i, -, \downarrow} = 1\rangle \right) |\Phi\rangle_{\text{cloud}} \left(\bigotimes_{i=1}^{n_2} |\hat{n}_{q_i, +, \uparrow} = 0\rangle |\hat{n}_{q_i, +, \downarrow} = 0\rangle \right) \quad (14.2)$$

$|\Phi\rangle_{\text{cloud}}$ will be obtained by diagonalising a small system of n_1 conduction bath k states. This completes the construction of the IR ground state $|\Psi\rangle_0$. The algorithm of reverse RG then involves applying unitary transformations on this IR wavefunction; the unitary transformations will be the inverse of those that were applied during the forward RG algorithm. Since the forwards RG transformations decoupled k states and transformed them into the IOMS, the inverse transformations will

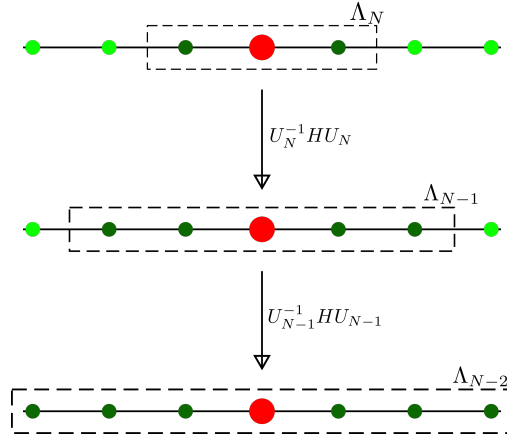


Figure 1.12: We start with a Hamiltonian with an impurity site (red) coupled with two conduction electrons (dark green), with four other decoupled electrons (bright green). The dotted rectangle represents the emergent window $(-\Lambda_j, \Lambda_j)$ at each step; the electrons inside that rectangle are still entangled with the impurity, while the ones outside have been decoupled. The next step of reverse RG involves applying the inverse transformation on the Hamiltonian, which will couple two more electrons from the IOMS (hence four dark green circles in the second step), leading to an enlargement of the emergent window. The unitary varies for each step, hence the notation U_j .

take the momentum states in the IOMs and re-entangle them with the impurity+cloud system, one at a time. This is shown in fig. 1.12.

The next step is to write down the unitaries that will take us from the IR ground state to the UV ground state. In the forward RG, we used the following unitary for decoupling the shell ϵ_q and spin β is

$$U_{q\beta} = \frac{1}{\sqrt{2}} \left[1 + \eta_{0\beta} + \eta_{1\beta}^\dagger \right]. \quad (14.3)$$

Here, the subscripts 0 and 1 indicate it decouples an electron above and below the Fermi surface respectively. The inverse transformation for re-entangling $q\beta$ is

$$U_{q\beta}^\dagger = \frac{1}{\sqrt{2}} \left[1 + \eta_{0\beta}^\dagger + \eta_{1\beta} \right]. \quad (14.4)$$

The wavefunction after reversing one step of the RG will thus be

$$|\Psi\rangle_1 = U_{q\uparrow}^\dagger U_{q\downarrow}^\dagger |\Psi\rangle_0 \quad (14.5)$$

Here, q is the first momentum state immediately outside the emergent cloud. Re-entangling further momentum states using their respective unitaries lead to the sequence of wavefunctions $|\Psi\rangle_2, |\Psi\rangle_3$ and so on. A straightforward way of calculating entanglement and correlations across the RG flow is to create a second-quantised prototypical form of the IR state $|\Psi_0\rangle$, and then apply the inverse unitary operators $\{U_i\}$ to this wavefunction in order to generate the family of states $|\Psi_i\rangle = U_i |\Psi_{i-1}\rangle, i > 0$. Quantities like entanglement entropy and correlations can then be calculated by using this family of states. If the unitary is of a specific form, there is instead a more efficient method of doing this, which is what we will describe in the next subsection.

1.14.2 Evolution equation for state tensor coefficients

In the following, we restrict ourselves to the reverse RG analysis of an impurity problem hybridising with the local degree of freedom of a conduction bath. At every step, we allow two states from the IOMS to interact with the already-entangled states (one above the Fermi energy and one below, in order to maintain particle-hole symmetry). After the j^{th} application of the reverse RG transformation, $2j$ number of states are entangled with the impurity site, and the state can be represented as

$$|\Psi_j\rangle = \sum_{v_d, v_1, \dots, v_{2j}} C_{v_d, v_1, \dots, v_{2j}, I_{2j+1}}^{(j)} |v_d\rangle |v_1\rangle |v_2\rangle \dots |v_{2j}\rangle \otimes |I_{2j+1}\rangle, \quad (14.6)$$

where $|v_d\rangle$ and $|v_i\rangle$ ($i \in [1, 2j]$) sum over the configurations of the impurity site and the effective zeroth site configuration at the j^{th} step. $|I_{2j+1}\rangle$ is the configuration of the integrals of motion from the $(2j+1)^{\text{th}}$ site and beyond. We define the product state $|\psi_{dz}(j)\rangle = \text{ket } v_d |v_1\rangle \dots |v_{2j}\rangle$ for future reference.

We consider a restricted form of RG evolution, where the unitary transformation from IR towards UV can be expressed at any step as

$$U_j^\dagger = 1 + \sum_{i \in [1, 2j]} \sum_{v_d, \mu_d, v_i, \mu_i} (J_{v_d, v_i}^{\mu_d, \mu_i} \mathcal{T}_{v_d}^{\mu_d} \mathcal{T}_{v_i}^{\mu_i} c_{2j+1} + K_{v_d, v_i}^{\mu_d, \mu_i} \mathcal{T}_{v_d}^{\mu_d} \mathcal{T}_{v_i}^{\mu_i} c_{2j+2}^\dagger), \quad (14.7)$$

where $\mathcal{T}_\alpha^v = |v\rangle \langle \alpha|$ is a transition operator and sums over all transition operators of the specific subspace (impurity or bath sites). The state at the $(j+1)^{\text{th}}$ step is

$$\begin{aligned} |\Psi_{j+1}\rangle &= \sum_{v_d, v_1, \dots, v_j} C_{v_d, v_1, \dots, v_{2j}, 1_{2j+1}, 0_{2j+2}, I_{j+1}}^{(j)} |\psi_{dz}(j)\rangle |1_{2j+1}\rangle |0_{2j+2}\rangle \otimes |I_{2j+3}\rangle \\ &+ \sum_i \sum_{\mu_d, \mu_i, v_d, v_i} C_{\mu_d, \mu_i, \dots, v_{2j}, 1_{2j+1}, 0_{2j+2}}^{(j)} J_{\mu_d, \mu_i}^{v_d, v_i} (-1)^{\mu_d - \mu_i + \sum_{k \leq i} v_k} |\psi_{dz}(j)\rangle |0_{2j+1}\rangle |0_{2j+2}\rangle |I_{2j+3}\rangle \\ &+ \sum_i \sum_{\mu_d, \mu_i, v_d, v_i} C_{\mu_d, \mu_i, \dots, v_{2j}, 1_{2j+1}, 0_{2j+2}}^{(j)} K_{\mu_d, \mu_i}^{v_d, v_i} (-1)^{\mu_d - \mu_i + \sum_{k \leq i} v_k} |\psi_{dz}(j)\rangle |1_{2j+1}\rangle |1_{2j+2}\rangle |I_{2j+3}\rangle \end{aligned} \quad (14.8)$$

Gathering similar terms and normalising the state, we can write it as

$$\begin{aligned} |\Psi_{j+1}\rangle &= \frac{1}{L_j} \sum_{v_d, \{v_i\}} |\psi_{dz}(j)\rangle \left[C_{v_d, \{v_i\}, 1_{2j+1}, 0_{2j+2}, I_{j+1}}^{(j)} |1_{2j+1}\rangle |0_{2j+2}\rangle + C_{+, v_d, \{v_i\}}^{(j)} |1_{2j+1}\rangle |1_{2j+2}\rangle \right. \\ &\quad \left. + C_{-, v_d, \{v_i\}}^{(j)} |0_{2j+1}\rangle |0_{2j+2}\rangle \right] |I_{2j+3}\rangle, \end{aligned} \quad (14.9)$$

where $L_j^2 = \sum_{v_d, \{v_i\}} \left(C_{v_d, \{v_i\}}^{(j)}{}^2 + C_{+, v_d, \{v_i\}}^{(j)}{}^2 + C_{-, v_d, \{v_i\}}^{(j)}{}^2 \right)$ defines the normalisation factor, and

$$\begin{aligned} C_{+, v_d, \{v_i\}} &= \sum_{i, \mu_d, \mu_i} C_{\mu_d, \mu_i, \dots, v_{2j}, 1_{2j+1}, 0_{2j+2}}^{(j)} K_{\mu_d, \mu_i}^{v_d, v_i} (-1)^{\mu_d - \mu_i + \sum_{k \leq i} v_k} \\ C_{-, v_d, \{v_i\}} &= \sum_{i, \mu_d, \mu_i} C_{\mu_d, \mu_i, \dots, v_{2j}, 1_{2j+1}, 0_{2j+2}}^{(j)} J_{\mu_d, \mu_i}^{v_d, v_i} (-1)^{\mu_d - \mu_i + \sum_{k \leq i} v_k}. \end{aligned} \quad (14.10)$$

We can now read off the renormalisation of the state tensor coefficients:

$$\begin{aligned}
 C_{\nu_d, \nu_1, \dots, \nu_{2j}, 1_{2j+1}, 0_{2j+2}, I_{2j+3}}^{(j+1)} &= \frac{1}{L_j} C_{\nu_d, \nu_1, \dots, \nu_{2j}, 1_{2j+1}, 0_{2j+2}, I_{2j+3}}^{(j)}, \\
 C_{\nu_d, \nu_1, \dots, \nu_{2j}, 1_{2j+1}, 1_{2j+2}, I_{2j+3}}^{(j+1)} &= \frac{1}{L_j} \sum_{i, \mu_d, \mu_i} C_{\mu_d, \mu_i, \dots, \nu_{2j}, 1_{2j+1}, 0_{2j+2}}^{(j)} K_{\mu_d, \mu_i}^{\nu_d, \nu_i} (-1)^{\mu_d - \mu_i + \sum_{k \leq i} \nu_k}, \\
 C_{\nu_d, \nu_1, \dots, \nu_{2j}, 1_{2j+1}, 0_{2j+2}, I_{2j+3}}^{(j+1)} &= \frac{1}{L_j} \sum_{i, \mu_d, \mu_i} C_{\mu_d, \mu_i, \dots, \nu_{2j}, 1_{2j+1}, 0_{2j+2}}^{(j)} J_{\mu_d, \mu_i}^{\nu_d, \nu_i} (-1)^{\mu_d - \mu_i + \sum_{k \leq i} \nu_k},
 \end{aligned} \tag{14.11}$$

One can also look at renormalisation of expectation values of operators; this is useful for tracking the evolution of correlations and entanglement measures. Consider an operator $O_{\mu_k}^{\nu_k}$ that acts on the k^{th} bath site. The expectation value of this operator at any step can be written as

$$\langle O_{\mu_k}^{\nu_k} \rangle_j = \sum_{\nu_d, \nu_1, \dots, \nu_{k-1}, \nu_{k+1}, \dots, \nu_{2j}} (-1)^{\nu_d - \mu_k + \sum_{i \leq k} \nu_i} C_{\nu_d, \nu_1, \dots, \nu_k, \dots, \nu_{2j}, I_{2j+1}}^{(j)} C_{\nu_d, \nu_1, \dots, \mu_k, \dots, \nu_{2j}, I_{2j+1}}^{(j)} \tag{14.12}$$

1.15 Zero temperature Greens function in frequency domain

1.15.1 Spectral representation of Greens function

The impurity retarded Green's function is defined as

$$G_{d\sigma}(t) = -i\theta(t) \langle \{O_\sigma(t), O_\sigma^\dagger\} \rangle \tag{15.1}$$

where the average $\langle \rangle$ is over a canonical ensemble at temperature T , and O_σ is the excitation whose spectral function we are interested in. What follows is a standard calculation where we write the Green's function in the spectral representation. The ensemble average for the operator O_σ can be written in terms of the exact eigenstates of the Hamiltonian:

$$H|n\rangle = E_n|n\rangle, \quad \langle O_\sigma \rangle \equiv \frac{1}{Z} \sum_n \langle n|O_\sigma|n\rangle e^{-\beta E_n} \tag{15.2}$$

where $Z = \sum_n e^{-\beta E_n}$ is the partition function and $\{|n\rangle\}$ is the set of eigenfunctions of the Hamiltonian. We can therefore write

$$\begin{aligned}
 & \langle \{O_\sigma(t), O_\sigma^\dagger\} \rangle \\
 &= \frac{1}{Z} \sum_m e^{-\beta E_m} \langle m | \{O_\sigma(t), O_\sigma^\dagger\} | m \rangle \\
 &= \frac{1}{Z} \sum_{m,n} e^{-\beta E_m} \langle m | \left(O_\sigma(t) | n \rangle \langle n | O_\sigma^\dagger + O_\sigma^\dagger | n \rangle \langle n | O_\sigma(t) \right) | m \rangle \quad \left[\sum_n |n\rangle \langle n| = 1 \right] \\
 &= \frac{1}{Z} \sum_{m,n} e^{-\beta E_m} \langle m | \left(e^{iH^*t} O_\sigma e^{-iH^*t} | n \rangle \langle n | O_\sigma^\dagger + O_\sigma^\dagger | n \rangle \langle n | e^{iH^*t} O_\sigma e^{-iH^*t} \right) | m \rangle \\
 &= \frac{1}{Z} \sum_{m,n} e^{-\beta E_m} \left(e^{i(E_m - E_n)t} \langle m | O_\sigma | n \rangle \langle n | O_\sigma^\dagger | m \rangle + e^{i(E_n - E_m)t} \langle m | O_\sigma^\dagger | n \rangle \langle n | O_\sigma | m \rangle \right) \\
 &= \frac{1}{Z} \sum_{m,n} e^{i(E_m - E_n)t} || \langle m | O_\sigma | n \rangle ||^2 \left(e^{-\beta E_m} + e^{-\beta E_n} \right)
 \end{aligned} \tag{15.3}$$

The time-domain impurity Green's function can thus be written as (this is the so-called Lehmann-Kallen representation)

$$G_{d\sigma} = -i\theta(t) \frac{1}{Z} \sum_{m,n} e^{i(E_m - E_n)t} || \langle m | O_\sigma | n \rangle ||^2 \left(e^{-\beta E_m} + e^{-\beta E_n} \right) \tag{15.4}$$

We are interested in the frequency domain form.

$$\begin{aligned}
 G_{dd}^\sigma(\omega) &= \int_{-\infty}^{\infty} dt e^{i(\omega)t} G_{dd}^\sigma(t) \\
 &= \frac{1}{Z} \sum_{m,n} || \langle m | O_\sigma | n \rangle ||^2 \left(e^{-\beta E_m} + e^{-\beta E_n} \right) (-i) \int_{-\infty}^{\infty} dt \theta(t) e^{i(\omega + E_m - E_n)t}
 \end{aligned} \tag{15.5}$$

To evaluate the time-integral, we will use the integral representation of the Heaviside function:

$$\theta(t) = -\frac{1}{2\pi i} \lim_{\eta \rightarrow 0^+} \int_{-\infty}^{\infty} \frac{1}{x + i\eta} e^{-ixt} dx \tag{15.6}$$

With this definition, the integral in $G_{d\sigma}(\omega)$ becomes

$$\begin{aligned}
 (-i) \int_{-\infty}^{\infty} dt \theta(t) e^{i(\omega + i0^+ + E_m - E_n)t} &= (-i) \frac{-1}{2\pi i} \lim_{\eta \rightarrow 0^+} \int_{-\infty}^{\infty} dx \frac{1}{x + i\eta} \int_{-\infty}^{\infty} dt e^{i(\omega + i0^+ + E_m - E_n - x)t} \\
 &= \frac{1}{2\pi} \lim_{\eta \rightarrow 0^+} \int_{-\infty}^{\infty} dx \frac{1}{x + i\eta} 2\pi \delta(\omega + i0^+ + E_m - E_n - x) \\
 &= \frac{1}{\omega + E_m - E_n + i0^+} .
 \end{aligned} \tag{15.7}$$

The frequency-domain Green's function is thus

$$G_{dd}^{\sigma}(\omega) = \frac{1}{Z} \sum_{m,n} ||\langle m | O_{\sigma} | n \rangle||^2 \left(e^{-\beta E_m} + e^{-\beta E_n} \right) \frac{1}{\omega + E_m - E_n + i0^+} \quad (15.8)$$

The infinitesimally positive imaginary part in the denominator shifts the poles onto the lower half of the complex plane, leaving it analytic in the upper half: this is necessary to make the retarded Greens function causal.

1.15.2 Real and imaginary parts - The Sokhotski-Plemelj theorem

In order to write down the real and imaginary parts of the Greens function, we first prove the *Sokhotski-Plemelj* formula:

$$\lim_{\eta \rightarrow 0^+} \frac{1}{x + i\eta} = P\left(\frac{1}{x}\right) - i\pi\delta(x). \quad (15.9)$$

where $P(f(x))$ is the Cauchy principal value, defined as

$$P\left[\int_{-\infty}^{\infty} \frac{dx}{x}\right] = \int_{-\infty}^{0^-} \frac{dx}{x} + \int_{0^+}^{\infty} \frac{dx}{x}. \quad (15.10)$$

To prove the above identity, we integrate the left-hand side using a test function $f(x)$:

$$\begin{aligned} \lim_{\eta \rightarrow 0^+} \int_{-\infty}^{\infty} dx \frac{f(x)}{x + i\eta} &= \lim_{\eta \rightarrow 0^+} \int_{-\infty}^{\infty} dx \frac{f(x)(x - i\eta)}{x^2 + \eta^2} \\ &= \lim_{\eta \rightarrow 0^+} \int_{-\infty}^{\infty} dx f(x) \left[\frac{x}{x^2 + \eta^2} - i\eta \frac{1}{x^2 + \eta^2} \right] \end{aligned} \quad (15.11)$$

The first integral can be split into three parts:

$$\lim_{\eta \rightarrow 0^+} \int_{-\infty}^{\infty} dx f(x) \frac{x}{x^2 + \eta^2} = \lim_{\eta \rightarrow 0^+} \lim_{\varepsilon \rightarrow 0^+} \left[\int_{-\infty}^{-\varepsilon} dx \frac{f(x)x}{x^2 + \eta^2} + \int_{\varepsilon}^{\infty} dx \frac{f(x)x}{x^2 + \eta^2} + \int_{-\varepsilon}^{\varepsilon} dx \frac{f(x)x}{x^2 + \eta^2} \right] \quad (15.12)$$

For the first two parts, it is safe to take the limit of η , because x is always non-vanishing there. For the third part, we can approximate $f(x)$ as $f(0)$ in the neighbourhood $x \in [-\varepsilon, \varepsilon]$, $\varepsilon \rightarrow 0$. This leaves an odd function $x/(x^2 + \eta^2)$ as the integrand, being integrated over a symmetric range. The third term therefore vanishes. In total, we get

$$\lim_{\eta \rightarrow 0^+} \int_{-\infty}^{\infty} dx f(x) \frac{x}{x^2 + \eta^2} = \int_{-\infty}^{-\varepsilon} dx \frac{f(x)}{x} + \int_{\varepsilon}^{\infty} dx \frac{f(x)}{x} = P\left(\int_{-\infty}^{\infty} dx \frac{f(x)}{x}\right). \quad (15.13)$$

To evaluate the second integral of eq. 15.11, we note that the only region in which the integrand is non-zero is when $|x| \simeq 0$; there, we again approximate $f(x)$ as $f(0)$. The remaining integral can then be evaluated easily:

$$-i \lim_{\eta \rightarrow 0^+} \eta \int_{-\infty}^{\infty} dx \frac{f(x)}{x^2 + \eta^2} = -i \lim_{\eta \rightarrow 0^+} \eta f(0) \int_{-\infty}^{\infty} \frac{dx}{x^2 + \eta^2} = -i\pi f(0), \quad (15.14)$$

where we used $\int \frac{dx}{x^2 + \eta^2} = \frac{1}{\eta} \arctan(x/\eta)$. Combining eqs. 15.11, 15.13 and 15.14, we get

$$\lim_{\eta \rightarrow 0^+} \int_{-\infty}^{\infty} dx \frac{f(x)}{x + i\eta} = P \left(\int_{-\infty}^{\infty} dx \frac{f(x)}{x} \right) - i\pi \int_{-\infty}^{\infty} dx f(x) \delta(x). \quad (15.15)$$

This proves eq. 15.9.

The Sokhotski-Plemelj formula allows us to split the spectral representation into a real and an imaginary part:

$$\begin{aligned} G'_{d\sigma} &= \frac{1}{Z} \sum_{m,n} ||\langle m | O_{\sigma} | n \rangle||^2 \left(e^{-\beta E_m} + e^{-\beta E_n} \right) P \left(\frac{1}{\omega + E_m - E_n} \right) \\ G''_{d\sigma} &= -\pi \frac{1}{Z} \sum_{m,n} ||\langle m | O_{\sigma} | n \rangle||^2 \left(e^{-\beta E_m} + e^{-\beta E_n} \right) \delta(\omega + E_m - E_n). \end{aligned} \quad (15.16)$$

The spectral function, specifically, is defined as follows:

$$A_{d\sigma}(\omega) = -\frac{1}{\pi} G'_{d\sigma} = \frac{1}{Z} \sum_{m,n} ||\langle m | O_{\sigma} | n \rangle||^2 \left(e^{-\beta E_m} + e^{-\beta E_n} \right) \delta(\omega + E_m - E_n) \quad (15.17)$$

1.15.3 Zero temperature spectral function

We specialise to zero temperature by taking the limit of $\beta \rightarrow \infty$. In both the partition function as well as inside the summation, the only term that will survive is the exponential of the ground state energy E_{GS} .

$$Z \equiv \sum_m e^{-\beta E_m} \implies \lim_{\beta \rightarrow \infty} Z = d_{GS} e^{-\beta E_{GS}}, \quad E_{GS} \equiv \min \{E_n\}$$

where d_{GS} is the degeneracy of the ground state. The spectral function then simplifies to

$$\begin{aligned} A_{d\sigma} &= \frac{1}{d_{GS} e^{-\beta E_{GS}}} \sum_{m,n} ||\langle m | O_{\sigma} | n \rangle||^2 \left[e^{-\beta E_m} \delta_{E_m, E_{GS}} + e^{-\beta E_n} \delta_{E_n, E_{GS}} \right] \delta(\omega + E_m - E_n) \\ &= \frac{1}{d_{GS}} \sum_{n, n_{GS}} \left[||\langle n_{GS} | O_{\sigma} | n \rangle||^2 \delta(\omega + E_{GS} - E_n) + ||\langle n | O_{\sigma} | n_{GS} \rangle||^2 \delta(\omega - E_{GS} + E_n) \right] \end{aligned} \quad (15.18)$$

The label n_{GS} sums over all states $|n_{GS}\rangle$ with energy E_{GS} . In practise, we evaluate this expression by replacing the formal Dirac delta functions with "nascent" ones, such as a Lorentzian function with a sufficiently sharp peak.

1.15.4 Reconstructing full Green function from spectral function: Kramers-Kronig relations

As mentioned earlier, the retarded Greens function $G_R(\omega + i0^+)$ has a complex pole in the lower half of the complex plane. In the following, the frequency ω is assumed to be complex. Consider the integral

$$I(\omega) = \oint_C d\omega' \frac{G_R(\omega')}{\omega' - \omega}, \quad \omega \in \mathbb{R}. \quad (15.19)$$

The contour C encloses the entirety of the upper half of the complex plane as well as almost all points of the real line but avoids the pole at $\omega' = \omega$ by forming a semicircle about that point that extends into the upper half of the plane. Since the integrand is completely analytic in the region enclosed by the contour (recall again that G_R is analytic in the upper half), the integral evaluates to zero (by Cauchy's theorem):

$$I(\omega) = 0 . \quad (15.20)$$

One can also evaluate the integral along each part of the contour. The semicircular part contributes zero: this is because while the length of the semicircular arc goes as $|\omega'|$, the integrand vanishes faster than $1/|\omega'|$ as $|\omega'| \rightarrow \infty$ (this assumes that the Green function vanishes in the limit of $\omega \rightarrow \infty$). The part of the contour along the real axis can now be evaluated. The two parts of the contour on either side of the pole contribute

$$\int_{-\infty}^{\omega-0^+} d\omega' \frac{G_R(\omega')}{\omega' - \omega} + \int_{\omega+0^+}^{\infty} d\omega' \frac{G_R(\omega')}{\omega' - \omega} = P \int_{-\infty}^{\infty} d\omega' \frac{G_R(\omega')}{\omega' - \omega} \quad (15.21)$$

To evaluate the part of the integral along the small semicircular around $\omega' = \omega$ (that extends into the upper half), we argue that this contribution should be equal to that obtained by taking the semicircle that goes instead into the lower half but circulates in the opposite direction. Each of these contributions should, in turn, be equal to that obtained from joining these two contours. But joining these contours leads to a closed integral about the pole at $\omega' = \omega$, which is equal to $2\pi G_R(\omega)$.

$$\int_{\text{upper}} d\omega' \frac{G_R(\omega')}{\omega' - \omega} = \int_{\text{lower}} d\omega' \frac{G_R(\omega')}{\omega' - \omega} = \frac{1}{2} \oint_{C(\omega)} d\omega' \frac{G_R(\omega')}{\omega' - \omega} = i\pi G_R(\omega) . \quad (15.22)$$

Combining the last two equations, we get

$$I(\omega) = P \int_{-\infty}^{\infty} d\omega' \frac{G_R(\omega')}{\omega' - \omega} + i\pi G_R(\omega) . \quad (15.23)$$

Comparing with eq. 15.20 and separating into real part G'_R and imaginary part G''_R gives

$$P \int_{-\infty}^{\infty} d\omega' \frac{[G'_R(\omega') + iG''_R(\omega')]}{\omega' - \omega} + \pi [-G''_R(\omega) + iG'_R(\omega)] = 0 . \quad (15.24)$$

Comparing the imaginary parts gives

$$G'_R(\omega) = \frac{1}{\pi} P \int_{-\infty}^{\infty} d\omega' \frac{G''_R(\omega')}{\omega' - \omega} = -\mathcal{H} [G''_R(\omega)] , \quad (15.25)$$

where $\mathcal{H} [g(\omega)] = \frac{1}{\pi} P \int_{-\infty}^{\infty} d\omega' \frac{g(\omega')}{\omega - \omega'}$ is the Hilbert transform of the function $g(\omega)$. The spectral function was defined above as $-1/\pi$ times the imaginary part of the Greens function. Exchanging G''_R for A gives

$$G'_R(\omega) = \pi \mathcal{H} [A(\omega)] . \quad (15.26)$$

This provides a method for calculating the real part of a Greens function if the spectral function is already known.

Chapter 2

Mapping From Auxiliary Model To Lattice: The Tiling Approach

2.1 The Philosophy of Auxiliary Models

In the previous chapter, we extended the Anderson impurity model to host a correlation-driven delocalisation-localisation transition on the impurity site, from a local Fermi liquid state to a local moment state. We demonstrated that this transition captures very well the phenomenology of the Hubbard dimension in infinite dimensions (as seen from, say, dynamical mean-field theory (DMFT)). In this chapter, we extend this approach by developing a formalism to study the more realistic case of finite-dimensional interacting models by using appropriately chosen impurity models as auxiliary models. The key to an *auxiliary model method* is to construct an appropriate auxiliary model that while being tractable captures some essential property of the full lattice model [28]. This involves separating the complete degrees of freedom into a system part (H_S) which one treats exactly and the rest of the system (H_R) that one might simplify in order to be able to solve the problem. The non-trivial nature of the problem arises of course from the hybridisation H_{SR} between the system and the bath:

$$H = \begin{bmatrix} H_R & H_{RS} \\ H_{RS}^* & H_S \end{bmatrix}. \quad (1.1)$$

This separation can always be done exactly, if one does not care about tractability. For example, one can take the Hubbard model that describes electrons hopping on a lattice and interacting when two electrons reside on the same lattice site:

$$H_{\text{hub}} = -t \sum_{\langle i,j \rangle, \sigma} (c_{i,\sigma}^\dagger c_{j,\sigma} + \text{h.c.}) + U \sum_i n_{i\uparrow} n_{i\downarrow}, \quad (1.2)$$

and separate this into parts, the system being the local orbitals of a specific site l and the rest of the system being all the other sites:

$$H_S = U n_{l\uparrow} n_{l\downarrow}, \quad H_R = -t \sum'_{\langle i,j \rangle, \sigma} (c_{i,\sigma}^\dagger c_{j,\sigma} + \text{h.c.}) + U \sum'_i n_{i\uparrow} n_{i\downarrow}, \quad H_{SR} + H_{RS} = -t \sum_{i,\sigma} (c_{l,\sigma}^\dagger c_{i,\sigma} + \text{h.c.}), \quad (1.3)$$

where the primed summation is carried out over all nearest-neighbour pairs that do not involve l and the summation in $H_{SR} + H_{RS}$ is over all sites adjacent to l .

The smaller system H_S (often called the *cluster*) is typically chosen such that its eigenstates are known exactly. Progress is then made by choosing a simpler form for H_R and its coupling H_{RS} with the smaller system. This combination of the cluster and the simpler bath is then called the *auxiliary system*. A tractable auxiliary system for the Hubbard model is the single-impurity Anderson model (SIAM); the correlated impurity site represents the set of local orbitals of any given site, while the conduction bath represents the rest of the lattice sites in an approximate fashion. Such a construction is shown in fig. 2.1.

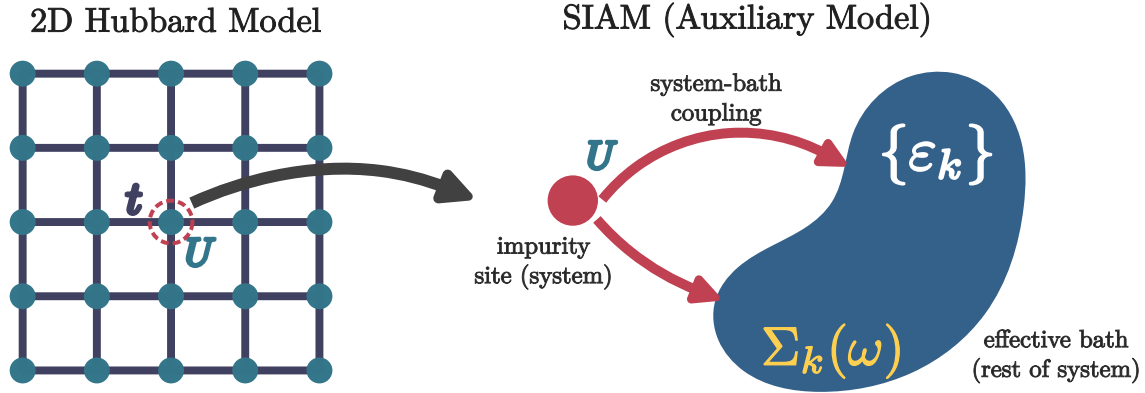


Figure 2.1: Left: Full Hubbard model lattice with onsite repulsion U^H on all sites and hopping between nearest neighbour sites with strength t^H . Right: Extraction of the auxiliary (cluster+bath) system from the full lattice. The central site on left becomes the impurity site (red) on the right (with an onsite repulsion ϵ_d), while the rest of the $N - 1$ sites on the left form a conduction bath (green circles) (with dispersion ϵ_k and correlation modelled by the self-energy $\Sigma_k(\omega)$) that hybridizes with the impurity through the coupling V .

It should be noted that any reasonable choice of the cluster and bath would break the translational symmetry of the full model: To allow computing quantities, one would need to make the bath (which is a much larger system) simpler than the cluster (which is a single site). This distinction breaks the translational symmetry of the Hubbard model. As a result, calculations that make use of the translation symmetry of the full model (such as k -space objects) will require a periodisation operation of an auxiliary model quantity.

The present work is aimed towards developing a new auxiliary model method to study correlated electronic systems. As described above, the first step is to identify the correct impurity model which can faithfully capture the physics of the lattice model we are trying to solve. The local behaviour of this impurity model should reflect the essential local physics of the lattice model. Typically, we will consider impurity model geometries where the real-space bath site connected directly to the impurity hosts some local interaction. We will henceforth refer to this site as the *zeroth site*. This model must be solved using an impurity solver. In the present work, we use the recently-developed unitary renormalisation group method [1–4, 29, 30]. The leap to the bulk model is then made by applying **manybody lattice translation operators** on the auxiliary model. This process, referred

to as *tiling* here, allows us to reconstruct the lattice model Hamiltonian from that of the auxiliary model and restores the translation invariance of the full lattice model. Once the Hamiltonians can be mapped onto one another, it is possible to relate the eigenstates and correlations functions between the auxiliary model and lattice model, using a manybody version of **Bloch's theorem**.

2.2 What Is The Correct Auxiliary Model For Correlated Electrons on a 2D Square Lattice?

As discussed in more detail in the previous chapter, it was necessary to extend the single-impurity Anderson model (SIAM) in order to obtain a phase transition on the impurity site. Without any modification, the standard Anderson model (at particle-hole symmetry) leads to the emergence of an $\omega = 0$ Kondo resonance in the impurity ($d\sigma$) spectral function at any value of the impurity correlation U_d , at sufficiently low temperatures:

$$H_{\text{SIAM}} = \sum_{k,\sigma} \varepsilon_k n_{k,\sigma} - V \sum_{\sigma} (c_{0,\sigma}^\dagger c_{d,\sigma} + \text{h.c.}) + \varepsilon_d \sum_{\sigma} n_{d,\sigma} + U_d n_{d,\uparrow} n_{d,\downarrow}. \quad (2.1)$$

The operator $c_{0,\sigma}$ acts on the local conduction bath density that couples with the impurity site. The low-energy is one of local Fermi liquid for all parameter regimes. Increasing U_d only serves to reduce the width of the central peak at the cost of the appearance of side bands at energy scales of the order of U_d , but the resonance never disappears. The extended Anderson model that we developed augments the SIAM by allowing for local correlation in the local conduction bath charge density that couples with the impurity spin:

$$H_{\text{ESIAM}} = H_{\text{SIAM}} + J_K \mathbf{S}_d \cdot \mathbf{S}_0 - \frac{U_b}{2} (n_{0,\uparrow}^2 - n_{0,\downarrow}^2); \quad (2.2)$$

tuning this bath correlation to large attractive values (see [31] for some recent findings of non-local effective attractive interactions within the Hubbard model) *does* lead to a localisation transition on the impurity where the local Fermi liquid excitations are replaced by the physics of a gapped local moment on the impurity. We note that the generation of such correlations in the bath is something that occurs in DMFT as well – the conduction bath acquires a *non-trivial self-energy* during the convergence to self-consistency.

While this modified impurity model captures well the Mott metal-insulator transition of the Hubbard model in infinite dimensions, it *lacks any knowledge of the underlying lattice geometry*, and is therefore ill-suited to address the phenomenology of more realistic low-dimensional models. Examples of such phenomenology include the pseudogap phase of the cuprates (with nodal-antinodal dichotomy), strange metal (with T -linear resistivity) and high- T_c superconducting phases observed in several families of quantum materials as well as graphene. This leads us to propose the framework of *lattice-embedded quantum impurity models*. Upon embedding the impurity site within the underlying lattice, the coupling between it and its environment must be consistent with the lowered symmetry of the lattice (see Fig. 2.2). Impurity-bath correlations calculated within such an impurity model are sensitive to the lattice geometry, and are much better suited to address the phenomenology mentioned above.

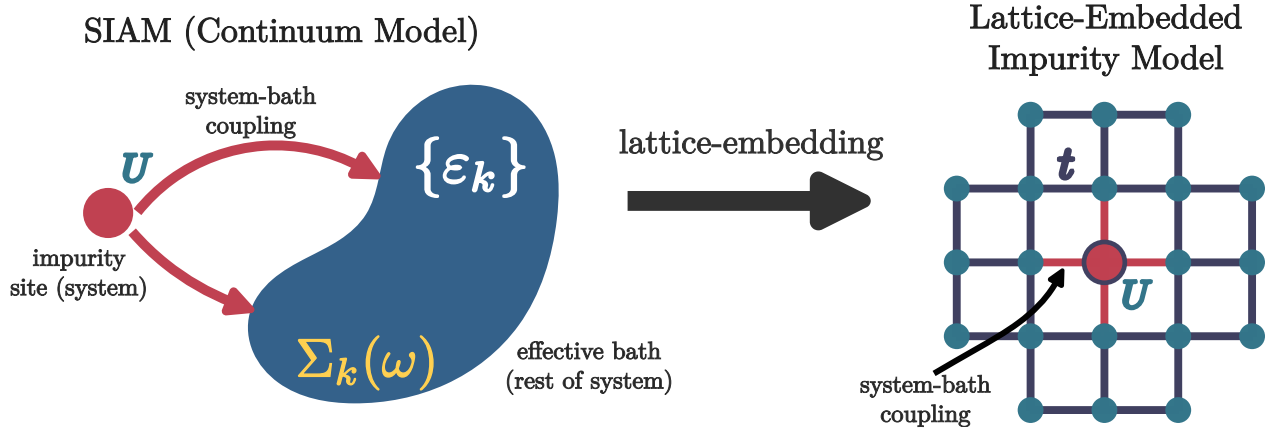


Figure 2.2

As a concrete example, we take the extended SIAM defined in eq. 2.2, and write down its lattice-embedded form on a 2D square lattice, with the impurity placed at the site r_d . This is done by allowing the impurity to hybridise with the four nearest-neighbours, through V and J_K . The same four sites also carry the local bath interaction. The complete auxiliary model is

$$\begin{aligned}
 H = & -t \sum_{\langle i,j \rangle, \sigma} (c_{i,\sigma}^\dagger c_{j,\sigma} + \text{h.c.}) + \sum_{\langle i,r_d \rangle} \left[-\frac{V}{Z} \sum_{\sigma} (c_{r_d,\sigma}^\dagger c_{i,\sigma} + \text{h.c.}) + \frac{J_K}{Z} \mathbf{S}_{r_d} \cdot \mathbf{S}_i - \frac{W}{2Z} (n_{i,\uparrow}^2 - n_{i,\downarrow}^2) \right] \\
 & + \varepsilon_d \sum_{\sigma} n_{r_d,\sigma} + U_d n_{r_d,\uparrow} n_{r_d,\downarrow},
 \end{aligned} \tag{2.3}$$

where Z is the coordination number of the lattice, and the second summation $\sum_{\langle i,r_d \rangle}$ is carried out over all sites i that are nearest-neighbour to the site r_d . Note that the one-particle hybridisation V and the spin-exchange coupling J_K have their *symmetry lowered* from the s -wave spherical symmetry of the continuum impurity models to a C_4 -symmetry consistent with the underlying lattice. This feature is crucial in explaining the phenomenology of the various low-dimensional materials.

Various enhancements can be applied to such a model. The lattice can be modified to study different materials. More complicated geometries such as a two-layer model can be studied using a two-impurity auxiliary model, where each impurity couples with its own conduction bath as well as to each other.

2.3 Mapping Auxiliary Model To Lattice Model

In this subsection, we provide an explicit example of constructing a lattice model from a lattice-embedded impurity model. This will involve the use of manybody lattice translation operators, so we begin with a brief discussion of them.

2.3.1 Manybody Translation Operators

In order to create the bulk model, we need to translate the local auxiliary model over the entire lattice. For this, we define *many-particle global* translation operators $T(\mathbf{a})$ that translate all positions by a vector $\mathbf{a}$. In terms of manybody states and operators, their action is defined as

$$\begin{aligned} T^\dagger(\mathbf{a}) |\mathbf{r}_1, \mathbf{r}_2, \dots\rangle &= |\mathbf{r}_1 + \mathbf{a}\rangle \otimes |\mathbf{r}_2 + \mathbf{a}\rangle \dots \otimes |\mathbf{r}_n + \mathbf{a}\rangle \\ T^\dagger(\mathbf{a}) O(\mathbf{r}_1, \mathbf{r}_2, \dots) T(\mathbf{a}) &= O(\mathbf{r}_1 + \mathbf{a}, \mathbf{r}_2 + \mathbf{a}, \dots) , \end{aligned} \quad (3.1)$$

where $|\mathbf{r}_1, \mathbf{r}_2, \dots\rangle$ is a state in the manyparticle Fock-space basis with the particles localised at the specified positions. For example, for a local fermionic creation operator $c^\dagger(\mathbf{r})$, we have

$$T^\dagger(\mathbf{a}) c^\dagger(\mathbf{r}) T(\mathbf{a}) = c^\dagger(\mathbf{r} + \mathbf{a}) . \quad (3.2)$$

It acts similarly on the auxiliary model Hamiltonian:

$$T^\dagger(\mathbf{a}) H_{\text{aux}}(\mathbf{r}_d) T(\mathbf{a}) = H_{\text{aux}}(\mathbf{r}_d + \mathbf{a}) , \quad (3.3)$$

translating all sites by the vector $\mathbf{a}$. By introducing the Fourier transform to momentum space,

$$|\mathbf{r}_1, \mathbf{r}_2, \dots\rangle = \otimes_{j=1}^N \int d\mathbf{k}_j e^{i\mathbf{r}_j \cdot \mathbf{k}_j} |\mathbf{k}_j\rangle , \quad (3.4)$$

it is easy to see that the total momentum states are eigenstates of the global translation operators:

$$\begin{aligned} T^\dagger(\mathbf{a}) |\mathbf{k}_1, \mathbf{k}_2, \dots\rangle &= \otimes_{j=1}^N \int d\mathbf{r}_j e^{-i\mathbf{r}_j \cdot \mathbf{k}_j} T^\dagger(\mathbf{a}) |\mathbf{r}_j\rangle , \\ &= \otimes_{j=1}^N \int d\mathbf{r}_j e^{-i\mathbf{r}_j \cdot \mathbf{k}_j} |\mathbf{r}_j + \mathbf{a}\rangle \\ &= \otimes_{j=1}^N e^{i\mathbf{a} \cdot \mathbf{k}_j} \int d\mathbf{r}_j e^{-i\mathbf{r}_j \cdot \mathbf{k}_j} |\mathbf{r}_j\rangle \\ &= e^{i\mathbf{a} \cdot \mathbf{k}_{\text{tot}}} |\mathbf{k}_1, \mathbf{k}_2, \dots\rangle , \end{aligned} \quad (3.5)$$

where $\mathbf{k}_{\text{tot}} = \sum_j \mathbf{k}_j$ is the total momentum. This gives a compact form for the translation operator:

$$T^\dagger(\mathbf{a}) = e^{i\mathbf{a} \cdot \hat{\mathbf{k}}_{\text{tot}}} , \quad (3.6)$$

where $\hat{\mathbf{k}}_{\text{tot}}$ is the total momentum operator.

We end this section by calculating the effect of the translation operators on the k -space annihilation operators. We have

$$c(\mathbf{k}) = \frac{1}{\sqrt{N}} \sum_{\mathbf{r}} e^{-i\mathbf{k} \cdot \mathbf{r}} c(\mathbf{r}) . \quad (3.7)$$

Applying the translation operator to this decomposition, we get

$$\begin{aligned} T(\vec{a}) c(\mathbf{k}) T^\dagger(\vec{a}) &= \frac{1}{\sqrt{N}} \sum_{\mathbf{r}} e^{-i\mathbf{k} \cdot \mathbf{r}} T(\vec{a}) c(\mathbf{r}) T^\dagger(\vec{a}) \\ &= \frac{1}{\sqrt{N}} \sum_{\mathbf{r}} e^{-i\mathbf{k} \cdot \mathbf{r}} c(\mathbf{r} - \vec{a}) \\ &= e^{-i\mathbf{k} \cdot \vec{a}} c(\mathbf{k}) . \end{aligned} \quad (3.8)$$

2.3.2 Periodisation of Auxiliary Model Into Extended Hubbard Model

We recall the lattice-embedded impurity model defined in eq. 2.3. As discussed above, the route to restoring translation symmetry and reconstructing the lattice model is to translate the auxiliary model using manybody translation operators. Accordingly, we identify the “unit cell” of translation. Since we expect the impurity site to capture most of the physics arising from the correlated electrons, a good choice for the unit cell is the impurity site and its surrounding correlated environment. In order to retain information of the lattice geometry, we convert the hybridisation and bath interaction terms from real space to k -space. The Hamiltonian for the unit cell is therefore

$$H_{\text{clust}}(\mathbf{r}_d) = - \sum_{\mathbf{k}, \sigma} e^{i\mathbf{k} \cdot \mathbf{r}_d} V(\mathbf{k}) (c_{\mathbf{r}_d, \sigma}^\dagger c_{\mathbf{k}, \sigma} + \text{h.c.}) + \sum_{\mathbf{k}_1, \mathbf{k}_2} e^{i(\mathbf{k}_1 - \mathbf{k}_2) \cdot \mathbf{r}_d} J_K(\mathbf{k}_1, \mathbf{k}_2) \mathbf{S}_{\mathbf{r}_d} \cdot \mathbf{S}_{\mathbf{k}_1, \mathbf{k}_2} + \varepsilon_d n_{\mathbf{r}_d} \\ + U_d n_{\mathbf{r}_d, \uparrow} n_{\mathbf{r}_d, \downarrow} - \frac{1}{2} \sum_{\{\mathbf{k}_i\}, \sigma, \sigma'} \sigma \sigma' e^{i(\mathbf{k}_1 - \mathbf{k}_2 + \mathbf{k}_3 - \mathbf{k}_4) \cdot \mathbf{r}_d} W(\{\mathbf{k}_i\}) c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \sigma'}^\dagger c_{\mathbf{k}_4, \sigma'} , \quad (3.9)$$

where $\mathbf{S}_{\mathbf{k}_1, \mathbf{k}_2} = \sum_{\alpha, \beta} \sigma_{\alpha, \beta} c_{\mathbf{k}_1, \alpha}^\dagger c_{\mathbf{k}_2, \beta}$, and the Fourier transforms are defined by the following relations:

$$\sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}_d} V(\mathbf{k}) c_{\mathbf{r}_d, \sigma}^\dagger c_{\mathbf{k}, \sigma} = \frac{V}{Z} \sum_{\mathbf{r} \rightarrow \text{NN of } \mathbf{r}_d} c_{\mathbf{r}_d, \sigma}^\dagger c_{\mathbf{r}, \sigma} , \\ \sum_{\mathbf{k}_1, \mathbf{k}_2} e^{i(\mathbf{k}_1 - \mathbf{k}_2) \cdot \mathbf{r}_d} J_K(\mathbf{k}_1, \mathbf{k}_2) \vec{S}_{\mathbf{r}_d} \cdot \vec{S}_{\mathbf{k}_1, \mathbf{k}_2} = \frac{J_K}{Z} \sum_{\mathbf{r} \rightarrow \text{NN of } \mathbf{r}_d} \mathbf{S}_{\mathbf{r}_d} \cdot \mathbf{S}_{\mathbf{r}} , \quad (3.10) \\ \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}_1, \mathbf{q}_2} e^{i(\mathbf{k}_1 - \mathbf{k}_2 + \mathbf{k}_3 - \mathbf{k}_4) \cdot \mathbf{r}_d} W(\{\mathbf{k}_i\}) c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \sigma'}^\dagger c_{\mathbf{k}_4, \sigma'} = \frac{W}{2Z} \sum_{\mathbf{r} \rightarrow \text{NN of } \mathbf{r}_d} n_{\mathbf{r}, \sigma} n_{\mathbf{r}, \sigma'} ,$$

We now translate this unit cell to all sites of the lattice, and reconstruct a translationally-invariant correlated lattice model:

$$H_{\text{latt}} = \sum_{\mathbf{r}} T^\dagger(\mathbf{r} - \mathbf{r}_d) H_{\text{clust}}(\mathbf{r}_d) T(\mathbf{r} - \mathbf{r}_d) . \quad (3.11)$$

We consider the effect of the translation operations on each part of the Hamiltonian. The purely local terms translate directly into a chemical potential term and a Hubbard interaction term:

$$\sum_{\mathbf{r}} T^\dagger(\mathbf{r} - \mathbf{r}_d) (\varepsilon_d \sum_{\sigma} n_{\mathbf{r}_d, \sigma} + U_d n_{\mathbf{r}_d, \uparrow} n_{\mathbf{r}_d, \downarrow}) T(\mathbf{r} - \mathbf{r}_d) = \varepsilon_d \sum_{\mathbf{r}, \sigma} n_{\mathbf{r}, \sigma} + U_d \sum_{\mathbf{r}} n_{\mathbf{r}, \uparrow} n_{\mathbf{r}, \downarrow} . \quad (3.12)$$

The one-particle hybridisation term maps onto a tight-binding hopping term:

$$\sum_{\mathbf{r}} T^\dagger(\mathbf{r} - \mathbf{r}_d) \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}_d} V(\mathbf{k}) c_{\mathbf{r}_d, \sigma}^\dagger c_{\mathbf{k}, \sigma} T(\mathbf{r} - \mathbf{r}_d) = \sum_{\mathbf{r}} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}} V(\mathbf{k}) c_{\mathbf{r}, \sigma}^\dagger c_{\mathbf{k}, \sigma} \\ = \frac{V}{Z} \sum_{\mathbf{r}} \sum_{\mathbf{r}' \rightarrow \text{NN of } \mathbf{r}} c_{\mathbf{r}, \sigma}^\dagger c_{\mathbf{r}', \sigma} \quad (3.13) \\ = \frac{2V}{Z} \sum_{\langle \mathbf{r}, \mathbf{r}' \rangle} c_{\mathbf{r}, \sigma}^\dagger c_{\mathbf{r}', \sigma} ,$$

where we used eq. 3.8 and the first equation from 3.10, and the fact that each nearest-neighbour pair $\mathbf{r}, \mathbf{r}'$ appears twice in the summation.

The Kondo terms map onto a nearest-neighbour spin-exchange term:

$$\begin{aligned} \sum_{\mathbf{r}, \mathbf{k}_1, \mathbf{k}_2} T^\dagger(\mathbf{r} - \mathbf{r}_d) e^{i(\mathbf{k}_1 - \mathbf{k}_2) \cdot \mathbf{r}_d} J_K(\mathbf{k}_1, \mathbf{k}_2) \vec{S}_{\mathbf{r}_d} \cdot \vec{S}_{\mathbf{k}_1, \mathbf{k}_2} T(\mathbf{r} - \mathbf{r}_d) &= \sum_{\mathbf{r}, \mathbf{k}_1, \mathbf{k}_2} e^{i(\mathbf{k}_1 - \mathbf{k}_2) \cdot \mathbf{r}} J_K(\mathbf{k}_1, \mathbf{k}_2) \vec{S}_{\mathbf{r}} \cdot \vec{S}_{\mathbf{k}_1, \mathbf{k}_2} \\ &= \frac{J_K}{Z} \sum_{\mathbf{r}} \sum_{\mathbf{r}' \rightarrow \text{NN of } \mathbf{r}} \vec{S}_{\mathbf{r}} \cdot \vec{S}_{\mathbf{r}'} , \\ &= \frac{2J_K}{Z} \sum_{\langle \mathbf{r}, \mathbf{r}' \rangle} \vec{S}_{\mathbf{r}} \cdot \vec{S}_{\mathbf{r}'} , \end{aligned} \quad (3.14)$$

where we used the second equation from 3.10. Finally, the bath interaction term W maps onto another Hubbard term

$$\begin{aligned} &\sum_{\mathbf{r}, \mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} T^\dagger(\mathbf{r} - \mathbf{r}_d) e^{i(\mathbf{k}_1 - \mathbf{k}_2 + \mathbf{k}_3 - \mathbf{k}_4) \cdot \mathbf{r}_d} W(\{\mathbf{k}_i\}) c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \sigma'}^\dagger c_{\mathbf{k}_4, \sigma'} T(\mathbf{r} - \mathbf{r}_d) \\ &= \sum_{\mathbf{r}, \mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} e^{i(\mathbf{k}_1 - \mathbf{k}_2 + \mathbf{k}_3 - \mathbf{k}_4) \cdot \mathbf{r}} W(\{\mathbf{k}_i\}) c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \sigma'}^\dagger c_{\mathbf{k}_4, \sigma'} \\ &= \frac{W}{2Z} \sum_{\mathbf{r}} \sum_{\mathbf{r}' \rightarrow \text{NN of } \mathbf{r}} n_{\mathbf{r}', \sigma} n_{\mathbf{r}', \sigma'} \\ &= \frac{W}{2} \sum_{\mathbf{r}} n_{\mathbf{r}, \sigma} n_{\mathbf{r}, \sigma'} , \end{aligned} \quad (3.15)$$

where we used the fact that each point $\mathbf{r}'$ (which we relabel as $\mathbf{r}$) appears Z times in the summation.

The complete lattice Hamiltonian, after applying the translation operators, is

$$H_{\text{latt}} = \varepsilon_d \sum_{\mathbf{r}, \sigma} n_{\mathbf{r}, \sigma} + U_d \sum_{\mathbf{r}} n_{\mathbf{r}, \uparrow} n_{\mathbf{r}, \downarrow} - \frac{2V}{Z} \sum_{\langle \mathbf{r}, \mathbf{r}' \rangle} c_{\mathbf{r}, \sigma}^\dagger c_{\mathbf{r}', \sigma} + \frac{2J_K}{Z} \sum_{\langle \mathbf{r}, \mathbf{r}' \rangle} \vec{S}_{\mathbf{r}} \cdot \vec{S}_{\mathbf{r}'} - \frac{W}{2} \sum_{\mathbf{r}} (n_{\mathbf{r}, \uparrow} - n_{\mathbf{r}, \downarrow})^2 . \quad (3.16)$$

This is clearly an extended Hubbard model, with both 1-particle and spin-exchange nearest-neighbour hybridisation. The general form of the model is

$$H_{\text{ex-Hub}} = -t \sum_{\langle i, j \rangle, \sigma} (c_{i, \sigma}^\dagger c_{j, \sigma} + \text{h.c.}) + J \sum_{\langle i, j \rangle} \mathbf{S}_i \cdot \mathbf{S}_j + U \sum_i n_{i, \uparrow} n_{i, \downarrow} - \mu \sum_{i, \sigma} n_{i, \sigma} , \quad (3.17)$$

such that the effective couplings resulting from the tiling operation is

$$t = \frac{2V}{Z}, \quad J = \frac{2J_K}{Z}, \quad U = U_d - W, \quad \mu = -\varepsilon_d + \frac{W}{2} . \quad (3.18)$$

2.3.3 Form Of The Eigenstates: Bloch's Theorem

The tiled Hamiltonian defined in eq. 3.11 is manifestly symmetric under discrete translation, given that it's expressed in terms of the translation operators. The fact that the Hamiltonian H_{latt} commutes

with the manybody translation operator implies that the total crystal momentum is a conserved quantity. We will now use this to write down a “tiling” relation between the eigenstates of the auxiliary model and those of the lattice model.

In the tight-binding approach to lattice problems, the full Hamiltonian is obtained by translating the dynamics of the localised orbitals at each site over the entire lattice. The full eigenstate $|\Psi\rangle$ is obtained by constructing linear combinations of the eigenstates $|\psi(i)\rangle$ of the local Hamiltonians. According to Bloch’s theorem, $|\Psi\rangle$ is of the form $|\Psi_{n,K}\rangle = \sum_i e^{i\mathbf{K}\cdot\mathbf{r}_i} |\psi_n(i)\rangle$, where $\mathbf{K}$ is the single-particle crystal momentum, n is the band-index that labels various states within the local orbital, and $\mathbf{r}_i$ sums over the positions of the local Hamiltonians. Bloch’s theorem ensures that eigenstates satisfy the following relation under a translation operation by an arbitrary number of lattice spacings $n\mathbf{a}$:

$$T^\dagger(l\mathbf{a}) |\Psi_{n,K}\rangle = \sum_i e^{i\mathbf{K}\cdot\mathbf{r}_i} |\psi_n(i+l)\rangle = e^{-il\mathbf{K}\cdot\mathbf{a}} |\Psi_{n,K}\rangle . \quad (3.19)$$

As discussed earlier, the eigenstates $|\Psi_{n,K}\rangle$ of the lattice Hamiltonian obtained using eq. 3.11 enjoy the discrete translation symmetry of the lattice, and therefore enjoy a *many-body* Bloch’s theorem [32]. This means that the *local* eigenstates $|\psi_n(\mathbf{r}_d)\rangle$ (with the impurity located at an arbitrary position $\mathbf{r}_d$) of the unit cell auxiliary model Hamiltonian $H_{\text{clust}}(\mathbf{r}_d)$ defined in eq. 3.9 can be used to construct eigenstates of the lattice Hamiltonian. The index $n(= 0, 1, \dots)$ in the subscript indicates that it is the n^{th} eigenstate of the auxiliary model. The following combination of the auxiliary model eigenstates satisfies a many-particle equivalent of Bloch’s theorem [32]:

$$|\Psi_s\rangle \equiv |\Psi_{K,n}\rangle = \frac{1}{\sqrt{N}} \sum_{\mathbf{r}_d} e^{i\mathbf{K}\cdot\mathbf{r}_d} |\psi_n(\mathbf{r}_d)\rangle , \quad (3.20)$$

where N is the total number of lattice sites and $\mathbf{r}_d$ is summed over all lattice spacings. The set of $n = 0$ states form the lowest band in the spectrum of the lattice, while higher values of n produce the more energetic bands.

2.4 Properties of the Bloch states

2.4.1 Translation invariance

It is easy to verify that $\Psi_{\vec{k}}(\{\mathbf{r}_k\})$ transforms like Bloch functions under translation by a displacement $\mathbf{r}$:

$$\Psi_{\vec{k}}(\{\mathbf{r}_k + \mathbf{r}\}) = \frac{1}{\lambda_{|\vec{k}\rangle} \mathcal{Z}N} \sum_{\mathbf{r}_i, \mathbf{a}} e^{i\vec{k}\cdot\vec{R}_i} \psi_{\text{aux}}(\mathbf{r}_i - \mathbf{r}, \mathbf{a}) = \frac{1}{\lambda_{|\vec{k}\rangle} \mathcal{Z}N} \sum_{\mathbf{r}_j, \mathbf{a}} e^{i\vec{k}\cdot(\vec{R}_j + \mathbf{r})} \psi_{\text{aux}}(\mathbf{r}_j, \mathbf{a}) = e^{i\vec{k}\cdot\vec{R}} \Psi_{\vec{k}}(\{\mathbf{r}_k\}) \quad (4.1)$$

In the last equation, we transformed $\mathbf{r}_i \rightarrow \mathbf{r}_j = \mathbf{r}_i - \mathbf{r}$. Note that the argument $\mathbf{a}$ does not change under a translation of the system, because that vector always represents the difference between the impurity lattice position and its nearest neighbours, irrespective of the absolute position of the impurity.

The wavefunction can be even brought into the familiar Bloch function form:

$$\Psi_{\vec{k}}^n(\{\mathbf{r}_k\}) = \sum_{\mathbf{r}_i, \mathbf{a}} \frac{e^{i\vec{k} \cdot \vec{R}_i}}{\lambda_{|\vec{k}\rangle} \mathcal{Z} N} \psi_{\text{aux}}(\{\mathbf{r}_d - \mathbf{r}_i, \mathbf{r}_0 - \mathbf{r}_i - \mathbf{a}\}) = \frac{e^{i\vec{k} \cdot \frac{1}{N} \sum_k \vec{r}_k}}{\lambda_{|\vec{k}\rangle} \mathcal{Z} N} \sum_{\mathbf{r}_i, \mathbf{a}} e^{-i\vec{k} \cdot (\frac{1}{N} \sum_k \mathbf{r}_k - \vec{R}_i)} \psi_{\text{aux}}(\{\mathbf{r}_d - \mathbf{r}_i, \mathbf{r}_0 - \mathbf{r}_i - \mathbf{a}\}) = e^{i\vec{k} \cdot \vec{r}_{\text{COM}}} \eta_{\vec{k}}(\{\mathbf{r}_k\}) \quad (4.2)$$

where $\mathbf{r}_{\text{COM}} = \frac{1}{N} \sum_k \mathbf{r}_k$ is the center-of-mass coordinate and $\eta_{\vec{k}}(\{\mathbf{r}_k\}) = \frac{1}{\lambda_{|\vec{k}\rangle} \mathcal{Z} N} \sum_{\mathbf{r}_i} e^{-i\vec{k} \cdot (\frac{1}{N} \sum_k \mathbf{r}_k - \vec{R}_i)} \psi_{\text{aux}}^n(\{\mathbf{r}_k - \mathbf{r}_i\})$ is the translation symmetric function. This form of the eigenstate allows the interpretation that tuning the Bloch momentum $\vec{k}$ corresponds to a translation of the center of mass of the system (or in this case, of the auxiliary models that comprise the system).

2.4.2 Orthonormality

It is straightforward to show that these states form an orthonormal basis. We start by writing down the inner product of two distinct such states:

$$\begin{aligned} \langle \Psi_{\vec{k}'} | \Psi_{\vec{k}} \rangle &= \frac{1}{\lambda_{|\vec{k}'\rangle}^* \lambda_{|\vec{k}\rangle} \mathcal{Z}^2 N^2} \sum_{\mathbf{r}_i, \mathbf{r}_j, \mathbf{a}, \mathbf{a}'} e^{i(\vec{k} \cdot \vec{R}_i - \vec{k}' \cdot \vec{R}_j)} \langle \psi_{\text{aux}}(\mathbf{r}_j, \mathbf{a}') | \psi_{\text{aux}}(\mathbf{r}_i, \mathbf{a}) \rangle \\ &= \frac{1}{\lambda_{|\vec{k}'\rangle}^* \lambda_{|\vec{k}\rangle} \mathcal{Z}^2 N^2} \sum_{\mathbf{r}_i, \mathbf{r}_j, \mathbf{a}, \mathbf{a}'} e^{i(\vec{k} \cdot \vec{R}_i - \vec{k}' \cdot \vec{R}_j)} \langle \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}') T^\dagger(\mathbf{r}_0 - \mathbf{r}_j) T(\mathbf{r}_0 - \mathbf{r}_i) | \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}) \rangle \end{aligned} \quad (4.3)$$

At this point, we insert a complete basis of momentum eigenkets $1 = \sum_{\vec{q}} |\vec{q}\rangle \langle \vec{q}|$ to resolve the translation operators:

$$\begin{aligned} \langle \Psi_{\vec{k}'} | \Psi_{\vec{k}} \rangle &= \frac{1}{\lambda_{|\vec{k}'\rangle}^* \lambda_{|\vec{k}\rangle} \mathcal{Z}^2 N^2} \sum_{\mathbf{r}_i, \mathbf{r}_j, \mathbf{a}, \mathbf{a}', \vec{q}} e^{i(\vec{k} \cdot \vec{R}_i - \vec{k}' \cdot \vec{R}_j)} \langle \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}') T(\mathbf{r}_j - \mathbf{r}_i) | \vec{q} \rangle \langle \vec{q} | \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}) \rangle \\ &= \frac{1}{\lambda_{|\vec{k}'\rangle}^* \lambda_{|\vec{k}\rangle} \mathcal{Z}^2 N^2} \sum_{\mathbf{r}_i, \mathbf{r}_j, \mathbf{a}, \mathbf{a}', \vec{q}} e^{i(\vec{k} \cdot \vec{R}_i - \vec{k}' \cdot \vec{R}_j)} \langle \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}') e^{i\vec{q} \cdot (\mathbf{r}_j - \mathbf{r}_i)} | \vec{q} \rangle \langle \vec{q} | \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}) \rangle \\ &= \frac{1}{|\lambda_{|\vec{k}\rangle}|^2 \mathcal{Z}^2} \delta_{\vec{k}, \vec{k}'} \sum_{\mathbf{a}, \mathbf{a}'} \langle \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}') | \vec{k} \rangle \langle \vec{k} | \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}) \rangle \end{aligned} \quad (4.4)$$

At the last step, we used the summation form of the Kronecker delta function: $\sum_{\mathbf{r}_i} e^{i\mathbf{r}_i \cdot (\vec{k} - \vec{q})} \sum_{\mathbf{r}_j} e^{i\mathbf{r}_j \cdot (\vec{q} - \vec{k}')} = N \delta_{\vec{k}, \vec{q}} N \delta_{\vec{q}, \vec{k}'}$. The final remaining step is to identify that the inner products inside the summation are actually independent of the direction $\mathbf{a}, \mathbf{a}'$ of the connecting vector, and both are equal to the normalisation factor $\lambda_{|\vec{k}\rangle}$:

$$\langle \Psi_{\vec{k}'} | \Psi_{\vec{k}} \rangle = \frac{1}{|\lambda_{|\vec{k}\rangle}|^2 \mathcal{Z}^2} \delta_{\vec{k}, \vec{k}'} \sum_{\mathbf{a}, \mathbf{a}'} |\lambda_{|\vec{k}\rangle}|^2 = \frac{1}{|\lambda_{|\vec{k}\rangle}|^2 \mathcal{Z}^2} \delta_{\vec{k}, \vec{k}'} w^2 |\lambda_{|\vec{k}\rangle}|^2 = \delta_{\vec{k}, \vec{k}'} \quad (4.5)$$

This concludes the proof of orthonormality.

2.4.3 Eigenstates

To demonstrate that these are indeed eigenstates of the bulk Hamiltonian, we will calculate the matrix elements of the full Hamiltonian between these states:

$$\begin{aligned}
 \langle \Psi_{\vec{k}} | H_{H-H} | \Psi_{\vec{k}} \rangle &= \frac{1}{\lambda_{|\vec{k}'\rangle}^* \lambda_{|\vec{k}\rangle} \mathcal{Z}^2 N^2} \sum_{\mathbf{r}_i, \mathbf{r}_j, \mathbf{r}_k, \mathbf{a}, \mathbf{a}'} e^{i(\vec{k} \cdot \vec{R}_i - \vec{k}' \cdot \vec{R}_j)} \langle \psi_{\text{aux}}(\mathbf{r}_j, \mathbf{a}') | H_{\text{aux}}(\mathbf{r}_k) | \psi_{\text{aux}}(\mathbf{r}_i, \mathbf{a}) \rangle \\
 &= \frac{1}{\lambda_{|\vec{k}'\rangle}^* \lambda_{|\vec{k}\rangle} \mathcal{Z}^2 N^2} \sum_{\mathbf{r}_i, \mathbf{r}_j, \mathbf{r}_k, \mathbf{a}, \mathbf{a}'} e^{i(\vec{k} \cdot \vec{R}_i - \vec{k}' \cdot \vec{R}_j)} \langle \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}') | T_{\mathbf{r}_0 - \mathbf{r}_j}^\dagger T_{\mathbf{r}_0 - \mathbf{r}_k} H_{\text{aux}}(\mathbf{r}_0) T_{\mathbf{r}_0 - \mathbf{r}_k}^\dagger T_{\mathbf{r}_0 - \mathbf{r}_i} | \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}) \rangle \\
 &= \frac{1}{\lambda_{|\vec{k}'\rangle}^* \lambda_{|\vec{k}\rangle} \mathcal{Z}^2 N^2} \sum_{\mathbf{r}_i, \mathbf{r}_j, \mathbf{r}_k, \mathbf{a}, \mathbf{a}'} e^{i(\vec{k} \cdot \vec{R}_i - \vec{k}' \cdot \vec{R}_j)} \langle \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}') | T_{\mathbf{r}_k - \mathbf{r}_j}^\dagger H_{\text{aux}}(\mathbf{r}_0) T_{\mathbf{r}_k - \mathbf{r}_i} | \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}) \rangle
 \end{aligned} \tag{4.6}$$

To resolve the translation operators, we will now insert the momentum eigenstates $|\vec{k}\rangle$ of the full model: $1 = \sum_{\vec{q}} |\vec{q}\rangle \langle \vec{q}|$ in between the operators.

$$\begin{aligned}
 \langle \Psi_{\vec{k}'} | H_{H-H} | \Psi_{\vec{k}} \rangle &= \sum_{\substack{\mathbf{r}_i, \mathbf{r}_j, \mathbf{r}_k, \\ \mathbf{a}, \mathbf{a}', \vec{q}, \vec{q}'}} \frac{e^{i(\vec{k} \cdot \vec{R}_i - \vec{k}' \cdot \vec{R}_j)}}{\lambda_{|\vec{k}'\rangle}^* \lambda_{|\vec{k}\rangle} \mathcal{Z}^2 N^2} \langle \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}') | \vec{q}' \rangle \langle \vec{q}' | T_{\mathbf{r}_k - \mathbf{r}_j}^\dagger H_{\text{aux}}(\mathbf{r}_0) T_{\mathbf{r}_k - \mathbf{r}_i} | \vec{q} \rangle \langle \vec{q} | \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}) \rangle \\
 &= \sum_{\substack{\mathbf{r}_i, \mathbf{r}_j, \mathbf{r}_k, \\ \mathbf{a}, \mathbf{a}', \vec{q}, \vec{q}'}} \frac{e^{i(\vec{k} \cdot \vec{R}_i - \vec{k}' \cdot \vec{R}_j)}}{\lambda_{|\vec{k}'\rangle}^* \lambda_{|\vec{k}\rangle} \mathcal{Z}^2 N^2} \langle \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}') | \vec{q}' \rangle e^{-i\vec{q}' \cdot (\mathbf{r}_k - \mathbf{r}_j)} e^{i\vec{q} \cdot (\mathbf{r}_k - \mathbf{r}_i)} \langle \vec{q}' | H_{\text{aux}}(\mathbf{r}_0) | \vec{q} \rangle \langle \vec{q} | \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}) \rangle \\
 &= \frac{N}{\lambda_{|\vec{k}'\rangle}^* \lambda_{|\vec{k}\rangle} \mathcal{Z}^2} \sum_{\mathbf{a}, \mathbf{a}', \vec{q}, \vec{q}'} \delta_{\vec{k}, \vec{q}} \delta_{\vec{q}, \vec{q}'} \delta_{\vec{q}', \vec{k}} \langle \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}') | \vec{q}' \rangle \langle \vec{q}' | H_{\text{aux}}(\mathbf{r}_0) | \vec{q} \rangle \langle \vec{q} | \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}) \rangle \\
 &= \frac{N}{|\lambda_{|\vec{k}\rangle}|^2 \mathcal{Z}^2} \delta_{kk'} \sum_{\mathbf{a}, \mathbf{a}'} \langle \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}') | \vec{k} \rangle \langle \vec{k} | H_{\text{aux}}(\mathbf{r}_0) | \vec{k} \rangle \langle \vec{k} | \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}) \rangle \\
 &= \frac{N}{|\lambda_{|\vec{k}\rangle}|^2 \mathcal{Z}^2} \delta_{kk'} \langle \vec{k} | H_{\text{aux}}(\mathbf{r}_0) | \vec{k} \rangle \underbrace{\sum_{\mathbf{a}, \mathbf{a}'} \langle \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}') | \vec{k} \rangle \langle \vec{k} | \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}) \rangle}_{w^2 |\lambda_{|\vec{k}\rangle}|^2} \\
 &= N \delta_{kk'} \langle \vec{k} | H_{\text{aux}}(\mathbf{r}_0) | \vec{k} \rangle
 \end{aligned} \tag{4.7}$$

The momentum eigenket $|\vec{k}\rangle$ can be expressed as a sum over position eigenkets at varying distances from $\mathbf{r}_0$. This form allows a systematic method of improving the energy eigenvalue estimate, because the interacting in the impurity model is extremely localised, so the overlap between two auxiliary models decreases rapidly with distance. The majority of the contribution will come from the state at $\mathbf{r}_0$, and improvements are then made by considering auxiliary models at further distances.

2.5 Mapping Greens Functions from Auxiliary Model to Lattice Model

2.5.1 One-particle k -space Greens functions

In the previous part, we proposed a form for the eigenstates $|\Psi_s\rangle$ of the bulk Hamiltonian in terms of the ground-states $|\psi_n\rangle$ of the auxiliary models. In this section, we will relate one-particle Greens functions of the lattice model to those of the auxiliary model. We define the retarded time-domain lattice k -space Greens function at zero temperature as

$$G(\mathbf{k}\sigma; t) = -i\theta(t) \langle \Psi_{\text{gs}} | \{c_{\mathbf{k}\sigma}(t), c_{\mathbf{k}\sigma}^\dagger\} | \Psi_{\text{gs}} \rangle . \quad (5.1)$$

where $|\Psi_{\text{gs}}\rangle$ is the lattice groundstate. Time evolution of the operators is governed by the lattice Hamiltonian H_{latt} :

$$c_{\mathbf{k}\sigma}(t) = e^{itH_{\text{latt}}} c_{\mathbf{k}\sigma} e^{-itH_{\text{latt}}} . \quad (5.2)$$

By using eq. 3.8 and the fact that the translation operators commute with H_{latt} , we have the relation

$$T(\vec{a}) c(\mathbf{k})(t) T^\dagger(\vec{a}) = e^{-i\mathbf{k}\cdot\vec{a}} c(\mathbf{k})(t) , \quad (5.3)$$

which combined with eq. 3.8 gives

$$T(\mathbf{a}) c(\mathbf{k})(t) c^\dagger(\mathbf{k}) T^\dagger(\mathbf{b}) = e^{i\mathbf{k}\cdot(\mathbf{b}-\mathbf{a})} c(\mathbf{k})(t) T^\dagger(\mathbf{b}-\mathbf{a}) c^\dagger(\mathbf{k}) . \quad (5.4)$$

In order to make use of the mapping between auxiliary model and lattice model Hamiltonians, we would like to express this lattice Greens function in terms of auxiliary model Greens functions – objects that can be computed purely from within a single auxiliary model. This will be a major simplification in our computations, because it is easier to work with a quantum impurity model, as compared to an interacting lattice model.

To make progress towards this, we express the lattice groundstate in terms of the auxiliary model groundstates (using eq. 3.20):

$$G(\mathbf{k}\sigma; t) = -i\theta(t) \frac{1}{N} \sum_{\mathbf{r}_1, \mathbf{r}_2} \langle \psi_{\text{gs}}(\mathbf{r}_2) | \{c_{\mathbf{k}\sigma}(t), c_{\mathbf{k}\sigma}^\dagger\} | \psi_{\text{gs}}(\mathbf{r}_1) \rangle . \quad (5.5)$$

We take one of the terms in the anticommutator and proceed to simplify it:

$$\begin{aligned} \sum_{\mathbf{r}_1, \mathbf{r}_2} \langle \psi_{\text{gs}}(\mathbf{r}_2) | c_{\mathbf{k}\sigma}(t) c_{\mathbf{k}\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_1) \rangle &= \sum_{\mathbf{r}_1, \mathbf{r}_2} \langle \psi_{\text{gs}}(\mathbf{r}_2) | T(\mathbf{r}_2 - \mathbf{r}_d) c_{\mathbf{k}\sigma}(t) c_{\mathbf{k}\sigma}^\dagger T^\dagger(\mathbf{r}_1 - \mathbf{r}_d) | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &= \sum_{\mathbf{r}_1, \mathbf{r}_2} e^{i\mathbf{k}\cdot(\mathbf{r}_1 - \mathbf{r}_2)} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{k}\sigma}(t) T^\dagger(\mathbf{r}_1 - \mathbf{r}_2) c_{\mathbf{k}\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &= N \sum_{\mathbf{r}} e^{i\mathbf{k}\cdot\mathbf{r}} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{k}\sigma}(t) T^\dagger(\mathbf{r}) c_{\mathbf{k}\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle . \end{aligned} \quad (5.6)$$

where $\mathbf{r}_d$ is a reference lattice site, and we used eq. 5.4 in the second step. The other term in the anticommutator gives

$$\begin{aligned}
 \sum_{\mathbf{r}_1, \mathbf{r}_2} \langle \psi_{\text{gs}}(\mathbf{r}_2) | c_{\mathbf{k}\sigma}^\dagger c_{\mathbf{k}\sigma}(t) | \psi_{\text{gs}}(\mathbf{r}_1) \rangle &= \sum_{\mathbf{r}_1, \mathbf{r}_2} \langle \psi_{\text{gs}}(\mathbf{r}_d) | T(\mathbf{r}_2 - \mathbf{r}_d) c_{\mathbf{k}\sigma}^\dagger c_{\mathbf{k}\sigma}(t) T^\dagger(\mathbf{r}_1 - \mathbf{r}_d) | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\
 &= \sum_{\mathbf{r}_1, \mathbf{r}_2} e^{-i\mathbf{k}(\mathbf{r}_1 - \mathbf{r}_2)} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{k}\sigma}^\dagger T^\dagger(\mathbf{r}_1 - \mathbf{r}_2) c_{\mathbf{k}\sigma}(t) | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\
 &= N \sum_{\mathbf{r}} e^{-i\mathbf{k} \cdot \mathbf{r}} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{k}\sigma}^\dagger T^\dagger(\mathbf{r}) c_{\mathbf{k}\sigma}(t) | \psi_{\text{gs}}(\mathbf{r}_d) \rangle .
 \end{aligned} \tag{5.7}$$

To resolve the time evolution of the operators, we insert $1 = \sum_m |\psi_m(\mathbf{r}_d)\rangle \langle \psi_m(\mathbf{r}_d)|$ into the equation. We again work with just the first term of the anticommutator for convenience:

$$\begin{aligned}
 \sum_{\mathbf{r}_1, \mathbf{r}_2} \langle \psi_{\text{gs}}(\mathbf{r}_2) | c_{\mathbf{k}\sigma}(t) c_{\mathbf{k}\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_1) \rangle &= N \sum_{\mathbf{r}, n_1, n_2} e^{i\mathbf{k} \cdot \mathbf{r}} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{k}\sigma}(t) | \psi_{n_1}(\mathbf{r}_d) \rangle \langle \psi_{n_1}(\mathbf{r}_d) | T^\dagger(\mathbf{r}) | \psi_{n_2}(\mathbf{r}_d) \rangle \langle \psi_{n_2}(\mathbf{r}_d) | c_{\mathbf{k}\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\
 &= N \sum_{\mathbf{r}, n_1, n_2} e^{i\mathbf{k} \cdot \mathbf{r}} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{k}\sigma}(t) | \psi_{n_1}(\mathbf{r}_d) \rangle \langle \psi_{n_1}(\mathbf{r}_d) | \psi_{n_2}(\mathbf{r}_d + \mathbf{r}) \rangle \langle \psi_{n_2}(\mathbf{r}_d) | c_{\mathbf{k}\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle .
 \end{aligned} \tag{5.8}$$

The overlap $\langle \psi_{n_1}(\mathbf{r}_d) | \psi_{n_2}(\mathbf{r}_d + \mathbf{r}) \rangle$ between the eigenstates of the auxiliary models at $\mathbf{r}_d$ and $\mathbf{r}_d + \mathbf{r}$ are appreciable only for $n_1 = n_2$. This is because, for small $\mathbf{r}$, the auxiliary models are very similar and the states are therefore almost orthogonal. For large $\mathbf{r}$, the overlap is again suppressed because the auxiliary model eigenstates are localised around the impurity. We therefore make the simplifying assumption: $\langle \psi_{n_1}(\mathbf{r}_d) | \psi_{n_2}(\mathbf{r}_d + \mathbf{r}) \rangle = \delta_{n_1, n_2} \langle \psi_{n_1}(\mathbf{r}_d) | \psi_{n_1}(\mathbf{r}_d + \mathbf{r}) \rangle = \delta_{n_1, n_2} F_{n_1}(\mathbf{r})$. Substituting this, we get

$$\sum_{\mathbf{r}_1, \mathbf{r}_2} \langle \psi_{\text{gs}}(\mathbf{r}_2) | c_{\mathbf{k}\sigma}(t) c_{\mathbf{k}\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_1) \rangle = N \sum_{\mathbf{n}} F_{\mathbf{n}}(\mathbf{k}) \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{k}\sigma}(t) | \psi_{\mathbf{n}}(\mathbf{r}_d) \rangle \langle \psi_{\mathbf{n}}(\mathbf{r}_d) | c_{\mathbf{k}\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle , \tag{5.9}$$

where we have defined the Fourier transform $F_{\mathbf{n}}(\mathbf{k}) = \sum_{\mathbf{r}} e^{i\mathbf{k} \cdot \mathbf{r}} F_{\mathbf{n}}(\mathbf{r})$ of the overlap function.

We now consider more carefully the transition operator $c_{\mathbf{k}\sigma}$ for the 1-particle excitation giving rise to the above Greens function. Within our auxiliary model approach, gapless excitations within the lattice model are represented by gapless excitations of the impurity site, specifically those that try to screen the impurity site. As a result, the uncoordinated scattering process $c_{\mathbf{k}\sigma}$ for the lattice model must be replaced by a coherent excitation $\mathcal{T}$ within the impurity model that captures those gapless excitations that occur in connection with the impurity, and projects out the uncorrelated excitations that take place even when the impurity site is decoupled from the bath.

In order to construct this auxiliary model $\mathcal{T}$ -matrix, we allow for all excitations of the impurity site in conjunction with those of the $\mathbf{k}$ -state:

$$\mathcal{T}_{\mathbf{k}, \sigma} = \left[c_{\mathbf{r}_d \sigma}^\dagger + S_{\mathbf{r}_d}^\sigma + C_{\mathbf{r}_d}^+ \right] c_{\mathbf{k}\sigma} , \tag{5.10}$$

where $S_{\mathbf{r}_d}^+$ and $C_{\mathbf{r}_d}^+$ are spin and charge flip operators on the impurity site that cause transition from $|\downarrow\rangle \rightarrow |\uparrow\rangle$ and $|0\rangle \rightarrow |\uparrow\downarrow\rangle$ respectively. The first term encodes the local component of the

k -space Greens function by creating an excitation on the impurity site (which is then translated across the lattice), and the rest of the terms encode the non-local components by tracking impurity-bath excitations.

Apart from modifying the excitation operator, we will now simplify the time-evolution as well. The first overlap on the right-hand side of eq. 5.9 expands to

$$\langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k},\sigma}(t) | \psi_n(\mathbf{r}_d) \rangle = \langle \psi_{\text{gs}}(\mathbf{r}_d) | e^{i(H_{\text{aux}}(\mathbf{r}_d) + H_{\text{cav}}(\mathbf{r}_d))t} \mathcal{T}_{\mathbf{k},\sigma} e^{-i(H_{\text{aux}}(\mathbf{r}_d) + H_{\text{cav}}(\mathbf{r}_d))t} | \psi_n(\mathbf{r}_d) \rangle, \quad (5.11)$$

where $H_{\text{aux}}(\mathbf{r}_d)$ (eq. 2.3) is the auxiliary model Hamiltonian consisting of the cluster Hamiltonian $H_{\text{clust}}(\mathbf{r}_d)$ along with all one-particle terms $H_{1\text{p}}$ in the Hamiltonian (in a typical lattice model, this will be the tight-binding hopping terms), and $H_{\text{cav}}(\mathbf{r}_d)$ is the “cavity” Hamiltonian consisting of all interaction (2-particle and beyond) terms in the Hamiltonian. Together, they create the full lattice Hamiltonian: $H_{\text{latt}} = H_{\text{aux}}(\mathbf{r}_d) + H_{\text{cav}}(\mathbf{r}_d)$. Using the Zassenhaus form of the Baker-Campbell-Hausdorff formula, we have

$$e^{i(H_{\text{aux}}(\mathbf{r}_d) + H_{\text{cav}}(\mathbf{r}_d))t} \approx e^{iH_{\text{aux}}(\mathbf{r}_d)t} e^{iH_{\text{cav}}(\mathbf{r}_d)t} e^{[H_{\text{aux}}(\mathbf{r}_d), H_{\text{cav}}(\mathbf{r}_d)]t^2/2}, \quad (5.12)$$

where we have dropped the higher-order terms because the commutators involve only the boundary degrees of freedom and become subsequently smaller as the order increases. One can think of the operator $e^{iH_{\text{cav}}(\mathbf{r}_d)t} e^{[H_{\text{aux}}(\mathbf{r}_d), H_{\text{cav}}(\mathbf{r}_d)]t^2/2}$ as renormalising the auxiliary model $\mathcal{T}$ -matrix through hybridisation with the rest of the system and formally leading to a new $\mathcal{T}$ -matrix:

$$\mathcal{T}_{\mathbf{k},\sigma}^* = e^{iH_{\text{cav}}(\mathbf{r}_d)t} e^{[H_{\text{aux}}(\mathbf{r}_d), H_{\text{cav}}(\mathbf{r}_d)]t^2/2} \mathcal{T}_{\mathbf{k},\sigma} e^{-[H_{\text{aux}}(\mathbf{r}_d), H_{\text{cav}}(\mathbf{r}_d)]t^2/2} e^{-iH_{\text{cav}}(\mathbf{r}_d)t}. \quad (5.13)$$

One can also absorb the modulating phase factor $F_n(\mathbf{k})$ (defined above and appearing in eq. 5.9) into this renormalisation:

$$\mathcal{T}_{\mathbf{k},\sigma}^* = e^{iH_{\text{cav}}(\mathbf{r}_d)t} e^{[H_{\text{aux}}(\mathbf{r}_d), H_{\text{cav}}(\mathbf{r}_d)]t^2/2} \mathcal{T}_{\mathbf{k},\sigma} F_n(\mathbf{k}) \mathcal{P}_n(\mathbf{r}_d) e^{-[H_{\text{aux}}(\mathbf{r}_d), H_{\text{cav}}(\mathbf{r}_d)]t^2/2} e^{-iH_{\text{cav}}(\mathbf{r}_d)t}, \quad (5.14)$$

where $\mathcal{P}_n(\mathbf{r}_d)$ projects onto the n^{th} eigenstate $|\psi_n(\mathbf{r}_d)\rangle$.

Since the overlap is calculated in the eigenstates of the auxiliary model at $\mathbf{r}_d$, we adopt the zeroth order approximation of ignoring this renormalisation. Later, we will specify a prescription of systematically improving our computation. The overlap expression becomes

$$\langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k},\sigma}(t) | \psi_n(\mathbf{r}_d) \rangle = \langle \psi_{\text{gs}}(\mathbf{r}_d) | e^{i(H_{\text{clust}}(\mathbf{r}_d) + H_{1\text{p}})t} \mathcal{T}_{\mathbf{k},\sigma} e^{-i(H_{\text{clust}}(\mathbf{r}_d) + H_{1\text{p}})t} | \psi_n(\mathbf{r}_d) \rangle. \quad (5.15)$$

To make the problem tractable, we treat the effect of $H_{1\text{p}}$ at the level of an expectation value and note that the effect of the transition operator is to create a hole of momentum $\mathbf{k}$ (eq. 5.10). Let the Hamiltonian $H_{1\text{p}}$ be diagonal in certain modes $c_{\nu,\sigma}$ with energies ϵ_{ν} , and the operator $c_{\mathbf{k},\sigma}$ can be expressed in terms of these modes: $c_{\mathbf{k},\sigma} = \sum_{\nu} C_{\nu,\mathbf{k}} c_{\nu,\sigma}$. A similar decomposition will hold for the $\mathcal{T}$ -matrix. Substituting it gives

$$\langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k},\sigma}(t) | \psi_n(\mathbf{r}_d) \rangle = \sum_{\nu} C_{\nu,\mathbf{k}} \langle \psi_{\text{gs}}(\mathbf{r}_d) | e^{i(H_{\text{clust}}(\mathbf{r}_d) + H_{1\text{p}})t} \mathcal{T}_{\nu,\sigma} e^{-i(H_{\text{clust}}(\mathbf{r}_d) + H_{1\text{p}})t} | \psi_n(\mathbf{r}_d) \rangle. \quad (5.16)$$

Each excitation $\mathcal{T}_{v,\sigma}$ therefore leads to an energy cost $-\epsilon_v$. Moreover, $|\psi_n(\mathbf{r}_d)\rangle$ and $|\psi_{gs}(\mathbf{r}_d)\rangle$ are eigenstates of $H_{\text{clust}}(\mathbf{r}_d)$ with energies E_n and E_0 . This leads to the following simplified expression:

$$\langle \psi_{gs}(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k},\sigma}(t) | \psi_n(\mathbf{r}_d) \rangle = \sum_v C_{v,\mathbf{k}} e^{-i(E_n - E_0 + \epsilon_v)t} \langle \psi_{gs}(\mathbf{r}_d) | \mathcal{T}_{v,\sigma} | \psi_n(\mathbf{r}_d) \rangle. \quad (5.17)$$

Treating the other overlap in eq. 5.9 in an identical fashion gives

$$\langle \psi_n(\mathbf{r}_d) | c_{\mathbf{k}\sigma}^\dagger | \psi_{gs}(\mathbf{r}_d) \rangle = \sum_v C_{v,\mathbf{k}} \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{v,\sigma}^\dagger | \psi_{gs}(\mathbf{r}_d) \rangle. \quad (5.18)$$

Substituting the expression back into 5.9 and then into 5.5, we get an expression for one half of the retarded Greens function:

$$\begin{aligned} & -i\theta(t) \langle \Psi_{gs} | c_{\mathbf{k}\sigma}(t) c_{\mathbf{k}\sigma}^\dagger | \Psi_{gs} \rangle \\ &= -i\theta(t) \sum_{v,n} C_{v,\mathbf{k}} e^{-i(E_n - E_0 + \epsilon_v)t} \langle \psi_{gs}(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k},\sigma} | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{v,\sigma}^\dagger | \psi_{gs}(\mathbf{r}_d) \rangle. \end{aligned} \quad (5.19)$$

Working through the other term with the same strategy, we get the full expression:

$$\begin{aligned} & -i\theta(t) \langle \Psi_{gs} | \{c_{\mathbf{k}\sigma}(t), c_{\mathbf{k}\sigma}^\dagger\} | \Psi_{gs} \rangle \\ &= -i\theta(t) \sum_{v,n} C_{v,\mathbf{k}} e^{-i(E_n - E_0 + \epsilon_v)t} \langle \psi_{gs}(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k},\sigma} | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{v,\sigma}^\dagger | \psi_{gs}(\mathbf{r}_d) \rangle \\ & \quad - i\theta(t) \sum_{v,n} C_{v,\mathbf{k}} e^{i(E_n - E_0 - \epsilon_v)t} \langle \psi_{gs}(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k},\sigma}^\dagger | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{v,\sigma} | \psi_{gs}(\mathbf{r}_d) \rangle. \end{aligned} \quad (5.20)$$

Changing from time to frequency domain using $g(\omega) = \int dt e^{i\omega T} f(t)$ gives

$$\begin{aligned} G(\mathbf{k}, \sigma; \omega) &= \sum_{v,n} C_{v,\mathbf{k}} \left[\frac{\langle \psi_{gs}(\mathbf{r}_d) | \mathcal{T}_{v,\sigma} | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{v,\sigma}^\dagger | \psi_{gs}(\mathbf{r}_d) \rangle}{\omega - (E_n - E_0 + \epsilon_v)} \right. \\ & \quad \left. + \frac{\langle \psi_{gs}(\mathbf{r}_d) | \mathcal{T}_{v,\sigma}^\dagger | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{v,\sigma} | \psi_{gs}(\mathbf{r}_d) \rangle}{\omega + (E_n - E_0 - \epsilon_v)} \right]. \end{aligned} \quad (5.21)$$

We recognise the two parts of the lattice model Greens function as the Greens functions of the operator $\mathcal{T}_{\mathbf{k},\sigma}$, calculated within the auxiliary model at $\mathbf{r}_d$, but at shifted frequencies $\omega - \epsilon_v$:

$$G(\mathbf{k}, \sigma; \omega) = \sum_v C_{v,\mathbf{k}} \mathcal{G}(\mathcal{T}_{v,\sigma}; \omega - \epsilon_v) \quad (5.22)$$

We now discuss how to systematically improve this estimate. One way is to include the effect of other auxiliary models in the overlap function $F_n(\mathbf{r})$. This will of course require us to calculate an off-diagonal Greens function. As a concrete example, upon including the effects of the auxiliary models nearest to the reference point $\mathbf{r}_d$, the expression for the Greens function becomes

$$\begin{aligned} G(\mathbf{k}, \sigma; \omega) &= \sum_{v,n} C_{v,\mathbf{k}} \left[\frac{\langle \psi_{gs}(\mathbf{r}_d) | \mathcal{T}_{v,\sigma}^* | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{v,\sigma}^\dagger | \psi_{gs}(\mathbf{r}_d) \rangle}{\omega - (E_n - E_0 + \epsilon_v)} \right. \\ & \quad \left. + \frac{\langle \psi_{gs}(\mathbf{r}_d) | \mathcal{T}_{v,\sigma}^\dagger | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{v,\sigma}^* | \psi_{gs}(\mathbf{r}_d) \rangle}{\omega + (E_n - E_0 - \epsilon_v)} \right], \end{aligned} \quad (5.23)$$

with $\tau_{\nu,\sigma}^* = \mathcal{T}_{\nu,\sigma}(F_n(0) + \sum_{\mathbf{a}} F_n(\mathbf{a})e^{i\mathbf{k}\cdot\mathbf{a}})$ where $\mathbf{a}$ sums over the primitive lattice vectors. Another way to include more non-local contributions is by considering the effects of the cavity Hamiltonian in eq. 5.14:

$$\mathcal{T}_{\mathbf{k},\sigma}^*(t) = e^{iH_{\text{cav}}(\mathbf{r}_d)t} (\mathcal{T}_{\mathbf{k},\sigma} + t^2/2 [H_{\text{aux}}(\mathbf{r}_d), H_{\text{cav}}(\mathbf{r}_d)], \mathcal{T}_{\mathbf{k},\sigma}) e^{-iH_{\text{cav}}(\mathbf{r}_d)t} . \quad (5.24)$$

The time-domain Greens function associated with this operator will then need to be calculated, which can then be used to calculate frequency-domain functions.

2.5.2 One-particle real-space Greens functions

We now work with Greens functions for real-space excitations. The retarded Greens function in this case is

$$G(\mathbf{r}\sigma; t) = -i\theta(t) \langle \Psi_{\text{gs}} | \{c_{\mathbf{r}_d+\mathbf{r},\sigma}(t), c_{\mathbf{r}_d,\sigma}^\dagger\} | \Psi_{\text{gs}} \rangle . \quad (5.25)$$

Since our lattice model has discrete translation symmetry, we instead calculate the equivalent quantity

$$G(\mathbf{r}\sigma; t) = -i\theta(t) \sum_{\mathbf{r}_d} \langle \Psi_{\text{gs}} | \{c_{\mathbf{r}_d+\mathbf{r},\sigma}(t), c_{\mathbf{r}_d,\sigma}^\dagger\} | \Psi_{\text{gs}} \rangle . \quad (5.26)$$

Similar to the k -space Greens functions, express the groundstates as a superposition of auxiliary model Greens functions. We take the first term in the anticommutator and proceed to simplify it:

$$\begin{aligned} \sum_{\mathbf{r}_d} \langle \Psi_{\text{gs}} | c_{\mathbf{r}_d+\mathbf{r},\sigma}(t) c_{\mathbf{r}_d,\sigma}^\dagger | \Psi_{\text{gs}} \rangle &= \frac{1}{N} \sum_{\mathbf{R}, \Delta, \mathbf{r}_d} \langle \psi_{\text{gs}}(\mathbf{R} + \Delta) | c_{\mathbf{r}_d+\mathbf{r},\sigma}(t) c_{\mathbf{r}_d,\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{R}) \rangle \\ &= \frac{1}{N} \sum_{\mathbf{R}, \Delta} \langle \psi_{\text{gs}}(-\mathbf{r}_d) | c_{\mathbf{r}-\mathbf{R}-\Delta,\sigma}(t) T(\Delta) c_{-\mathbf{R},\sigma}^\dagger | \psi_{\text{gs}}(-\mathbf{r}_d) \rangle . \end{aligned} \quad (5.27)$$

Local Greens function: $\mathbf{r} = 0$

We first consider the case of local Greens function, by setting $\mathbf{r} = 0$. We also change the dummy variable $\mathbf{r}_d$, $\mathbf{R}$ and Δ to $-\mathbf{r}_d$, $\mathbf{R}$ and $-\Delta$ to make the expressions tidier:

$$\sum_{\mathbf{r}_d} \langle \Psi_{\text{gs}} | c_{\mathbf{r}_d,\sigma}(t) c_{\mathbf{r}_d,\sigma}^\dagger | \Psi_{\text{gs}} \rangle = \frac{1}{N} \sum_{\mathbf{R}, \Delta, \mathbf{r}_d} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{R}+\Delta,\sigma}(t) T^\dagger(\Delta) c_{\mathbf{R},\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle . \quad (5.28)$$

We split the full summation into four parts, based on whether $\mathbf{R} = \mathbf{r}_d$ and whether $\Delta + \mathbf{R} = \mathbf{r}_d$:

$$\begin{aligned} \sum_{\mathbf{r}_d} \langle \Psi_{\text{gs}} | c_{\mathbf{r}_d,\sigma}(t) c_{\mathbf{r}_d,\sigma}^\dagger | \Psi_{\text{gs}} \rangle &= \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{r}_d,\sigma}(t) c_{\mathbf{r}_d,\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &\quad + \sum_{\Delta \neq 0} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{r}_d+\Delta,\sigma}(t) T^\dagger(\Delta) c_{\mathbf{r}_d,\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &\quad + \sum_{\mathbf{R} \neq \mathbf{r}_d} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{r}_d,\sigma}(t) T^\dagger(\mathbf{r}_d - \mathbf{R}) c_{\mathbf{R},\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &\quad + \sum_{\mathbf{R} \neq \mathbf{r}_d, \Delta \neq \mathbf{r}_d} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\Delta,\sigma}(t) T^\dagger(\Delta - \mathbf{R}) c_{\mathbf{R},\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle , \end{aligned} \quad (5.29)$$

where we have used the fact that the location of the auxiliary model does not affect the correlation calculated within the auxiliary model in order to cancel out the summation over $\mathbf{r}_d$ with the factor of $1/N$. The reason for splitting the expression into these parts is to explicitly write out which operators act on the impurity site $\mathbf{r}_d$ and which act on other real-space sites away from the impurity site. We will now replace the real-space excitation operators away from the impurity site with the associated T -matrices, following the discussion above eq. 5.10:

$$\mathcal{T}_{\mathbf{r},\sigma} = \left[\left(\sum_{\sigma'} c_{\mathbf{r}_d\sigma'}^\dagger + \text{h.c.} \right) + \left(S_{\mathbf{r}_d}^+ + \text{h.c.} \right) + \left(C_{\mathbf{r}_d}^+ + \text{h.c.} \right) \right] c_{\mathbf{r}\sigma} = \sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}} \mathcal{T}_{\mathbf{k},\sigma} . \quad (5.30)$$

We also insert a complete basis of states of the auxiliary model at $\mathbf{r}_d$, $1 = \sum_n |\psi_n(\mathbf{r}_d)\rangle \langle \psi_n(\mathbf{r}_d)|$, generating an overlap factor F_n as before:

$$\begin{aligned} \sum_{\mathbf{r}_d} \langle \Psi_{\text{gs}} | c_{\mathbf{r}_d,\sigma}(t) c_{\mathbf{r}_d,\sigma}^\dagger | \Psi_{\text{gs}} \rangle &= \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{r}_d,\sigma}(t) c_{\mathbf{r}_d,\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &+ \sum_{\mathbf{k},n} F'_n(\mathbf{k}) \langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k},\sigma}(t) | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | c_{\mathbf{r}_d,\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &+ \sum_{\mathbf{k},n} F'_n(\mathbf{k})^* \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{r}_d,\sigma}(t) | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k},\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &+ \sum_{\mathbf{k}_1, \mathbf{k}_2} F'_n(\mathbf{k}_1) F'_n(\mathbf{k}_2)^* \langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k}_1,\sigma}(t) | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k}_2,\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle , \end{aligned} \quad (5.31)$$

where $F'_n(\mathbf{k})$ is the truncated Fourier transform $\sum_{\mathbf{r} \neq 0} e^{i\mathbf{k}\cdot\mathbf{r}} F_n(\mathbf{r})$ of the overlap factor.

We next treat the time-evolution of the operators exactly as in the k -space Greens function. Without repeating the details, we state the final result here, for the frequency-domain real-space local Greens function:

$$\begin{aligned} G_{\text{loc}}(\omega) &= \mathcal{G}(d\sigma; \omega - \epsilon_{\text{loc}}) + \sum_{\nu} C_{\nu} \left[\mathcal{G}(\mathcal{T}_{\nu,\sigma}, d\sigma; \omega - \epsilon_{\nu}) + \sum_{\mathbf{k}} \mathcal{G}(\mathcal{T}_{\nu,\sigma}, \mathcal{T}_{\mathbf{k},\sigma}; \omega - \epsilon_{\nu}) \right] \\ &+ \sum_{\mathbf{k}} \mathcal{G}(d\sigma, \mathcal{T}_{\mathbf{k},\sigma}; \omega) , \end{aligned} \quad (5.32)$$

where $\mathcal{G}(d\sigma; \omega)$ is the retarded Greens function of the impurity site, $\mathcal{G}^>(\mathcal{T}_{\nu,\sigma}, d\sigma; \omega)$ is the off-diagonal retarded Greens function between the impurity site and the bath state $\nu\sigma$, and $\mathcal{G}^>(\mathcal{T}_{\nu,\sigma}, \mathcal{T}_{\mathbf{k},\sigma}; \omega)$ is the same but between multiple states in the bath. $\epsilon_{\text{loc}} = \langle n_{d\sigma} = 0 | H_{\text{lp}} | n_{d\sigma} = 0 \rangle - \langle n_{d\sigma} = 1 | H_{\text{lp}} | n_{d\sigma} = 1 \rangle$, and the coefficient C_{ν} is given by $C_{\nu} = \sum_{\mathbf{k}} C_{\nu,\mathbf{k}}$.

Non-local Greens function: $\mathbf{r} \neq 0$

We next consider the case of non-local Greens function, where the excitations propagate a non-zero distance $\mathbf{r}$:

$$\sum_{\mathbf{r}_d} \langle \Psi_{\text{gs}} | c_{\mathbf{r}_d+\mathbf{r},\sigma}(t) c_{\mathbf{r}_d,\sigma}^\dagger | \Psi_{\text{gs}} \rangle = \frac{1}{N} \sum_{\mathbf{R}, \Delta, \mathbf{r}_d} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{R}+\Delta+\mathbf{r},\sigma}(t) T^\dagger(\Delta) c_{\mathbf{R},\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle . \quad (5.33)$$

To bring this expression closer to the one for the local Greens function, we redefine $\Delta + \mathbf{r} \rightarrow \Delta$:

$$\sum_{\mathbf{r}_d} \langle \Psi_{\text{gs}} | c_{\mathbf{r}_d + \mathbf{r}, \sigma}(t) c_{\mathbf{r}_d, \sigma}^\dagger | \Psi_{\text{gs}} \rangle = \frac{1}{N} \sum_{\mathbf{R}, \Delta, \mathbf{r}_d} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{R} + \Delta, \sigma}(t) T^\dagger(\Delta) T(\mathbf{r}) c_{\mathbf{R}, \sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle . \quad (5.34)$$

This allows to again split the full summation based on whether $\mathbf{R} = \mathbf{r}_d$ and whether $\Delta + \mathbf{R} = \mathbf{r}_d$:

$$\begin{aligned} \sum_{\mathbf{r}_d} \langle \Psi_{\text{gs}} | c_{\mathbf{r}_d + \mathbf{r}, \sigma}(t) c_{\mathbf{r}_d, \sigma}^\dagger | \Psi_{\text{gs}} \rangle &= \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{r}_d, \sigma}(t) T(\mathbf{r}) c_{\mathbf{r}_d, \sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &+ \sum_{\Delta \neq 0} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{r}_d + \Delta, \sigma}(t) T^\dagger(\Delta - \mathbf{r}) c_{\mathbf{r}_d, \sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &+ \sum_{\mathbf{R} \neq \mathbf{r}_d} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{r}_d, \sigma}(t) T^\dagger(\mathbf{r}_d - \mathbf{R} - \mathbf{r}) c_{\mathbf{R}, \sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &+ \sum_{\mathbf{R} \neq \mathbf{r}_d, \Delta \neq \mathbf{r}_d} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\Delta, \sigma}(t) T^\dagger(\Delta - \mathbf{R} - \mathbf{r}) c_{\mathbf{R}, \sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle . \end{aligned} \quad (5.35)$$

Introducing Fourier transforms of the operators and inserting eigenstates of the auxiliary models leads to

$$\begin{aligned} &\sum_{\mathbf{r}_d} \langle \Psi_{\text{gs}} | c_{\mathbf{r}_d + \mathbf{r}, \sigma}(t) c_{\mathbf{r}_d, \sigma}^\dagger | \Psi_{\text{gs}} \rangle \\ &= \sum_n F_n^*(\mathbf{r}) \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{r}_d, \sigma}(t) | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | c_{\mathbf{r}_d, \sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &+ \sum_{\mathbf{k}, n} F'_n(\mathbf{k}) F_n^*(\mathbf{r}) \langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k}, \sigma}(t) | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | c_{\mathbf{r}_d, \sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &+ \sum_{\mathbf{k}, n} F'_n(\mathbf{k})^* F_n^*(\mathbf{r}) \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{r}_d, \sigma}(t) | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k}, \sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &+ \sum_{\mathbf{k}_1, \mathbf{k}_2} F'_n(\mathbf{k}_1) F'_n(\mathbf{k}_2)^* F_n^*(\mathbf{r}) \langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k}_1, \sigma}(t) | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k}_2, \sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle , \end{aligned} \quad (5.36)$$

The net difference from the diagonal Greens function is the presence of the overlap factor $F_n^*(\mathbf{r})$.

2.5.3 Equal-Time Ground State Correlators

Apart from dynamical correlations such as Greens functions, we would also like to use our approach to calculate static correlation functions in terms of auxiliary model quantities. We first consider a real-space correlation that tracks the propagation of a general excitation described by the operator $O(\mathbf{r})$ over a distance Δ :

$$C_O(\mathbf{r}) = \langle \Psi_{\text{gs}} | O(\mathbf{r}_d + \mathbf{r}) O^\dagger(\mathbf{r}_d) | \Psi_{\text{gs}} \rangle . \quad (5.37)$$

We first consider a local correlation: $\mathbf{r} = 0$. To obtain a tractable expression for this, we note that the quantity being computed here is almost identical to the one computed in eq. 5.27, the only difference being the time-evolution of the operator. For the local correlation, we therefore assume the result

of eq. 5.30, dropping the time evolution because the correlators in this section are calculated at equal time. We also transform from k -space back to real-space because there are no time-evolution operators which need extended states in order to be resolved. Putting all this together, we get

$$\begin{aligned}
 C_O^{\text{loc}} &= \langle \psi_{\text{gs}}(\mathbf{r}_d) | O(\mathbf{r}_d) O^\dagger(\mathbf{r}_d) | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\
 &+ \sum_{\mathbf{r} \neq 0, n} F_n(\mathbf{r}) \langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{O(\mathbf{r}+\mathbf{r}_d)} | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | O^\dagger(\mathbf{r}_d) | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\
 &+ \sum_{\mathbf{r} \neq 0, n} F_n(\mathbf{r}) \langle \psi_{\text{gs}}(\mathbf{r}_d) | O(\mathbf{r} + \mathbf{r}_d) | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{O(\mathbf{r}_d)}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\
 &+ \sum_{\mathbf{r}_1 \neq 0, \mathbf{r}_2 \neq 0, n} F_n(\mathbf{r}) \langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{O(\mathbf{r}_1+\mathbf{r}_d)} | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{O(\mathbf{r}_2+\mathbf{r}_d)}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle ,
 \end{aligned} \tag{5.38}$$

where the transition operator $\mathcal{T}_{O(\mathbf{r}+\mathbf{r}_d)}$ represents impurity-bath correlation:

$$\mathcal{T}_{O(\mathbf{r}+\mathbf{r}_d), \sigma} = \left[c_{\mathbf{r}_d \sigma}^\dagger + S_{\mathbf{r}_d}^\sigma + C_{\mathbf{r}_d}^+ \right] O(\mathbf{r} + \mathbf{r}_d) , \tag{5.39}$$

We next consider non-local correlations over a distance Λ . Adapting eq. 5.36 for the non-local real-space Greens function in the same manner, we get

$$\begin{aligned}
 C_O(\Lambda) &= \sum_n F_n^*(\Lambda) \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{r}_d, \sigma}(t) | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | c_{\mathbf{r}_d, \sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\
 &+ \sum_{\mathbf{r} \neq 0, n} F_n(\mathbf{r} + \Lambda) \langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{O(\mathbf{r}+\Lambda+\mathbf{r}_d)} | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | O^\dagger(\mathbf{r}_d) | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\
 &+ \sum_{\mathbf{r} \neq 0, n} F_n(\mathbf{r} + \Lambda) \langle \psi_{\text{gs}}(\mathbf{r}_d) | O(\mathbf{r} + \Lambda + \mathbf{r}_d) | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{O(\mathbf{r}_d)}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\
 &+ \sum_{\mathbf{r}_1 \neq 0, \mathbf{r}_2 \neq 0, n} F_n(\mathbf{r} + \Lambda) \langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{O(\mathbf{r}_1+\Lambda+\mathbf{r}_d)} | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{O(\mathbf{r}_2+\mathbf{r}_d)}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle ,
 \end{aligned} \tag{5.40}$$

2.6 Entanglement Measures

We will now describe the prescription of calculating entanglement measures of the lattice model from within our auxiliary model treatment. In this section, we are interested mainly in two such measures, the entanglement entropy and the mutual information. Given a pure state $|\Psi\rangle$ describing the complete system, the entanglement entropy $S_{\text{EE}}(\nu)$ of a subsystem ν quantifies the entanglement of ν with the rest of the subsystem, and is defined as

$$S_{\text{EE}}(\nu) = -\text{Tr} [\rho(\nu) \log \rho(\nu)] , \quad \rho(\nu) = \text{Tr}_\nu [|\Psi\rangle \langle \Psi|] \tag{6.1}$$

where $\text{Tr} [\cdot]$ is the trace operator, and $\rho(\nu)$ is the reduced density matrix (RDM) for the subsystem ν obtained by taking the partial trace Tr_ν (over the states of ν) of the full density matrix $\rho = |\Psi\rangle \langle \Psi|$. If the subsystem ν describes local regions in real space (or states in k -space), we might be interested

in the entanglement between two such subsystems ν_1 and ν_2 . The correct measure to quantify such entanglement is the mutual information:

$$I_2(\nu_1, \nu_2) = S_{\text{EE}}(\nu_1) + S_{\text{EE}}(\nu_2) - S_{\text{EE}}(\nu_1 \cup \nu_2), \quad (6.2)$$

where $\nu_1 \cup \nu_2$ is a larger subsystem formed by combining ν_1 and ν_2 .

We start by recalling the decomposition of entanglement entropy in terms of multipartite information as laid out in Subsection 1.13.2. Given a partitioning of the system into mutually exclusive subsystems $\{\nu_i\}$, the average entanglement entropy $S_{\text{EE}}^{(1)}$ over all subsystems can be expressed as

$$S_{\text{EE}}^{(1)} = \frac{1}{N} \sum_{n=2}^N (-1)^n \binom{N}{n} \bar{I}_n, \quad (6.3)$$

$$\bar{I}_n = \frac{(N-n)!n!}{N!} \sum_{S_n \in C_n(S_N)} I_n(S_n),$$

where $\bar{I}_n$ is the average multipartite information I_n over all distinct combinations of n subsystems.

We first take a look at real-space entanglement. Real-space entanglement measures can be used to probe delocalisation-localisation transitions. The simplest such measure is the entanglement of a lattice site in real-space. Since the local entanglement entropy will be uniform at each lattice site for a system with translation invariance, it suffices to calculate the real-space averaged entanglement entropy $S_{\text{EE}}^{(1)}$. The corresponding multipartite measures $\bar{I}_n$ involve collections of lattice sites. We write such a measure as

$$\bar{I}_n = \frac{(N-n)!n!}{N!} \frac{1}{N} \sum_{I(\mathbf{r}_d)} \sum_{B^{n-1}(\mathbf{r}_d)} I_n(I(\mathbf{r}_d); B^{n-1}(\mathbf{r}_d)), \quad (6.4)$$

where $I(\mathbf{r}_d)$ sums over all lattice sites in the form of an impurity site and $B^{n-1}(\mathbf{r}_d)$ considers all possible subsets of size $n-1$ of the remaining bath sites. Together, this choice corresponds to an auxiliary model with the impurity placed at $\mathbf{r}_d$. The additional factor of $1/N$ takes care of the overcounting arising from each choice appearing in N auxiliary models. We can therefore write the multipartite information as an average over auxiliary models, each auxiliary model contributing an average I_n over its bath sites:

$$\bar{I}_n = \frac{n}{N^2} \sum_{\mathbf{r}_d} I_n^{(\text{tot})}(\mathbf{r}_d). \quad (6.5)$$

The quantity $I_n^{(\text{tot})}(\mathbf{r}_d)$ is defined as $I_n^{(\text{tot})}(\mathbf{r}_d) = \frac{(N-n)!(n-1)!}{(N-1)!} \sum_{B^{n-1}(\mathbf{r}_d)} I_n(I(\mathbf{r}_d); B^{n-1}(\mathbf{r}_d))$. By translation invariance, such a measure will be uniform across auxiliary models and it suffices to calculate the measure for a single auxiliary model located at a reference point $\mathbf{r}_d$; the whole expression therefore simplifies to

$$S_{\text{EE}}^{(1)} = \frac{1}{N} \sum_{n=2}^N (-1)^n \binom{N-1}{n-1} I_n^{(\text{tot})}(\mathbf{r}_d) = \frac{1}{N} \sum_{n=2}^N (-1)^n \sum_{B^{n-1}(\mathbf{r}_d)} I_n(I(\mathbf{r}_d); B^{n-1}(\mathbf{r}_d)). \quad (6.6)$$

For example, truncating the summation to $n=3$, we have the explicit expression

$$S_{\text{EE}}^{(1)} = \frac{N-1}{N} I_2^{(\text{tot})}(\mathbf{r}_d) - \frac{(N-1)(N-2)}{2N} I_3^{(\text{tot})}(\mathbf{r}_d) \quad (6.7)$$

Chapter 3

Kondo Breakdown in a Heavy-Fermion Model

3.1 Bilayer extended Hubbard model: Impurity model

[33] We approach the heavy-fermion problem by starting from a bilayer extended Hubbard model, consisting of two layers (f and c). Towards studying this lattice model, we adopt a two-layer impurity problem that hosts a correlated impurity site in each layer (S_f and S_d):

$$H_{\text{aux}} = H_{\text{lp}} + H_f + H_d + H_{fd} , \quad (1.1)$$

where H_{lp} is the Hamiltonian for the non-interacting itinerant electrons of either layer,

$$H_{\text{lp}} = -t \sum_{\sigma, \alpha} \sum_{\langle i, j \rangle} \left(c_{i, \sigma, \alpha}^\dagger c_{j, \sigma, \alpha} + \text{h.c.} \right) - \sum_{i, \sigma, \alpha} \mu_\alpha n_{i, \sigma, \alpha} , \quad (1.2)$$

such that α sums over the two layers f and d . H_f and H_d describe the dynamics of the correlated impurity sites (and their local neighbourhood) in each layer:

$$\begin{aligned} H_f = & -\frac{U_f}{2} (n_{f, \uparrow} - n_{f, \downarrow})^2 + \sum_{Z \in \text{NN}} \left[-V \sum_{\sigma} \left(c_{f, \sigma}^\dagger c_{Z, \sigma, f} + \text{h.c.} \right) + \frac{J}{2} \sum_{\alpha, \beta} \mathbf{S}_f \cdot \boldsymbol{\sigma}_{\alpha\beta} c_{Z, \alpha, f}^\dagger c_{Z, \beta, f} \right. \\ & \left. - \frac{W_f}{2} (n_{Z, \uparrow, f} - n_{Z, \downarrow, f})^2 \right] , \\ H_d = & -\eta_d n_d - \frac{U_d}{2} (n_{d, \uparrow} - n_{d, \downarrow})^2 + \sum_{Z \in \text{NN}} \left[-V \sum_{\sigma} \left(c_{d, \sigma}^\dagger c_{Z, \sigma, d} + \text{h.c.} \right) + \frac{J}{2} \sum_{\alpha, \beta} \mathbf{S}_d \cdot \boldsymbol{\sigma}_{\alpha\beta} c_{Z, \alpha, d}^\dagger c_{Z, \beta, d} \right] , \end{aligned} \quad (1.3)$$

where $c_{v, \sigma, \alpha}^\dagger$ can refer to creation operator for either the f -layer. Finally, H_{fd} represents the inter-layer hybridisation:

$$H_{fc} = -V_\perp \sum_{\sigma} \left(c_{d, \sigma}^\dagger c_{f, \sigma} + \text{h.c.} \right) . \quad (1.4)$$

3.2 Tiling Into Bilayer Extended Hubbard Model

Tiling the impurity model leads to a bilayer extended Hubbard model on the 2D square lattice. In order to tile our auxiliary model to restore translation invariance, we identify the cluster which acts as the “unit” cell of translation. As discussed in the text surrounding eq. 3.9, a good choice for the unit cell is the impurity site and its surrounding correlated environment. Placing the impurity sites at $\mathbf{r}_d$ in each layer, we get

$$H_{\text{clust}}(\mathbf{r}_d) = H_f(\mathbf{r}_d) + H_d(\mathbf{r}_d) + H_{fc}(\mathbf{r}_d) . \quad (2.1)$$

The details of tiling a cluster Hamiltonian were already shown in Sec. 2.3.2; the general idea is to use manybody translation operators (details in Sec. 2.3.1) to center the cluster Hamiltonian at various sites of the lattice, such that combining all such terms leads to a periodised Hamiltonian. we write down the final result here:

$$\begin{aligned} H_{\text{latt}} &= \sum_{\mathbf{r}} T^\dagger(\mathbf{r}) H_{\text{clust}}(\mathbf{r}_d) T(\mathbf{r}) \\ &= -\frac{2V}{Z} \sum_{\langle i,j \rangle, \sigma} \left(c_{i,\sigma}^\dagger c_{j,\sigma} + \text{h.c.} \right) - \sum_i \left[\eta_d n_{i,d} + \frac{U_d}{2} (n_{i,\uparrow,d} - n_{i,\downarrow,d})^2 \right] + \frac{2J}{Z} \sum_{\langle i,j \rangle} \mathbf{S}_{i,d} \cdot \mathbf{S}_{j,d} \\ &\quad - \frac{2V}{Z} \sum_{\langle i,j \rangle, \sigma} \left(f_{i,\sigma}^\dagger f_{j,\sigma} + \text{h.c.} \right) - \frac{U_f - W_f}{2} \sum_i (n_{i,\uparrow,f} - n_{i,\downarrow,f})^2 + \frac{2J}{Z} \sum_{\langle i,j \rangle} \mathbf{S}_{i,f} \cdot \mathbf{S}_{j,f} \\ &\quad - V_\perp \sum_{i,\sigma} (f_{i,\sigma}^\dagger c_{i,\sigma} + \text{h.c.}) \end{aligned} \quad (2.2)$$

Tiling the impurity model therefore leads to a bilayer extended Hubbard model, with effective couplings that depend on the auxiliary model couplings. The value of η_d defines the effective chemical potential of the d -layer; a non-zero value moves the layer away from half-filling. The f -layer is pinned at half-filling.

Following Sec. 2.5.1, the retarded k -space Greens function in either layer can be expressed in terms of auxiliary model Greens functions:

$$G(\mathbf{k}, \sigma, \alpha; \omega) = \sum_{\nu} C_{\nu, k\alpha} \mathcal{G}(\mathcal{T}_{\nu, \sigma, \alpha}; \omega - \mathcal{E}_{\nu}(\mathbf{k})) \quad (2.3)$$

where $\alpha = f, d$ denotes the layer in which the Greens function is being calculated, the modes ν indicate the operators which diagonalise the Hamiltonian H_{lp} , with energies $\mathcal{E}_{\nu}(\mathbf{k})$:

$$H_{\text{lp}} = -\frac{2V}{Z} \sum_{\langle i,j \rangle, \sigma} \left(c_{i,\sigma}^\dagger c_{j,\sigma} + \text{h.c.} + f_{i,\sigma}^\dagger f_{j,\sigma} + \text{h.c.} \right) - V_\perp \sum_{i,\sigma} (f_{i,\sigma}^\dagger c_{i,\sigma} + \text{h.c.}) - \sum_i \eta_d n_{i,d} , \quad (2.4)$$

and the $\mathcal{T}$ -matrices are defined in eq. 5.10:

$$\mathcal{T}_{\nu, \sigma, \alpha} = \left[\left(\sum_{\sigma'} c_{\alpha\sigma'}^\dagger + \text{h.c.} \right) + (S_\alpha^+ + \text{h.c.}) + (C_\alpha^+ + \text{h.c.}) \right] c_{\nu, \sigma} , \quad (2.5)$$

In order to calculate $\mathcal{E}_v(\mathbf{k})$, we introduce the k -space modes via a Fourier transform:

$$c_{i,\sigma}^\dagger = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{r}_i} c_{\mathbf{k},\sigma}^\dagger, \quad f_{i,\sigma}^\dagger = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{r}_i} f_{\mathbf{k},\sigma}^\dagger. \quad (2.6)$$

In terms of these modes, the 1-particle Hamiltonian decouples in k -space:

$$H_{1p} = \sum_{\mathbf{k},\sigma} \left[(\epsilon_{\mathbf{k}} - \eta_d)(n_{\mathbf{k},\sigma,d} + n_{\mathbf{k},\sigma,f}) - V_\perp (c_{\mathbf{k},\sigma}^\dagger f_{\mathbf{k},\sigma} + \text{h.c.}) \right]. \quad (2.7)$$

The diagonal modes are given by

$$v_{\mathbf{k},\sigma,\pm}^\dagger = C_{\pm,k} c_{\mathbf{k},\sigma}^\dagger + \sqrt{1 - C_{\pm,k}^2} f_{\mathbf{k},\sigma}^\dagger, \quad (2.8)$$

the coefficients obtained from diagonalising the matrix

$$H_{1p}(\mathbf{k}) = \begin{pmatrix} \epsilon_{\mathbf{k}} & -V_\perp \\ -V_\perp & \epsilon_{\mathbf{k}} - \eta_d \end{pmatrix}. \quad (2.9)$$

The energies are given by $\mathcal{E}_\pm(\mathbf{k}) = \epsilon_{\mathbf{k}} - \eta_d/2 \pm \Delta$, with $\epsilon_{\mathbf{k}}$ being $-\frac{4V}{Z}(\cos k_x + \cos k_y)$ for a 2D square lattice and $\Delta = \sqrt{V_\perp^2 + \eta_d^2/4}$. Substituting into the expression for the lattice Greens function, we get

$$G(\mathbf{k}, d; \omega) = C_{+,k} \mathcal{G}(\mathcal{T}_{\mathbf{k},+,d}; \omega - (\epsilon_{\mathbf{k}} - \eta_d/2 + \Delta)) + C_{-,k} \mathcal{G}(\mathcal{T}_{\mathbf{k},-,d}; \omega - (\epsilon_{\mathbf{k}} - \eta_d/2 - \Delta)). \quad (2.10)$$

Following eq. 5.32, we can similarly write down an expression the local lattice Greens function, in real-space:

$$\begin{aligned} G_{\text{loc}}(\omega) &= \mathcal{G}(d\sigma; \omega) + \sum_{\mathbf{k}} \mathcal{G}(d\sigma, \mathcal{T}_{\mathbf{k},\sigma}; \omega) \\ &+ \sum_{\mathbf{k},\alpha=\pm} C_{\alpha,k} \left[\mathcal{G}^>(\mathcal{T}_{\mathbf{k},\alpha,d}, d\sigma; \omega - \epsilon_{\mathbf{k}} + \eta_d/2 + \alpha\Delta) + \mathcal{G}^<(\mathcal{T}_{\mathbf{k},\alpha,d}, d\sigma; \omega + \epsilon_{\mathbf{k}} - \eta_d/2 - \alpha\Delta) \right] \\ &+ \sum_{\mathbf{k},\mathbf{q},\alpha=\pm} C_{\alpha,k} \left[\mathcal{G}^>(\mathcal{T}_{\mathbf{k},\alpha,d}, \mathcal{T}_{\mathbf{q},\sigma,d}; \omega - \epsilon_{\mathbf{k}} + \eta_d/2 + \alpha\Delta) + \mathcal{G}^<(\mathcal{T}_{\mathbf{k},\alpha,d}, \mathcal{T}_{\mathbf{q},\sigma,d}; \omega + \epsilon_{\mathbf{k}} - \eta_d/2 - \alpha\Delta) \right] \end{aligned} \quad (2.11)$$

3.3 Unitary RG Analysis of The Bilayer Lattice-Embedded ES-IAM

In the limit of large U_α , we carry out a Schrieffer-Wolff transformation and work with the following low-energy Hamiltonian:

$$H_{\text{aux}} = \sum_{\mathbf{k},\sigma,\alpha} \epsilon_{\mathbf{k},\alpha} n_{\mathbf{k},\sigma,\alpha} + \sum_{\alpha} \sum_{Z \in \text{NN}} \left[\frac{1}{2} J_\alpha \sum_{\sigma,\sigma'} \mathbf{S}_\alpha \cdot \boldsymbol{\sigma}_{\alpha\beta} c_{Z,\sigma,\alpha}^\dagger c_{Z,\sigma',\alpha} - \frac{W_\alpha}{2} (n_{Z,\uparrow,\alpha} - n_{Z,\downarrow,\alpha})^2 \right] + J \mathbf{S}_f \cdot \mathbf{S}_d. \quad (3.1)$$

The Kondo coupling and bath interaction acquire k -dependence:

$$\begin{aligned} J(\mathbf{k}, \mathbf{q}) &= \frac{J}{2} [\cos(\mathbf{k}_x - \mathbf{q}_x) + \cos(\mathbf{k}_y - \mathbf{q}_y)] , \\ W(\mathbf{k}, \mathbf{q}, \mathbf{k}', \mathbf{q}') &= \frac{W}{2} [\cos(\mathbf{k}_x - \mathbf{q}_x + \mathbf{k}'_x - \mathbf{q}'_x) + \cos(\mathbf{k}_y - \mathbf{q}_y + \mathbf{k}'_y - \mathbf{q}'_y)] . \end{aligned} \quad (3.2)$$

The particle-hole transformed state $\bar{\mathbf{q}}$ associated with $\mathbf{q}$ is defined as $\bar{\mathbf{q}} = \mathbf{q} + (\pi, \pi)$ if $\mathbf{q}$ is in the first or third quadrant and $\mathbf{q} + (\pi, -\pi)$ is in the second or fourth quadrant.

In order to study the low-energy physics of the impurity model, we iteratively integrate out high-energy degrees of freedom using the unitary RG method. Let the momentum states being decoupled from the two layers be $|\mathbf{q}, \sigma, f\rangle$ and $|\mathbf{q}, \sigma, d\rangle$ if they are above the Fermi energy (unoccupied in the low-energy configuration) and $|\bar{\mathbf{q}}, \sigma, f\rangle$ and $|\bar{\mathbf{q}}, \sigma, d\rangle$ if they are below the Fermi energy (occupied in the low-energy configuration). For every shell of conduction bath states of width ΔD integrated out, the Hamiltonian couplings renormalise by folding in the effects of virtual excitations to these states. We already have the renormalisation group equations for the couplings J_α in the case of $J = 0$:

$$\frac{\Delta J_f(\mathbf{k}_1, \mathbf{k}_2)}{\Delta D} = \int_{UV} d\mathbf{q} \rho(\mathbf{q}) [J_f(\mathbf{k}_1, \mathbf{q})J_f(\mathbf{q}, \mathbf{k}_2) + J_f(\mathbf{q}, \bar{\mathbf{q}})W_f(\mathbf{k}_1, \mathbf{q}, \bar{\mathbf{q}}, \mathbf{k}_2)] G_f^{(0)}(\mathbf{q}) , \quad (3.3)$$

where $\mathbf{q}$ sums over the UV states being integrated out, $\rho(\mathbf{q})$ is the density of states for the momentum states being integrated out and $G_f^{(0)}$ is the propagator for the excited intermediate state (for $J = 0$):

$$G_f^{(0)}(\mathbf{q}) = \frac{1}{\omega - |\epsilon_f(\mathbf{q})|/2 + J_f(\mathbf{q})/4 + W_f(\mathbf{q})/2} . \quad (3.4)$$

An identical equation holds for ΔJ_d , with all f -quantities replaced with d -quantities. Switching on J modifies the excitation energy in the propagator:

$$G_f(\mathbf{q}) = \frac{1}{1/G_f^{(0)} - J\vec{S}_f \cdot \vec{S}_d} . \quad (3.5)$$

This expression can be simplified by noting that the operator in the denominator has only two eigenvalues, $-3/4$ in the singlet sector and $1/4$ in the triplet sector. Defining the projectors $\mathcal{P}_0$ and $\mathcal{P}_1$ for the two sectors, we have

$$\vec{S}_f \cdot \vec{S}_d = \frac{-3}{4}\mathcal{P}_0 + \frac{1}{4}\mathcal{P}_1, \quad \mathcal{P}_0 + \mathcal{P}_1 = \mathbb{I} . \quad (3.6)$$

Using these relations, we can write

$$\begin{aligned} G_f(\mathbf{q}) &= \frac{1}{1/G_f^{(0)} + 3J/4}\mathcal{P}_0 + \frac{1}{1/G_f^{(0)} - J/4}\mathcal{P}_1 \\ &= \left(\frac{1/4}{1/G_f^{(0)} + 3J/4} + \frac{3/4}{1/G_f^{(0)} - J/4} \right) \mathbb{I} + \left(\frac{1}{1/G_f^{(0)} + 3J/4} - \frac{1}{1/G_f^{(0)} - J/4} \right) \vec{S}_f \cdot \vec{S}_d . \end{aligned} \quad (3.7)$$

Noting that only the first operator renormalises the Kondo interaction, we get the following modified RG equation for the Kondo coupling J_α :

$$\frac{\Delta J_\alpha}{\Delta D} = \int_{UV} d\mathbf{q} \rho(\mathbf{q}) [J_\alpha(\mathbf{k}_1, \mathbf{q}) J_\alpha(\mathbf{q}, \mathbf{k}_2) + J_\alpha(\mathbf{q}, \bar{\mathbf{q}}) W_\alpha(\mathbf{k}_1, \mathbf{q}, \bar{\mathbf{q}}, \mathbf{k}_2)] \left(\frac{1}{4} \frac{1}{1/G_\alpha^{(0)}(\mathbf{q}) + 3J/4} + \frac{3}{4} \frac{1}{1/G_\alpha^{(0)}(\mathbf{q}) - J/4} \right), \quad (3.8)$$

We now turn to the renormalisation of J , arising from hybridisation of f - and d -layers. In order to capture coherent scattering processes between the two layers, we project onto the following rotated basis of excited states:

$$|H\rangle_\pm = \frac{1}{\sqrt{2}} \left(|H(\mathbf{q})\rangle_f \pm |H(\mathbf{q})\rangle_d \right), \quad (3.9)$$

$$\mathcal{P}(\mathbf{q})_H = |H\rangle_+ \langle H|_+ + |H\rangle_- \langle H|_- ,$$

where $|H(\mathbf{q})\rangle_\alpha$ is the state obtained upon exciting both states in the layer α : $|H(\mathbf{q})\rangle_\alpha = c_{\mathbf{q},\alpha}^\dagger c_{\bar{\mathbf{q}},\alpha} |L\rangle_f |L\rangle_d$, and $\mathcal{P}(\mathbf{q})_H$ projects onto this rotated excited basis. We focus on the spin-flip component $J S_f^+ S_d^-$ of the Hamiltonian. There are two kinds of processes that renormalise this coupling. We first consider one that involves a spin-flip of the d -layer followed by a spin-flip of the f -layer:

$$\Delta H = \frac{1}{N^2} \sum_{\mathbf{q} \in UV} \langle L | \frac{1}{2} J_f(\bar{\mathbf{q}}, \mathbf{q}) S_f^+ c_{\mathbf{q}-\downarrow, f}^\dagger c_{\mathbf{q}+\uparrow, f} \mathcal{P}(\mathbf{q})_H G \mathcal{P}(\mathbf{q})_H \frac{1}{2} J_d(\mathbf{q}, \bar{\mathbf{q}}) S_d^- c_{\mathbf{q}\uparrow, d}^\dagger c_{\bar{\mathbf{q}}\downarrow, d} | L \rangle, \quad (3.10)$$

where $|L\rangle = |L\rangle_f |L\rangle_d$ is the initial configuration for both layers. G is the propagator for the excited state:

$$G = \frac{1}{\omega - H_D}, \quad (3.11)$$

where H_D is the diagonal part of the Hamiltonian corresponding to the excited(intermediate) state. The diagonal part consists of the kinetic energy and the Ising part of the Kondo interaction:

$$H_D = \sum_{\mathbf{q}, \sigma \in \mathbf{q}_\pm, \alpha} \epsilon_\alpha(\mathbf{q}) \tau_{\mathbf{q}, \sigma} + \frac{1}{2} \sum_{\alpha} J_\alpha S_\alpha^z \sum_{\mathbf{q}, \sigma} \sigma n_{\mathbf{q}, \sigma, \alpha} + J \vec{S}_f \cdot \vec{S}_d. \quad (3.12)$$

Calculating the matrix elements of the propagator in the rotated basis gives

$$\mathcal{P}(\mathbf{q})_H G \mathcal{P}(\mathbf{q})_H = \frac{1}{2} (G_f + G_d) (|H\rangle_+ \langle H|_+ + |H\rangle_- \langle H|_-) + \frac{1}{2} (G_f - G_d) (|H\rangle_+ \langle H|_- + |H\rangle_- \langle H|_+), \quad (3.13)$$

where

$$G_\alpha(\mathbf{q}) = \frac{1}{\omega - \frac{1}{2} [\epsilon_\alpha(\mathbf{q}) - \epsilon_\alpha(\bar{\mathbf{q}})] + \frac{1}{2} J_\alpha(\mathbf{q}) + W_\alpha(\mathbf{q}) - \frac{1}{4} J}. \quad (3.14)$$

In eq. 3.10, since the first process is an excitation into a d -state and the second process is a de-excitation from an f -state, the expression can be simplified into

$$\begin{aligned} \langle L | S_f^+ c_{\mathbf{q}-\downarrow, f}^\dagger c_{\mathbf{q}+\uparrow, f} \mathcal{P}(\mathbf{q})_H G \mathcal{P}(\mathbf{q})_H S_d^- c_{\mathbf{q}+\uparrow, d}^\dagger c_{\mathbf{q}-\downarrow, d} | L \rangle &= S_f^+ S_d^- \langle H_f | \mathcal{P}(\mathbf{q})_H G \mathcal{P}(\mathbf{q})_H | H_d \rangle \\ &= \frac{1}{2} (G_f + G_d) S_f^+ S_d^- . \end{aligned} \quad (3.15)$$

The net Hamiltonian renormalisation is finally a sum over the contribution from each mode:

$$\Delta H = \frac{1}{8N^2} \sum_{\mathbf{q} \in \text{UV}} J_f(\bar{\mathbf{q}}, \mathbf{q}) J_d(\mathbf{q}, \bar{\mathbf{q}}) (G_f(\mathbf{q}) + G_d(\mathbf{q})) S_f^+ S_d^- . \quad (3.16)$$

Converting the sums to integrals (assuming a density of states $\rho(\mathbf{q})$) gives the final expression

$$\frac{\Delta H}{\Delta D} = \frac{1}{8N} S_f^+ S_d^- \int_{\text{UV}} d\mathbf{q} \rho(\mathbf{q}) J_f(\bar{\mathbf{q}}, \mathbf{q}) J_d(\mathbf{q}, \bar{\mathbf{q}}) (G_f(\mathbf{q}) + G_d(\mathbf{q})) . \quad (3.17)$$

The time-reversed of this process can be obtained simply by exchanging the d -interaction with the f -interaction. Since the renormalisation itself is symmetric, the contribution is equal to what we obtained here. The total renormalisation in the coupling for the term $S_f^+ S_d^-$ is therefore

$$\begin{aligned} \frac{\Delta J}{\Delta D} &= \frac{1}{2} \int_{\text{UV}} d\mathbf{q} \rho(\mathbf{q}) J_f(\bar{\mathbf{q}}, \mathbf{q}) J_d(\mathbf{q}, \bar{\mathbf{q}}) (G_f(\mathbf{q}) + G_d(\mathbf{q})) \\ &= \frac{1}{2} \int_{\text{UV}} d\mathbf{q} \rho(\mathbf{q}) \left[\frac{J_f(\bar{\mathbf{q}}, \mathbf{q}) J_d(\mathbf{q}, \bar{\mathbf{q}})}{\omega - \frac{1}{2} \bar{\epsilon}_f(\mathbf{q}) + \frac{1}{2} J_f(\mathbf{q}) + W_f(\mathbf{q}) - \frac{1}{4} J} + \frac{J_f(\bar{\mathbf{q}}, \mathbf{q}) J_d(\mathbf{q}, \bar{\mathbf{q}})}{\omega - \frac{1}{2} \bar{\epsilon}_d(\mathbf{q}) + \frac{1}{2} J_d(\mathbf{q}) + W_d(\mathbf{q}) - \frac{1}{4} J} \right] , \end{aligned} \quad (3.18)$$

where $\bar{\epsilon}_\alpha(\mathbf{q}) = \epsilon_f(\mathbf{q}) - \epsilon_f(\bar{\mathbf{q}})$ is the total excitation energy for the pair of states $\mathbf{q}, \bar{\mathbf{q}}$.

Since the bath interaction doesn't renormalise, it's functional form remains unchanged and can be substituted into the RG equations. In fact, we can write the expressions in a compact matrix form for efficient computation:

$$\frac{\Delta J_\alpha}{\Delta D} = J_\alpha \mathcal{G}_\alpha J_\alpha^\dagger + \mathcal{W}_\alpha \text{Tr} [\mathcal{G}'_\alpha] , \quad (3.19)$$

where all objects are matrices. J_α is the Kondo coupling matrix. $\mathcal{G}_\alpha$ is a diagonal matrix with matrix elements

$$\mathcal{G}_\alpha[q, q] = \begin{cases} \rho(q) \left(\frac{1}{8} \frac{1}{1/G_\alpha^{(0)}(\mathbf{q}) + \mu/2 + 3J/4} + \frac{3}{8} \frac{1}{1/G_\alpha^{(0)}(\mathbf{q}) + \mu/2 - J/4} \right), & \mathbf{q} \in \text{PS} , \\ \rho(q) \left(\frac{1}{8} \frac{1}{1/G_\alpha^{(0)}(\mathbf{q}) - \mu/2 + 3J/4} + \frac{3}{8} \frac{1}{1/G_\alpha^{(0)}(\mathbf{q}) - \mu/2 - J/4} \right), & \mathbf{q} \in \text{HS} , \end{cases} \quad (3.20)$$

if q is in the UV subspace and zero otherwise. $\mathcal{W}_\alpha$ is the bath interaction matrix whose matrix elements are given by $\mathcal{W}_\alpha[k, k'] = \frac{W}{2} [\cos(k_x - k'_x) + \cos(k_y - k'_y)]$. Finally, $\mathcal{G}'_\alpha$ is another diagonal matrix whose matrix elements are defined as $\mathcal{G}'_\alpha[q, q] = \mathcal{G}[q, q] J_\alpha[q, \bar{q}]$.

The RG equation for the inter-layer coupling J can be written as

$$\frac{\Delta J}{\Delta D} = \frac{1}{2} \mathbf{J} \cdot \mathbf{G} , \quad (3.21)$$

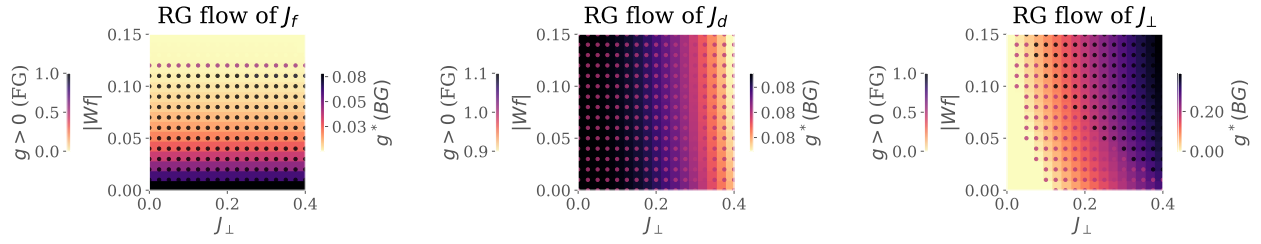


Figure 3.1: RG phase diagrams for various Kondo couplings: f -layer (left), d -layer (middle) and inter-layer (right). For each plot, the background color (quantified by the right colorbar) tracks the fixed point value of the coupling as a function of the inter-layer couplings $J_\perp$ and the f -layer bath interaction $|W_f|$. The marker color complements this by tracking whether the coupling is relevant, intermediate-coupling or irrelevant in the RG sense, by taking the values 1, 1/2 or 0 respectively.

where $\mathbf{J}$ and $\mathbf{G}$ are vectors defined by the elements $\mathbf{J}[q] = J_f(\bar{\mathbf{q}}, \mathbf{q})J_d(\mathbf{q}, \bar{\mathbf{q}})$ and

$$\mathbf{G}[q] = \rho(\mathbf{q}) \left[\frac{1}{\omega - \frac{1}{2}\bar{\epsilon}_f(\mathbf{q}) + \frac{1}{2}J_f(\mathbf{q}) + W_f(\mathbf{q}) - \frac{1}{4}J} + \frac{1}{\omega - \frac{1}{2}\bar{\epsilon}_d(\mathbf{q}) + \frac{1}{2}J_d(\mathbf{q}) + W_d(\mathbf{q}) - \frac{1}{4}J} \right]. \quad (3.22)$$

3.4 Renormalisation Group Flows of Kondo Couplings: RG Phase Diagram

We first analyse the model by obtaining unitary RG flows of three kinds of Kondo couplings. Fig. 3.1 shows the fixed point values of the couplings under renormalisation to a low-energy theory. We view these values as a function of the f -layer bath interaction W_f and inter-layer Kondo coupling $J_\perp$. In each panel, the right (left) colorbar quantifies the background (marker) color of the plot. The background displays the fixed point value of each coupling whereas the marker color tracks whether the couplings is relevant (1), intermediate coupling (0.5) or irrelevant (0).

The Kondo coupling J_d (middle panel) of the d -layer remains relevant for the entirety of the chosen parameter regime. The Kondo coupling J_f (left panel) of the f -layer becomes RG-irrelevant for larger values of $|W_f|$; this leads to an in-plane Mott transition of the kind observed in our earlier work, where the local Fermi liquid excitations of the f -impurity give way to a pseudogapped non-Fermi liquid Mott metal phase which ultimately destabilises into a local moment phase. The RG flow of J_f does not appear to be significantly affected by the variation of $J_\perp$. Concomitant with the irrelevance of J_f under increasing $|W_f|$, the inter-plane coupling $J_\perp$ (right panel) becomes RG-relevant, as seen from the darkening markers as we go to higher $|W_f|$.

These simulations motivate an RG-based phase diagram with the following regimes: (i) weak $J_\perp$, weak $|W_f|$: decoupled metallic layers, each layer participating in transport independently, (ii) strong $J_\perp$, strong $|W_f|$: Kondo insulator regime, where the local moments of the f -layer hybridise with the itinerant electrons of the d -layer to form a band insulator with the chemical potential situated in the hybridisation gap, and (iii) weak $J_\perp$, strong $|W_f|$: the so-called RKKY regime, where the f -layer

forms local moments which decouple from the d -layer and are then free to order magnetically, and (iv) strong $J_\perp$, weak $|W_f|$: this is similar to the Kondo insulator regime, except here the f -electrons remain itinerant and it is they who hybridise with the d -electrons to again form two bands separated by a hybridisation gap.

3.5 Results at half-filling

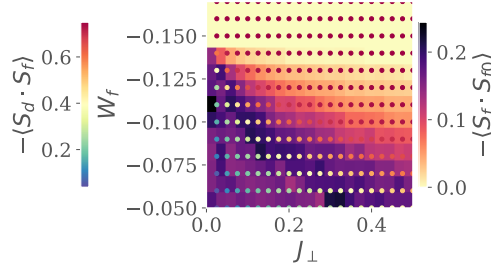


Figure 3.2: Left: Real-space spin correlations for the auxiliary model. Background colour (right colourbar) represents that between the f -impurity and its nearest-neighbour sites, while the marker colour (left colourbar) represents that between the f -impurity and the d -impurity. While the former shrinks as $J_\perp$ or $|W_f|$ is increased, the latter grows. This marks a transfer of entanglement from intra-plane channel into inter-plane channel. Right:

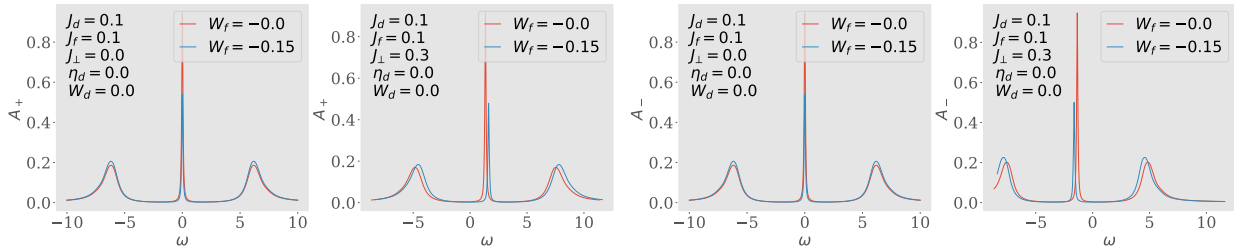


Figure 3.3: Spectral functions of the tiled model, for excitations in the bonding (two on the left) and antibonding (two on the right) basis, defined by the Greens function $G_{k,\pm} = G(k_d \pm k_f; \omega)$. There is no hybridisation for $J_\perp = 0$ (first and third plots), and the only effect of W_f is to shrink the height of the central resonance. Introducing a non-zero $J_\perp$ splits the central resonance into an upper band (second from left) and a lower band (first from right). This leads to a Kondo insulator, where $\omega = 0$ resides inside the hybridisation gap.

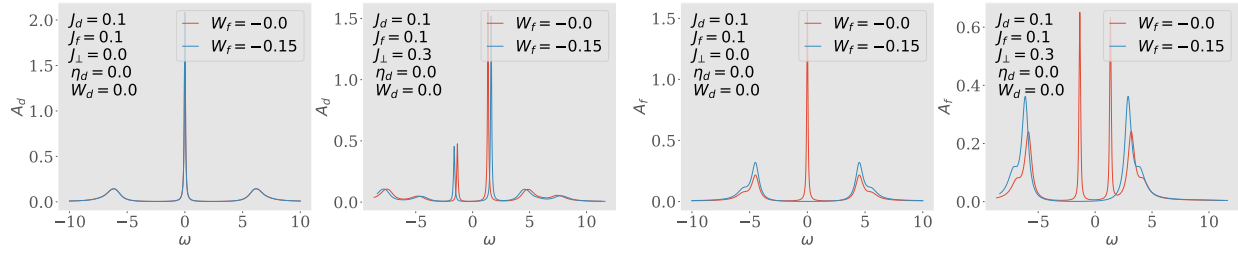


Figure 3.4: Spectral functions A_d and A_f of the auxiliary model, for excitations in the d - and f -layers respectively. Both show a central Kondo resonance for $J_\perp = 0$ which splits into two bands upon adding a non-zero $J_\perp$. Increasing $|W_f|$ to large values gaps out the local Fermi liquid excitations of A_f .

Bibliography

- [1] Anirban Mukherjee and Siddhartha Lal. Holographic unitary renormalisation group for correlated electrons-i: a tensor network approach. *Nuclear Physics B*, 960:115170, 2020.
- [2] Anirban Mukherjee and Siddhartha Lal. Holographic unitary renormalisation group for correlated electrons-ii: insights on fermionic criticality. *Nuclear Physics B*, 960:115163, 2020.
- [3] Anirban Mukherjee and Siddhartha Lal. Scaling theory for mott–hubbard transitions: I. $t = 0$ phase diagram of the 1/2-filled hubbard model. *New Journal of Physics*, 22(6):063007, jun 2020.
- [4] Anirban Mukherjee and Siddhartha Lal. Scaling theory for mott–hubbard transitions-II: quantum criticality of the doped mott insulator. *New Journal of Physics*, 22(6):063008, jun 2020.
- [5] Kenji Suzuki and Ryoji Okamoto. General structure of effective interaction in degenerate perturbation theory. *Progress of Theoretical Physics*, 71(6):1221–1238, 1984.
- [6] PW Anderson. A poor man’s derivation of scaling laws for the kondo problem. *Journal of Physics C: Solid State Physics*, 3(12):2436, 1970.
- [7] FDM Haldane. Scaling theory of the asymmetric anderson model. *Physical Review Letters*, 40(6):416, 1978.
- [8] Tathagata Chowdhury and Kevin Ingersent. Critical charge fluctuations in a pseudogap anderson model. *Phys. Rev. B*, 91:035118, Jan 2015.
- [9] Rukhsan-Ul-Haq Wani and Vidhyadhiraja N S. Scaling analysis of the extended single impurity anderson model: Renormalization due to valence fluctuations. 10 2017.
- [10] J H Jefferson. a renormalisation group approach to the mixed valence problem. *Journal of Physics C: Solid State Physics*, 10(18):3589–3599, sep 1977.
- [11] Stefan Kehrein. *The flow equation approach to many-particle systems*, volume 217. Springer, 2007.
- [12] S. D Glazek and Kenneth G Wilson. Renormalization of hamiltonians. *Physical Review D*, 48(12):5863, 1993.
- [13] Franz Wegner. Flow-equations for hamiltonians. *Annalen der physik*, 506(2):77–91, 1994.

- [14] Barbara M. Terhal, Michael M. Wolf, and Andrew C. Doherty. Quantum entanglement: A modern perspective. *Physics Today*, 56(4):46–52, 2003.
- [15] Ryszard Horodecki, Paweł Horodecki, Michał Horodecki, and Karol Horodecki. Quantum entanglement. *Reviews of modern physics*, 81(2):865, 2009.
- [16] J. Eisert, M. Cramer, and M. B. Plenio. Colloquium: Area laws for the entanglement entropy. *Rev. Mod. Phys.*, 82:277–306, Feb 2010.
- [17] Nicolas Laflorencie. Quantum entanglement in condensed matter systems. *Physics Reports*, 646:1–59, 2016.
- [18] Bei Zeng, Xie Chen, Duan-Lu Zhou, and Xiao-Gang Wen. *Quantum information meets quantum matter*. Springer, 2019.
- [19] Manuel Erhard, Mario Krenn, and Anton Zeilinger. Advances in high-dimensional quantum entanglement. *Nature Reviews Physics*, 2(7):365–381, Jul 2020.
- [20] X. G. Wen. Topological orders in rigid states. *International Journal of Modern Physics B*, 04(02):239–271, 1990.
- [21] X. G. Wen and Q. Niu. Ground-state degeneracy of the fractional quantum hall states in the presence of a random potential and on high-genus riemann surfaces. *Phys. Rev. B*, 41:9377–9396, May 1990.
- [22] Xiao-Gang Wen. Topological orders and edge excitations in fractional quantum hall states. *Advances in Physics*, 44(5):405–473, 1995.
- [23] Qian Niu, Ds J Thouless, and Yong-Shi Wu. Quantized hall conductance as a topological invariant. *Physical Review B*, 31(6):3372, 1985.
- [24] Saurya Das and S. Shankaranarayanan. How robust is the entanglement entropy-area relation? *Phys. Rev. D*, 73:121701, Jun 2006.
- [25] M. Cramer, J. Eisert, M. B. Plenio, and J. Dreißig. Entanglement-area law for general bosonic harmonic lattice systems. *Phys. Rev. A*, 73:012309, Jan 2006.
- [26] Samuel L Braunstein and Carlton M Caves. Statistical distance and the geometry of quantum states. *Physical Review Letters*, 72(22):3439, 1994.
- [27] Philipp Hauke, Markus Heyl, Luca Tagliacozzo, and Peter Zoller. Measuring multipartite entanglement through dynamic susceptibilities. *Nature Physics*, 12(8):778–782, Aug 2016.
- [28] Richard M. Martin, Lucia Reining, and David M. Ceperley. *Interacting Electrons: Theory and Computational Approaches*. Cambridge University Press, 2016.
- [29] Siddhartha Patra and Siddhartha Lal. Origin of topological order in a cooper-pair insulator. *Phys. Rev. B*, 104:144514, Oct 2021.

- [30] Santanu Pal, Anirban Mukherjee, and Siddhartha Lal. Correlated spin liquids in the quantum kagome antiferromagnet at finite field: a renormalization group analysis. *New Journal of Physics*, 21(2):023019, feb 2019.
- [31] Daria Gazizova and J. P. F. LeBlanc. Emergent nearest-neighbor attraction in the fully renormalized interactions of the single-band repulsive hubbard model at weak coupling, arxiv:2307.02360, 2023.
- [32] Alexandrina Stoyanova. *Delocalized and correlated wave functions for excited states in extended systems*. PhD thesis, University of Groningen, 2006.
- [33] Abhirup Mukherjee, N S Vidhyadhiraja, A Taraphder, and Siddhartha Lal. Kondo frustration via charge fluctuations: a route to mott localisation. *New Journal of Physics*, 25(11):113011, nov 2023.